

Asymptotically Optimal Mixed-State Cloning

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Abstract

Unknown quantum states can only be cloned approximately. Although the optimal pure-state cloner has long been known, the mixed-state problem remains challenging. We determine the asymptotically optimal global n -to- m_n cloning root fidelity on every fixed-rank stratum with a simple positive spectrum, in fixed dimension and with $m_n/n \rightarrow \gamma > 1$. If the rank r and the positive spectrum $p_1 > \dots > p_r > 0$ are known, the minimax fidelity on the corresponding unitary orbit is

$$F_{\text{orb}}^{(r,d)}(\gamma, p) = \gamma^{-r(d-r)/2} \prod_{1 \leq i < j \leq r} \frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}}, \quad q_{ij} = \frac{p_j}{p_i}.$$

If the spectrum is unknown but the rank is known, a single rank-aware cloner guarantees

$$F_{\text{univ}}^{(r,d)}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(r-1)/2} F_{\text{orb}}^{(r,d)}(\gamma, p),$$

and this gives the minimax value on every regular compact subset of the rank- r spectral chamber. The case $r = d$ recovers the full-rank interior case without a separate argument. We also solve the positively degenerate flat-spectrum orbit: for $\rho = P/r$, $P \in \text{Gr}(r, d)$, the optimal root fidelity is $\gamma^{-r(d-r)/2}$. Unlike fidelity-threshold copy-complexity results, our theorems make no near-unit-fidelity assumption: the fixed gain $\gamma > 1$ is arbitrary, and the displayed formulas give the explicit, generally nonunit, asymptotic fidelity. We further upper-bound the actual mixed-system output of purify-clone-trace and prove that our cloner has strictly larger asymptotic fidelity for every simple positive spectrum of rank $r > 1$ and every flat projector orbit with $r > 1$; for $r = 1$, the protocols coincide. The proofs combine Schur–Weyl label processing, an exact rank-compression identity, Cartan-sector channels, and fixed-rank local asymptotic normality in its interior and boundary forms.

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1 Introduction

The no-cloning theorem rules out perfect copying of an arbitrary unknown quantum state [WZ82, Die82]. Approximate cloning asks instead how well a quantum channel can transform n identical

inputs into an output resembling m ideal copies. For pure states, early universal qubit machines were followed by exact solutions for the standard global and single-clone criteria [BH96, GM97, BDE+98]: Werner solved the global universal problem in arbitrary dimension [Wer98], while Keyl–Werner established optimality when individual clones are tested [KW99]. State-dependent global cloning was developed in [BDE+98, CB99]; see also the reviews [SIGA05, FWJ+14].

The choice of figure of merit is essential. At the level of individual clones, the many-output limit is closely tied to state estimation [MP95, BEM98, BA06, CD06]. When all outputs are examined jointly, the relation is subtler: global asymptotic cloning has been analyzed for broad pure-state families, and an optimal measure-and-prepare output need not consist of independent copies of the estimated state [CY14]. The dependence of the optimum on a joint, marginal, or process-level criterion is already visible for coherent-state and phase-covariant families [CKNWW05, KC22]. Here we use root fidelity with the entire product target $\rho^{\otimes m}$. This global criterion is stronger than reproducing one-site marginals and is therefore distinct from broadcasting, which permits correlations among the outputs [BCFJS96].

Mixed states do not evade the exact no-cloning obstruction: the general no-broadcasting theorem rules out a single channel that clones every member of a noncommuting family of density matrices, with an entropic proof given by Kalev–Hen [BCFJS96, KH08]. Approximate mixed-state cloning poses an additional structural difficulty: both the eigenvalues and the eigenbasis may be unknown, and the two kinds of information behave differently under repeated sampling. In contrast to the pure-state case, earlier mixed-state results either did not identify an optimal cloner with a matching fixed-gain value or applied only to special state families or different criteria. Work directly on mixed-state cloning derived global-fidelity, relative-error, and state-dependent bounds for prescribed finite ensembles [Ras03a, Ras03b, Ras03c]; constructed cloners for symmetric-subspace inputs and identical mixed qubits under one-site shrinking-factor criteria [Fan03, FLS07]; studied a restricted mixed-qubit construction under an averaged Hilbert–Schmidt criterion [Kaz10]; and established general entropic limitations on simultaneous approximate cloning and broadcasting of mixed states [LW17]. Closely related work solved optimal mixed-qubit purification and a local-fidelity cloning/estimation problem, rather than the joint product-fidelity task considered here [CEM99]. Multi-input mixed-state work has also focused on broadcasting and superbroadcasting, which optimize marginals rather than the joint product target [DMP05, BDMP05, BDMP06, DF07]. More recently, the random-purification channel of Tang–Wright–Zhandry was combined with pure-state cloning to give a general all-output guarantee, rather than an exact fixed-gain optimum, for rank-constrained mixed states [TWZ25, LTHC26]. Concurrent work has since asked how many inputs are needed to guarantee squared fidelity at least $1 - \epsilon$, uniformly over states of rank at most r , and shown that this purify–clone–trace construction has optimal copy complexity up to constants [FGSS26, JSO26]. The resulting $1/\epsilon$ behavior describes the near-unit-fidelity boundary. Our primary question imposes no such accuracy requirement: we fix an arbitrary $\gamma = m_n/n > 1$, determine the explicit asymptotic fidelity curve, which may remain bounded away from one, and prove that it lies strictly above the actual PCT fidelity in the cases covered by our comparison theorem. See Section 1.3.

The mixed-state cloning problem admits two models. In the known-spectrum problem, the eigenvalues are supplied and the state ranges over one unitary orbit; cloning must reproduce only the unknown eigenbasis. In the unknown-spectrum problem, a single rank-aware but otherwise spectrum-independent sequence of channels must handle both the positive eigenvalues and the eigenbasis, uniformly over compact subsets of a fixed-rank spectral chamber. The second problem is not merely the first preceded by spectrum estimation: the measured spectrum retains fluctuations of order $n^{-1/2}$, which must themselves be converted from the n -copy to the m_n -copy scale.

Here we resolve the mixed-state cloning problem in the proportional-output asymptotic regime

on every fixed-rank stratum with distinct positive eigenvalues. For every fixed dimension and $m_n/n \rightarrow \gamma > 1$, we determine the optimal global cloning fidelity on each such orbit and, when the positive spectrum is unknown but its rank is known, uniformly over regular compact spectral sets. We additionally solve the flat positive spectrum on the Grassmannian projector orbit. Thus our primary result is a fixed-gain fidelity-curve theorem, not merely a threshold statement near fidelity one: γ need not be close to one, and the limiting fidelity may be nonunit. For every rank $r > 1$ state with simple positive spectrum, and for every flat projector state with $r > 1$, our achievable fidelity is also strictly larger than an upper bound on the actual post-trace PCT output.

Two complementary decompositions drive the solution. Schur–Weyl duality separates the spectral information, carried by the Young label, from the eigenbasis information, carried by the conditional irreducible state [KW01]. Our cloners process the label classically and apply a multiplicity-one Cartan-sector channel to the irreducible state [PRV67, FH91]. Two-way local asymptotic normality (LAN) gives the corresponding asymptotic separation into $r - 1$ classical spectral coordinates, $\binom{r}{2}$ displaced thermal modes within the support, and $r(d - r)$ coherent support–kernel modes [GJ07, GK06, KG09, LN24]. It transfers any putative improvement to a Gaussian dilation or amplification problem. Quantum amplifier noise limits [Cav82] and the least-noise mixed-Gaussian cloner [GM06] identify the quantum benchmark; a unified flat-prior fidelity argument gives the classical and quantum converses used here.

1.1 Results

Fix positive integers m_n such that

$$\gamma_n := \frac{m_n}{n} \longrightarrow \gamma > 1, \quad r_d := \binom{d}{2}.$$

In particular, $m_n > n$ for all sufficiently large n . For the finitely many indices with $m_n \leq n$, every cloner below is defined by the marginal channel, with the empty partial trace interpreted as the identity. Thus all nontrivial formulas concern $m_n > n$.

For $1 \leq r \leq d$, write

$$\mathbf{A}_r := \left\{ x \in \mathbb{R}^r : \sum_{i=1}^r x_i = 1 \right\}, \quad \Delta_r^\circ := \{ p \in \mathbf{A}_r : p_1 > \cdots > p_r > 0 \},$$

and, for $p \in \Delta_r^\circ$, set

$$\bar{p} = (p_1, \dots, p_r, 0, \dots, 0) \in \mathbb{R}^d, \quad \rho_{\bar{p}} = \text{diag}(\bar{p}), \quad \rho_{\bar{p},g} = g \rho_{\bar{p}} g^*.$$

The active orbital roots are

$$\mathcal{I}_{r,d} := \{(i, j) : 1 \leq i \leq r, i < j \leq d\}, \quad q_{ij} := \frac{\bar{p}_j}{\bar{p}_i} = \begin{cases} p_j/p_i, & j \leq r, \\ 0, & j > r. \end{cases} \quad (1)$$

Thus positive–positive roots give thermal modes, support–kernel roots give vacuum modes, and kernel–kernel roots are absent because they belong to the stabilizer. When $r = d$, we suppress the bar and write ρ_p and $\rho_{p,g}$. For positive trace-class operators A, C , not necessarily normalized, we write

$$F(A, C) = \|\sqrt{A}\sqrt{C}\|_1.$$

For states this is the unsquared Uhlmann fidelity. For nonnegative functions or measures, the same notation denotes their Hellinger affinity; in particular, $F(P, Q) = \sum_x \sqrt{P(x)Q(x)}$ on a countable space. The target in every statement below is the entire product state. Define

$$f_\gamma(q) = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q}, \quad c_\gamma = \frac{2\sqrt{\gamma}}{1 + \gamma},$$

and, for $p \in \Delta_r^\circ$,

$$F_{\text{orb}}^{(r,d)}(\gamma, p) = \prod_{(i,j) \in \mathcal{I}_{r,d}} f_\gamma(q_{ij}) = \gamma^{-r(d-r)/2} \prod_{1 \leq i < j \leq r} f_\gamma\left(\frac{p_j}{p_i}\right), \quad (2)$$

$$F_{\text{univ}}^{(r,d)}(\gamma, p) = c_\gamma^{(r-1)/2} F_{\text{orb}}^{(r,d)}(\gamma, p). \quad (3)$$

When $r = d$, we occasionally use the abbreviations

$$F_{\text{orb}}(\gamma, p) := F_{\text{orb}}^{(d,d)}(\gamma, p), \quad F_{\text{cl}}(\gamma, d) := c_\gamma^{(d-1)/2}, \quad F_{\text{univ}}(\gamma, p) := F_{\text{univ}}^{(d,d)}(\gamma, p). \quad (4)$$

The first product in (2) is the unified form of the orbital answer. Each positive–positive root contributes its thermal factor $f_\gamma(p_j/p_i)$, while every support–kernel root has $q_{ij} = 0$ and contributes $f_\gamma(0) = \gamma^{-1/2}$. The additional factor $c_\gamma^{(r-1)/2}$ is the cost of dilating the $r - 1$ unknown positive spectral fluctuations.

Known spectrum Fixing p isolates the genuinely noncommutative part of the problem. The target distribution of Young labels is then known and can be sampled exactly at size m_n ; only the conditional representation sectors, which encode the eigenbasis, need to be cloned.

Theorem 1.1 (Known-spectrum fixed-rank optimum). *For every $1 \leq r \leq d$ and $p \in \Delta_r^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{orb}}^{(r,d)}(\gamma, p), \quad (5)$$

where the supremum is over all quantum channels from n to m_n d -level systems. The explicit Schur–Cartan channel described in [Section 4.2](#) asymptotically attains the displayed value.

Thus each positive–positive pair contributes one displaced-thermal factor, while each support–kernel pair contributes $f_\gamma(0) = \gamma^{-1/2}$. When $r = d$ there are no support–kernel factors, while $r = 1$ recovers Werner’s pure-state value.

Unknown spectrum If p is not supplied, the target Young-label distribution can no longer be sampled directly. The measured input label determines the spectrum only at the $n^{-1/2}$ scale, and converting n copies into $m_n \sim \gamma n$ copies requires dilating this classical fluctuation. This produces the extra, spectrum-independent factor $c_\gamma^{(r-1)/2}$.

Theorem 1.2 (Unknown-spectrum achievability and compact-set optimum). *For every fixed $1 \leq r \leq d$, there exists a sequence of rank-aware, spectrum-independent, $\mathcal{U}(d)$ -covariant channels $\mathcal{C}_n^{(r)}$ with the following properties.*

(a) *For every compact $K \subseteq \Delta_r^\circ$,*

$$\liminf_{n \rightarrow \infty} \inf_{\substack{p \in K \\ g \in \mathcal{U}(d)}} \left[F\left(\mathcal{C}_n^{(r)}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) - F_{\text{univ}}^{(r,d)}(\gamma, p) \right] \geq 0. \quad (6)$$

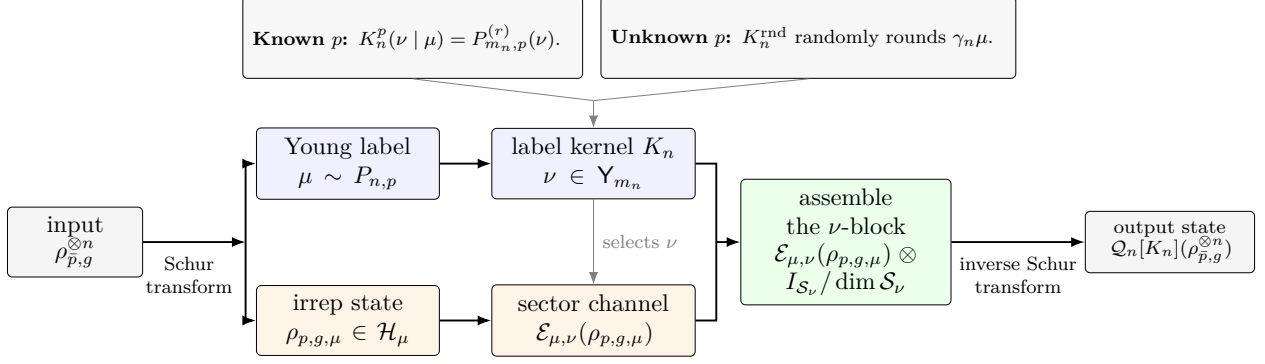


Figure 1: **Cloner Construction.** The Schur transform separates the classical Young label from the conditional irrep state. A Markov kernel selects the output label, while the generic sector channel $\mathcal{E}_{\mu,\nu}$ maps the conditional state into that sector; a maximally mixed Specht factor is then restored. Here $\mathcal{Q}_n[K_n]$ denotes the lifted channel associated with the selected kernel. For known spectrum the target label is sampled independently of μ , whereas for unknown spectrum a randomized dilation is used; these choices produce our cloners for $m_n > n$, namely the known-spectrum and rank-aware universal channels, respectively. On the label pairs carrying asymptotically all probability, the sector fidelity tends to $F_{\text{orb}}^{(r,d)}(\gamma, p)$. Labels have at most r active rows and are padded by $d - r$ zeros; the randomized dilation acts only on those active rows.

(b) Let $K \subseteq \Delta_r^\circ$ be nonempty and satisfy $K = \overline{\text{int}_{A_r} K}^{A_r}$. Then

$$\limsup_{n \rightarrow \infty} \inf_{\substack{\mathcal{M}_n \\ p \in K \\ g \in \text{U}(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \inf_{p \in K} F_{\text{univ}}^{(r,d)}(\gamma, p), \quad (7)$$

where the supremum is over all quantum channels from n to m_n d -level systems. The same sequence $\mathcal{C}_n^{(r)}$, described in [Section 4.2](#), asymptotically attains the displayed value.

Here spectrum independent means independent of p within the specified rank- r stratum; the channel is allowed to know r . When $r = d$, we also write $\mathcal{C}_n^{(d)} = \mathcal{C}_n^{\text{univ}}$.

Theorem 1.3 (Proportional cloning on the Grassmannian projector orbit). *For fixed $1 \leq r < d$,*

$$\limsup_{n \rightarrow \infty} \inf_{P \in \text{Gr}(r,d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) = \gamma^{-r(d-r)/2}. \quad (8)$$

Equivalently, the squared-fidelity value is $\gamma^{-r(d-r)}$. For $r = d$, the orbit is a singleton and the value is trivially one.

The common architecture of the known-spectrum and rank-aware cloners is summarized in [Fig. 1](#). After the Schur transform, the Young label μ carries the spectral information, while the conditional irrep state carries the unknown eigenbasis. A classical kernel chooses an output label ν , and the common covariant sector channel maps the conditional state into the ν -sector before the m_n -system output is reconstructed. For rank r , the relevant labels have at most r rows and are padded by zeros. For known p , the kernel samples the exact target law in effective dimension r ; the unknown-spectrum kernel randomly dilates only the active r rows. Exact rank compression supplies the support-kernel factor. The details are developed in [Sections 2](#) and [4](#).

The minimax meaning of (7) is as follows. The set K may be known when the channel is designed, but the actual spectrum $p \in K$ and eigenbasis g are not. The order of optimization is

important: the supremum first allows a channel sequence tailored to K , and the inner infimum then tests that single sequence against every $p \in K$ and $g \in \mathcal{U}(d)$. Since the unknown-spectrum benchmark $F_{\text{univ}}^{(r,d)}(\gamma, p)$ varies with p , the resulting worst-case scalar is governed by its least favorable value, hence the infimum over p on the right-hand side. No further infimum over g appears there because unitary covariance makes the benchmark constant along the orbit of each fixed spectrum. Thus, even with knowledge of K , no K -adapted sequence of channels can asymptotically beat $\inf_{p \in K} F_{\text{univ}}^{(r,d)}(\gamma, p)$. The single rank- r construction $\mathcal{C}_n^{(r)}$, which does not depend on K , guarantees $F_{\text{univ}}^{(r,d)}(\gamma, p)$ pointwise, uniformly for $p \in K$, and hence asymptotically attains this worst-case value.

The condition $K = \overline{\text{int}_{\mathbf{A}_r} K}^{\mathbf{A}_r}$ is a clean sufficient regularity assumption for full-dimensional spectral uncertainty relative to the fixed-rank face. Its interior contains small neighborhoods in all $r - 1$ spectral directions, and the closure condition allows the infimum over interior points in the converse to be replaced, by continuity of $F_{\text{univ}}^{(r,d)}$, with the infimum over all of K . We do not claim that this condition is necessary. When $r \geq 2$, however, some thickness assumption cannot be removed from a geometry-independent statement of this form. For a singleton $K = \{p\}$ with $r \geq 2$, the channel may be tailored to p , so the problem has the known-spectrum value $F_{\text{orb}}^{(r,d)}(\gamma, p)$, not $F_{\text{univ}}^{(r,d)}(\gamma, p)$. More generally, lower-dimensional spectral classes encode partial spectral knowledge and may have different local Gaussian penalties. Thus (7) is the broadest geometry-independent full-dimensional converse proved here; no statement with the same right-hand side can extend to all compact K when $r \geq 2$. For $r = 1$, the spectral chamber is itself a singleton and $F_{\text{orb}}^{(1,d)} = F_{\text{univ}}^{(1,d)}$, so this distinction disappears.

For example, a natural admissible class keeps the smallest positive eigenvalue and every adjacent positive spectral gap uniformly bounded away from zero. For $r \geq 2$ and $0 < \kappa < 2/[r(r+1)]$,¹ set

$$\Delta_{r,\kappa} = \{p \in \mathbf{A}_r : p_r \geq \kappa, p_i - p_{i+1} \geq \kappa \ (i < r)\}.$$

This is a nonempty compact subset of Δ_r° and satisfies $\Delta_{r,\kappa} = \overline{\text{int}_{\mathbf{A}_r} \Delta_{r,\kappa}}^{\mathbf{A}_r}$, so (7) applies with $K = \Delta_{r,\kappa}$. Indeed, choose $\kappa' \in (\kappa, 2/[r(r+1)])$ and set $p_i = 1/r + ((r+1)/2 - i)\kappa'$. Then $p_i - p_{i+1} = \kappa' > \kappa$ and $p_r > 2/[r(r+1)] > \kappa$, so this is a relative-interior point. The set is closed in the probability simplex, hence compact; being convex with nonempty relative interior gives the stated closure property.

1.2 Remarks

Dependence on spectral mixedness. The product formula makes precise one sense in which a more mixed positive spectrum is easier to clone. For fixed $\gamma > 1$,

$$f'_\gamma(q) = \frac{(\gamma - 1)^2}{2\sqrt{q(\gamma - 1 + q)}(\sqrt{\gamma(\gamma - 1 + q)} + \sqrt{q})^2} > 0, \quad 0 < q < 1.$$

Consequently, if $p, p' \in \Delta_r^\circ$ satisfy

$$\frac{p'_j}{p'_i} \geq \frac{p_j}{p_i} \quad \text{for every } 1 \leq i < j \leq r,$$

then

$$F_{\text{orb}}^{(r,d)}(\gamma, p') \geq F_{\text{orb}}^{(r,d)}(\gamma, p), \quad F_{\text{univ}}^{(r,d)}(\gamma, p') \geq F_{\text{univ}}^{(r,d)}(\gamma, p),$$

¹At $\kappa = 2/[r(r+1)]$, the identity $1 = rp_r + \sum_{i=1}^{r-1} i(p_i - p_{i+1}) \geq \frac{\kappa r(r+1)}{2}$ forces a unique spectrum, so the class is a singleton and its optimum is $F_{\text{orb}}^{(r,d)}$, not $F_{\text{univ}}^{(r,d)}$. The compact-set theorem is not asserted uniformly as the spectrum approaches an eigenvalue collision or the boundary of the probability simplex.

with strict inequality if any ratio is strictly larger. The support–kernel factor is independent of p . Ratio-wise flattening is stronger than majorization when $r \geq 3$; majorization alone need not increase the cloning fidelity.

Limitations and remaining degeneracies. The fixed-rank theorems allow zero eigenvalues but require the positive eigenvalues to be distinct. The Grassmannian theorem separately treats the maximally degenerate positive spectrum $p_i = 1/r$. We do not resolve arbitrary intermediate multiplicity patterns, rank-changing local neighborhoods, or uniformity as $p_r \downarrow 0$, a positive spectral gap closes, or $\gamma \downarrow 1$. We also do not determine the exact finite-copy optimum. The rank-aware universal channel is allowed to know r ; no theorem here gives a single channel uniformly over changing rank. The lack of uniformity as $\gamma \downarrow 1$ concerns triangular joint small-error limits only; it does not restrict any theorem at fixed $\gamma > 1$.

Conjecture for general positive degeneracies. Suppose a rank- r spectrum has distinct positive values $\lambda_1 > \dots > \lambda_s > 0$ with multiplicities $r_1, \dots, r_s$, where $\sum_a r_a = r$. The exact compression identity proved below isolates the zero eigenspace, leaving only the unresolved positive-degeneracy problem in effective dimension r . We conjecture the known-spectrum orbital value

$$\gamma^{-r(d-r)/2} \prod_{a < b} f_\gamma \left(\frac{\lambda_b}{\lambda_a} \right)^{r_a r_b}.$$

For an unknown positive spectrum in a rank-preserving neighborhood that may split the blocks, the conjectural additional classical factor is $c_\gamma^{(\sum_a r_a^2 - 1)/2}$. Rotations within a degenerate block lie in the stabilizer and create no orbital mode, whereas block-splitting directions become classical. The simple-positive-spectrum formulas above and the flat known-spectrum projector theorem are the two proved endpoints; the intermediate positive-degeneracy cases remain open.

1.3 Comparison to purify-and-clone

Concurrent work of Fanizza–Grinko–Scharnhorst–Spilecki and Jeon–Sohn–Oh asks how many inputs are needed to guarantee a prescribed squared fidelity $1 - \epsilon$, uniformly over all states of rank at most r [FGSS26, JSO26]. To put their fidelity-threshold statements and our fixed-gain statements in a common convention, let N be the number of inputs, let M be the number of *additional* outputs, and write

$$m = N + M, \quad \delta = \frac{M}{N}, \quad \gamma = 1 + \delta, \quad t = \frac{N}{M} = \delta^{-1}, \quad \mathcal{F} := F^2.$$

Thus $\mathcal{F}$ is the squared Uhlmann fidelity used in those papers. Uniformly over states of rank at most r , they establish the optimal scaling

$$N = \Theta \left(\frac{Mrd}{\epsilon} \right).$$

Their lower bounds already use flat projector families of rank comparable to r , while their matching upper bounds use purify–clone–trace (PCT). This is a uniform worst-case theorem about the resources required to cross a fidelity threshold; its $1/\epsilon$ behavior describes the near-unit boundary. It does not determine the fidelity as a function of a fixed nonzero output/input ratio, and copy-complexity optimality of PCT does not imply its pointwise fidelity optimality. Our primary theorems instead fix an arbitrary $t > 0$, or equivalently an arbitrary $\gamma > 1$, and determine the exact, generally

nonunit, limiting fidelity at a fixed spectrum. No near-unit-fidelity hypothesis is imposed. The sample-complexity statements below are secondary inversions of these fixed-gain curves.

The PCT protocol first applies random purification [TWZ25], then Werner’s pure-state cloner on the dr -dimensional purification space [Wer98], and finally discards the purifying registers [LTHC26, FGSS26, JSO26]. Its published data-processing guarantee is

$$\mathcal{F}_{\text{PCT}} \geq \frac{\binom{N+dr-1}{dr-1}}{\binom{N+M+dr-1}{dr-1}} \longrightarrow (1+t^{-1})^{-(dr-1)}.$$

The binomial ratio is the exact squared fidelity of the pure Werner step; after the purifying registers are traced out it is only a lower bound on the mixed-system fidelity. Thus comparing this guarantee with our exact values does not by itself order the actual channels.

Appendix A makes that stronger comparison. For every rank- r state with simple positive spectrum and fixed $\gamma > 1$, it proves an explicit upper bound $B_{\text{PCT}}^{(r,d)}(\gamma, p)$ on the actual PCT root fidelity and, for $r > 1$,

$$\limsup_{n \rightarrow \infty} F_{\text{PCT}} \leq B_{\text{PCT}}^{(r,d)}(\gamma, p) < F_{\text{univ}}^{(r,d)}(\gamma, p) < F_{\text{orb}}^{(r,d)}(\gamma, p).$$

This is the central comparison: $B_{\text{PCT}}^{(r,d)}$ upper-bounds the fidelity of the actual mixed-system PCT output and lies strictly below the fidelity achieved by our rank-aware cloner. Because $\gamma > 1$ is arbitrary, the ordering holds across the full fixed-gain regime, including when the limiting fidelity is bounded away from one; it is not inferred by comparing our value with PCT’s published lower guarantee. We next translate this ordering into the two sample-complexity statements most relevant here.

Projector states. Let $\rho = P/r$, with $1 \leq r < d$, and put $a = r(d-r) = \dim_{\mathbb{C}} \text{Gr}(r, d)$. Our exact Grassmannian theorem gives

$$\mathcal{F}_{\text{ours}}^{\text{proj}}(t) = (1+t^{-1})^{-a}.$$

This is the complete proportional-limit curve for every fixed $t > 0$, not a near-unit expansion. The parameter ϵ enters only when we invert that curve into copy-complexity language. In particular, the curve reaches squared fidelity $1 - \epsilon$ at the exact ratio

$$\frac{N}{M} = t_{\text{ours}}^{\text{proj}}(\epsilon) := \frac{1}{(1-\epsilon)^{-1/a} - 1}.$$

The appendix bounds the actual PCT squared fidelity from above by

$$U_{r,d}(t) := (1+t^{-1})^{-a} \left(\frac{\sqrt{1+2t^{-1}}}{1+t^{-1}} \right)^{r-1}.$$

Let $t_B(\epsilon) := \inf\{t > 0 : U_{r,d}(t) \geq 1 - \epsilon\}$, and let

$$t_{\text{guar}}(\epsilon) := \frac{1}{(1-\epsilon)^{-1/(rd-1)} - 1}$$

be the ratio certified by the published PCT lower guarantee. For $1 < r < d$,

$$t_{\text{ours}}^{\text{proj}}(\epsilon) < t_B(\epsilon) \leq t_{\text{guar}}(\epsilon).$$

In words, the upper bound on the *actual* PCT output forces PCT to use at least $Mt_B(\epsilon)$ inputs in the proportional limit, whereas the published guarantee certifies that $Mt_{\text{guar}}(\epsilon)$ inputs suffice. The three small-error expansions make the size of our advantage explicit:

$$\begin{aligned} t_{\text{ours}}^{\text{proj}}(\epsilon) &= \frac{a}{\epsilon} - \frac{a+1}{2} + O(\epsilon), \\ t_B(\epsilon) &= \frac{a}{\epsilon} - \frac{a+1}{2} + \frac{r-1}{2a} + O(\epsilon), \\ t_{\text{guar}}(\epsilon) &= \frac{rd-1}{\epsilon} - \frac{rd}{2} + O(\epsilon). \end{aligned}$$

Thus, relative to the published PCT certificate, our exact projector result improves the leading coefficient from $rd-1$ to $r(d-r)$. Relative to the actual PCT channel, the upper fidelity bound forces PCT to use at least $Mt_B(\epsilon)$ inputs. Hence our cloner saves at least $(r-1)M/[2r(d-r)] + O(M\epsilon)$ inputs within the iterated comparison. The present bounds do not determine whether PCT's actual leading coefficient equals $r(d-r)$ or is larger. For $r=1$ the two protocols coincide, while $r=d$ is the singleton state I_d/d and needs no input copies.

A fixed simple full-rank state. Now let $r=d$, fix $p_1 > \dots > p_d > 0$, set $q_{ij} = p_j/p_i$, and write $\sigma_{n,m_n}^{\text{PCT}}(p)$ for the actual PCT output. For every fixed $\delta > 0$, with $m_n/n \rightarrow 1 + \delta$, the actual-channel comparison already gives

$$\limsup_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}\right)^2 \leq B_{\text{PCT}}(1 + \delta, p)^2 < F_{\text{univ}}(1 + \delta, p)^2.$$

Thus the fidelity advantage does not depend on a near-unit limit. To translate it into small-error copy-complexity notation, we expand our unknown-spectrum fidelity and the actual-PCT upper bound at $\delta = M/N \downarrow 0$:

$$\begin{aligned} F_{\text{univ}}(1 + \delta, p)^2 &= 1 - A_{\text{ours}}(p)\delta^2 + O_p(\delta^3), \\ B_{\text{PCT}}(1 + \delta, p)^2 &= 1 - A_{\text{PCT}}(p)\delta^2 + O_p(\delta^3), \end{aligned}$$

where

$$\begin{aligned} A_{\text{ours}}(p) &= \frac{d-1}{8} + \frac{1}{4} \sum_{i < j} \frac{1}{q_{ij}}, \\ A_{\text{PCT}}(p) &= \frac{d-1}{2} + \frac{1}{4} \sum_{i < j} \frac{(1 + q_{ij})^2}{q_{ij}}, \\ A_{\text{PCT}}(p) - A_{\text{ours}}(p) &= \frac{3(d-1)}{8} + \frac{1}{4} \sum_{i < j} (2 + q_{ij}) > 0. \end{aligned}$$

Since the actual PCT limiting fidelity is at most B_{PCT} , inversion yields

$$\frac{N_{\text{ours}}}{M} \sim \sqrt{\frac{A_{\text{ours}}(p)}{\epsilon}}, \quad \frac{N_{\text{PCT}}}{M} \gtrsim \sqrt{\frac{A_{\text{PCT}}(p)}{\epsilon}}.$$

Thus our cloner has the iterated pointwise scale $N/M \sim \sqrt{A_{\text{ours}}(p)/\epsilon}$, whereas any PCT threshold must satisfy $N/M \gtrsim \sqrt{A_{\text{PCT}}(p)/\epsilon}$. Since $A_{\text{PCT}}(p) > A_{\text{ours}}(p)$, this proves a strict coefficient-level advantage for our cloner. The available bounds do not determine PCT's exact pointwise ϵ -scaling: its necessary $M/\sqrt{\epsilon}$ bound and its published sufficient Md^2/ϵ bound do not match. If the spectrum is known, the still smaller orbital coefficient is obtained by deleting $(d-1)/8$ from $A_{\text{ours}}(p)$. The matching worst-case lower bound over rank-at-most- d states is driven by lower-rank projector states of rank comparable to $d/2$, not by this fixed interior spectrum.

Scope of the sample-complexity translation. The strict fixed-gain fidelity comparisons above hold for every fixed $\gamma > 1$ and require neither $\epsilon \downarrow 0$ nor $\gamma \downarrow 1$. Only their small- ϵ copy-complexity translations are iterated-limit consequences: first $N, M \rightarrow \infty$ at fixed $t = N/M$, and only then $t \rightarrow \infty$. Our present proportional theorem is not uniform for a triangular sequence $\gamma_N \downarrow 1$, so these inversions are not yet finite-sample theorems in the joint regime treated by the two concurrent papers. This qualification does not affect the pointwise fixed-gain separation from actual PCT. The inversions isolate a next-order improvement for projector states and a strict quadratic-coefficient separation from the PCT fidelity upper bound for every fixed simple full-rank spectrum. Full-rank spectra with positive degeneracies remain outside the proved simple-spectrum comparison.

1.4 Earlier and related work

Pure-state cloning and state estimation. Universal qubit cloners were constructed in [BH96, GM97, BDE+98]; the arbitrary-dimensional global and single-clone optima were obtained in [Wer98, KW99], while links to finite-copy estimation were developed in [MP95, BEM98]. At the single-clone level, cloning with infinitely many outputs reduces to estimation [BA06, CD06]; global all-output problems have also been studied for broad pure-state families [CB99, CY14, KC22]. These settings differ from our proportional-output mixed-state problem; see [SIGA05, FWJ+14] for reviews.

Finite-dimensional mixed-state cloning. The earlier finite-dimensional literature summarized above provides bounds or constructions for restricted state families and criteria, rather than the exact proportional-output minimax value under joint product fidelity [CEM99, Ras03a, Ras03b, Ras03c, Fan03, FLS07, Kaz10, LW17]. More recently, random purification followed by pure-state cloning gave a universal all-output guarantee for rank-constrained mixed states, but not an exact optimum [TWZ25, LTHC26]; companion work instead amplifies a selected eigenstate [LTHC26b]. Concurrent work proves that PCT has optimal copy complexity up to constants for attaining a target squared fidelity $1 - \epsilon$, uniformly over rank-at-most- r states [FGSS26, JSO26]. This worst-case fidelity-threshold optimality does not identify an exact fixed-gain fidelity or imply pointwise optimality of PCT. Our results impose no near-unit-fidelity condition: for every fixed $\gamma > 1$, they determine the explicit limiting fidelity on the proved fixed-rank strata. By analyzing the actual post-trace PCT output, rather than only its published data-processing guarantee, we further prove that its asymptotic fidelity is strictly smaller for every simple positive spectrum of rank $r > 1$ and every flat projector orbit with $r > 1$. Conversely, the concurrent copy-complexity results are uniform in their joint small-error regime, whereas our present $\gamma \downarrow 1$ translation is an iterated limit. See Section 1.3 for the quantitative comparison.

Broadcasting and superbroadcasting. Single-input families are broadcastable precisely when they commute [BCFJS96]; an input-independent-fidelity criterion likewise obstructs universal broadcasting for arbitrary mixed qubits [CC07]. Multiple-input superbroadcasting can nevertheless purify individual mixed-qubit marginals; universal, phase-covariant, and optimal maps are known [DMP05, BDMP05, BDMP06, DF07], as is a continuous-variable version for displaced thermal states [DPS06]. These marginal criteria allow correlated outputs, so they do not determine full tensor-power fidelity. Virtual broadcasting is a related nonphysical extension linking universal cloning and optimal physical approximation, without changing this distinction [PFBC24].

Gaussian cloning and LAN. Continuous-variable cloning grew from coherent-state amplifier-beam-splitter schemes [CIR00, BCI+01]; under joint rather than single-clone fidelity, a non-Gaussian cloner can be optimal [CKNWW05]. Caves established the phase-insensitive amplifier noise limit

[Cav82], and Guță–Matsumoto reduced displaced mixed-Gaussian cloning to least-noise amplification [GM06]. Quantum LAN was formulated statistically in [GJ07] and developed through likelihood-ratio and explicit-channel formulations in [YFG13, GK06, KG09], yielding sharp purification, dilution, and reversal results [BGA11, BGA12]; related Gaussian tasks include amplifying and purifying noisy coherent states [ZC17]. This LAN–amplification lineage underlies our converse. Kahn–Guță supplies the interior antecedent, while the fixed-rank theorem of Lahiry–Nussbaum gives the unified experiment used here, including both its $r = d$ specialization and the support–kernel boundary modes when $r < d$ [KG09, LN24].

Other cloning and replication models. Further variants include probabilistic exact cloning [DG98]; economical, phase-covariant, asymmetric, and telecloning schemes [NG98, BCDM00, Cer00, MJPV99]; and cloning of transformations and superreplication [CDP08, CYY13, GCMBC14, DSS15, CYH15, CY16]. A higher-order framework for general channel cloning includes state cloning [SGKS25]; no-signalling yields general cloning bounds [SSD15, HT24], while optimal pure-state cloning and transposition can be complementary channels [BGSQ26]. These models differ from ours in the success rule, family, target object, or replication rate.

1.5 Organization

Section 2 develops the intrinsic positive-support Schur–Cartan and label-processing ingredients. These are dimension-generic active-support lemmas used in the fixed-rank proof. Section 3 similarly isolates the faithful thermal Gaussian and LAN-transfer tools. Section 4 then proves Theorems 1.1 and 1.2 directly for every $1 \leq r \leq d$: exact compression supplies the ambient support–kernel factor, active-row label processing gives achievability, and one thermal–vacuum fixed-rank LAN model gives both converses. Setting $r = d$ simply removes the vacuum modes. Section 5 proves the exceptional flat projector orbit directly. The PCT regimes are compared in Section 1.3, which separates the near-unit fidelity-threshold question from our arbitrary fixed-gain fidelity curves. The actual-channel bounds and strict fixed- γ ordering are proved in Appendix A. The remaining appendices supply the PBW positive-support calculation, the active-label local limit and randomized rounding estimate, and the unified flat-prior Gaussian converse.

Acknowledgments

We would like to thank Patrick Hayden for discussions and early collaborations.

Generative AI usage

The original idea for the cloner construction was due to the authors. We then used OpenAI’s ChatGPT Sol 5.6 and Anthropic’s Fable 5 to assist with the analysis. Suggestions from these systems informed the evaluation of the cloning fidelity and the converse argument. We also used them for language editing, manuscript organization, and figure preparation. All AI-assisted material was reviewed and revised by the authors, who take full responsibility for the accuracy and content of the manuscript.

Glossary of notation

We collect the notation that recurs across sections. Symbols local to a single proof are defined where they first appear.

Spaces and maps. $\mathbb{C}$ and $\mathbb{R}$ denote the complex and real numbers, $U(d)$ the unitary group, and $\mathcal{T}_1(\mathcal{H})$ the trace-class operators on a Hilbert space $\mathcal{H}$. We write A^* for the adjoint, $\text{Tr } A$ for the trace, $\|A\|_1$ for the trace norm, and $\otimes$ for tensor products. A channel is a completely positive trace-preserving map.

Asymptotic scale. The ambient dimension d and rank r are fixed, n is the number of inputs, m_n the number of outputs, and $\gamma_n = m_n/n \rightarrow \gamma > 1$. The number of pairs $i < j$ is $r_d = \binom{d}{2}$. A capital N denotes a generic sample size.

Spectra and states. $A_r = \{x \in \mathbb{R}^r : \sum_i x_i = 1\}$, and Δ_r° is its strictly ordered, positive spectral chamber. For $p \in \Delta_r^\circ$, $\bar{p} = (p, 0_{d-r})$, $\rho_{\bar{p}} = \text{diag}(\bar{p})$, and $\rho_{\bar{p},g} = g\rho_{\bar{p}}g^*$. Bars are suppressed when $r = d$. The notation $K \subseteq \Delta_r^\circ$ means that K is a compact subset of this open chamber; relative interiors and closures are taken in A_r .

Fidelity and optimal values. The unadorned $F(A, C)$ denotes root (unsquared) fidelity; for classical measures it denotes Hellinger affinity. With $q_{ij} = \bar{p}_j/\bar{p}_i$, including $q_{ij} = 0$ when $j > r$, the one-mode factor is $f_\gamma(q_{ij})$. The rank-aware values $F_{\text{orb}}^{(r,d)}$ and $F_{\text{univ}}^{(r,d)}$ are defined in (2)–(3); F_{orb} , F_{cl} , and F_{univ} denote the $r = d$ abbreviations in (4).

Schur–Weyl notation. $\mathcal{Y}_N$ is the set of partitions $\lambda \vdash N$ with at most d rows. The corresponding unitary and symmetric-group spaces are $\mathcal{H}_\lambda$ and $\mathcal{S}_\lambda$. The Young-label law under $\rho_p^{\otimes N}$ is $P_{N,p}$, its conditional irreducible state is $\rho_{p,\lambda}$, and $\mathcal{T}_N(p)$ is the typical-label set. On the rank- r face these objects carry a superscript (r) ; all active labels have at most r rows and are padded by zeros in dimension d .

Cloners and label kernels. $\mathcal{M}_n$ denotes a generic competing channel. A Markov kernel $K_n(\nu \mid \mu)$ on Young labels is lifted to the covariant channel $\mathcal{Q}_n[K_n]$. For compatible labels, $\mathcal{C}_{\mu,\nu}$ is the Cartan-sector channel; $\mathcal{E}_{\mu,\nu}$ extends it to all label pairs by a fallback channel. The rank-aware unknown-spectrum cloner is $\mathcal{C}_n^{(r)}$; when $r = d$ we also write $\mathcal{C}_n^{\text{univ}}$. Thus K denotes a spectral set, while K_n denotes a label kernel.

Local coordinates. $\mathbf{H}_r = \{h \in \mathbb{R}^r : \sum_i h_i = 0\}$ is the positive-spectral tangent hyperplane. The active root set is $\mathcal{I}_{r,d} = \{(i, j) : 1 \leq i \leq r, i < j \leq d\}$. The variables $h \in \mathbf{H}_r$ and $z \in \mathbb{C}^{\mathcal{I}_{r,d}}$ describe local spectral and orbital perturbations, respectively; $X_p(z)$ is the source-normalized skew-Hermitian generator in the fixed-rank LAN theorem, $\Omega_L^{(r,d)}$ is a bounded local box, and $v_{n,L}^{(r)}(p)$ is its minimax value. Alternative physical rotation coordinates are related by invertible, spectrum-dependent linear maps; whenever such a coordinate is used, the Gaussian displacement is reparametrized by the same map.

Gaussian notation. The calligraphic $\mathcal{F}$ denotes one-mode Fock space and is distinct from the roman F for fidelity. The fixed-rank model has $\binom{r}{2}$ thermal positive–positive modes and $r(d-r)$ vacuum support–kernel modes. The state τ_p is the thermal product in the input and ideal target, while $\tau_p^{(\gamma)}$ is the noisy amplifier output at zero displacement. The states $\Phi_z^{(p)}$ and $\Psi_z^{(p,\gamma)}$ are the displaced input and ideal target built from τ_p . The symbols Θ^{in} and Θ^{tar} include the classical Gaussian factor. The number operator is $\hat{N}$, distinct from the sample size N .

Limits and uniformity. Unless stated otherwise, asymptotics are as $n \rightarrow \infty$ with d and γ fixed. An $o(1)$ term is uniform only over the parameters specified in the accompanying statement.

The fixed-rank results are not uniform when the rank changes, a positive spectral gap closes, or γ varies.

2 Schur–Cartan ingredients on the positive support

Both fixed-rank cloning protocols use the same finite-dimensional architecture. This section establishes the ingredients intrinsic to a simple positive spectrum. The working dimension is denoted by d , but the statements are dimension-uniform: in the ambient rank- r problem of [Section 4](#), they are invoked with the working dimension replaced by r . Thus these are dimension-generic active-support lemmas used in the fixed-rank proof.

We first recall the Schur–Weyl decomposition in [Section 2.1](#) and lift an arbitrary Markov kernel on Young labels to a covariant quantum channel in [Section 2.2](#). We then choose the exact target-law kernel for known spectrum and the randomized dilation kernel for unknown spectrum in [Sections 2.3](#) and [2.4](#), respectively, and combine their label estimates with the common sector fidelity in [Section 2.5](#). Exact rank compression and the ambient fixed-rank assembly are carried out in [Sections 4.1](#) and [4.2](#). We now develop the active-support components summarized in [Fig. 1](#).

2.1 Schur–Weyl decomposition

Let

$$\mathbf{Y}_N = \{\lambda \vdash N : \ell(\lambda) \leq d\}.$$

Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\lambda \in \mathbf{Y}_N} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda,$$

where $\mathcal{H}_\lambda$ is the irreducible $U(d)$ -module of highest weight λ and $\mathcal{S}_\lambda$ is the corresponding Specht module. For $\rho_p = \text{diag}(p_1, \dots, p_d)$,

$$\rho_p^{\otimes N} = \bigoplus_{\lambda \in \mathbf{Y}_N} P_{N,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (9)$$

For $\rho_{p,g} = g\rho_p g^*$, replace $\rho_{p,\lambda}$ by $\rho_{p,g,\lambda} = U_\lambda(g)\rho_{p,\lambda}U_\lambda(g)^*$. The Young law $P_{N,p}$ depends on p , but not on g .

Define

$$\mathbf{T}_N(p) = \left\{ \lambda \in \mathbf{Y}_N : \left\| \frac{\lambda}{N} - p \right\|_\infty \leq N^{-1/3} \right\}.$$

Lemma 2.1 (Uniform Young concentration). *There is a constant $c_d > 0$, depending only on d , such that*

$$\sup_{p \in \Delta_d^\circ} P_{N,p}(\mathbf{T}_N(p)^c) \leq e^{-c_d N^{1/3}}$$

for all sufficiently large N .

Proof. Write $q = \lambda/N$. The Weyl dimension formula gives $\dim \mathcal{H}_\lambda \leq (N+1)^{d(d-1)/2}$. Every weight ξ of $\mathcal{H}_\lambda$ has $\lambda - \xi$ equal to a nonnegative integral combination of positive roots. Since $p_j/p_i < 1$ for $i < j$, this gives $p^\xi \leq p^\lambda$. Hence $s_\lambda(p) \leq \dim \mathcal{H}_\lambda p^\lambda$. The standard bounds $\dim \mathcal{S}_\lambda \leq \binom{N}{\lambda_1, \dots, \lambda_d}$ and $\binom{N}{\lambda_1, \dots, \lambda_d} p^\lambda \leq \exp[-ND(q\|p)]$ give

$$P_{N,p}(\lambda) \leq (N+1)^{d(d-1)/2} \exp[-ND(q\|p)], \quad (10)$$

which is the finite-sample estimate underlying the Keyl–Werner spectrum large-deviation theorem [KW01].

There are at most $(N+1)^{d-1}$ partitions of N with at most d rows. If $\lambda \notin \mathsf{T}_N(p)$, then $\|q - p\|_1 > N^{-1/3}$, and Pinsker’s inequality gives $D(q\|p) \geq \frac{1}{2}N^{-2/3}$. Therefore

$$P_{N,p}(\mathsf{T}_N(p)^c) \leq (N+1)^{d(d-1)/2+d-1} e^{-N^{1/3}/2}.$$

For all sufficiently large N , the polynomial factor is at most $e^{N^{1/3}/4}$. Hence the asserted bound holds with $c_d = 1/4$. $\square$

2.2 Cartan-sector channel and lifting of a label kernel

Let λ, θ be dominant weights, with unit highest-weight vectors v_λ, v_θ . The Cartan component $\mathcal{H}_{\lambda+\theta}$ occurs with multiplicity one in $\mathcal{H}_\lambda \otimes \mathcal{H}_\theta$. Let

$$V_{\lambda,\theta} : \mathcal{H}_{\lambda+\theta} \longrightarrow \mathcal{H}_\lambda \otimes \mathcal{H}_\theta, \quad V_{\lambda,\theta} v_{\lambda+\theta} = v_\lambda \otimes v_\theta, \quad (11)$$

be the normalized intertwining isometry. For $\mu \in \mathsf{Y}_n$ and $\nu \in \mathsf{Y}_m$, write $\nu - \mu \vdash m - n$ when the coordinatewise difference $\nu - \mu$ is a partition of $m - n$. If this holds, put $\omega = \nu - \mu$ and define

$$\mathcal{C}_{\mu,\nu}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^* (X \otimes I_{\mathcal{H}_\omega}) V_{\mu,\omega}. \quad (12)$$

The partial trace $\mathrm{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*)$ commutes with the irreducible $\mathrm{U}(d)$ -action on $\mathcal{H}_\mu$; hence Schur’s lemma and a trace comparison give $\mathrm{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*) = \frac{\dim \mathcal{H}_\nu}{\dim \mathcal{H}_\mu} I_{\mathcal{H}_\mu}$. Thus (12) is completely positive by tensoring with $I_{\mathcal{H}_\omega}$, applying $Y \mapsto V_{\mu,\omega}^* Y V_{\mu,\omega}$, and positive rescaling; the identity above makes it trace preserving, while equivariance of $V_{\mu,\omega}$ gives $\mathrm{U}(d)$ -covariance.

For every pair $(\mu, \nu) \in \mathsf{Y}_n \times \mathsf{Y}_m$, define a covariant channel $\mathcal{E}_{\mu,\nu} : \mathcal{T}_1(\mathcal{H}_\mu) \rightarrow \mathcal{T}_1(\mathcal{H}_\nu)$ by

$$\mathcal{E}_{\mu,\nu}(X) = \begin{cases} \mathcal{C}_{\mu,\nu}(X), & \nu - \mu \vdash m - n, \\ \mathrm{Tr}(X) I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu, & \text{otherwise.} \end{cases}$$

Let $K_n : \mathsf{Y}_n \rightsquigarrow \mathsf{Y}_{m_n}$, with transition probabilities $K_n(\nu \mid \mu)$, be any Markov kernel. Its Schur–Cartan lifting is the following channel, denoted by $\mathcal{Q}_n[K_n]$: apply the Schur transform, measure μ , discard the input Specht factor, sample ν from $K_n(\cdot \mid \mu)$, apply $\mathcal{E}_{\mu,\nu}$, adjoin $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$, and apply the inverse Schur transform. Each step is completely positive and trace preserving, so their composition is a channel. Because the label measurement and sampling are invariant and every map $\mathcal{E}_{\mu,\nu}$ is covariant, $\mathcal{Q}_n[K_n]$ is $\mathrm{U}(d)$ -covariant.

Proposition 2.2 (Fidelity of a lifted label kernel). *Fix $p \in \Delta_d^\circ$. Let*

$$\Pi_{n,p}(\mu, \nu) = P_{n,p}(\mu) K_n(\nu \mid \mu), \quad Q_{m_n,p}(\nu) = \sum_{\mu} \Pi_{n,p}(\mu, \nu).$$

Suppose $G \subset \mathsf{Y}_n \times \mathsf{Y}_{m_n}$ satisfies $\nu - \mu \vdash m_n - n$ for every $(\mu, \nu) \in G$, and

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) \geq b \quad ((\mu, \nu) \in G)$$

for some $b \in [0, 1]$. Then, for every $g \in \mathrm{U}(d)$,

$$F\left(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) \geq b \left(F(Q_{m_n,p}, P_{m_n,p}) - \sqrt{\Pi_{n,p}(G^c)}\right). \quad (13)$$

Proof. Covariance makes every sector fidelity independent of g . In the output Schur decomposition, the actual state in the ν -block is the mixture of $\mathcal{E}_{\mu,\nu}(\rho_{p,g,\mu})$ with weights $\Pi_{n,p}(\mu, \nu)$, whereas the ideal block is $P_{m_n,p}(\nu)\rho_{p,g,\nu}$. The Specht factors are identical and therefore cancel by multiplicativity of the trace norm under tensor products.

For $Q_{m_n,p}(\nu) > 0$, set

$$r(\nu) = \sum_{\mu: (\mu,\nu) \notin G} \frac{\Pi_{n,p}(\mu, \nu)}{Q_{m_n,p}(\nu)},$$

and set $r(\nu) = 0$ otherwise. The direct-sum identity for fidelity, its scaling under positive scalar multiples, and joint concavity give

$$\begin{aligned} F(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) \\ \geq b \left(F(Q_{m_n,p}, P_{m_n,p}) - \sum_{\nu} \sqrt{Q_{m_n,p}(\nu) P_{m_n,p}(\nu)} r(\nu) \right). \end{aligned}$$

Since $0 \leq r \leq 1$ and $\sum_{\nu} Q_{m_n,p}(\nu) r(\nu) = \Pi_{n,p}(G^c)$, Cauchy–Schwarz yields

$$\sum_{\nu} \sqrt{Q_{m_n,p}(\nu) P_{m_n,p}(\nu)} r(\nu) \leq \sqrt{\Pi_{n,p}(G^c)} \left(\sum_{\nu} P_{m_n,p}(\nu) r(\nu) \right)^{1/2} \leq \sqrt{\Pi_{n,p}(G^c)}.$$

This proves (13). $\square$

We now specialize the lifted channel by choosing its output-label kernel. The two protocols differ only in this choice: the known-spectrum construction samples the exact target law, whereas the unknown-spectrum construction uses a randomized dilation of the measured input label.

2.3 Known spectrum: exact target-label sampling

Fix $p \in \Delta_d^\circ$. For every n such that $m_n > n$, define

$$K_n^p(\nu \mid \mu) = P_{m_n,p}(\nu).$$

Thus ν is independent of μ and has the exact target law. Define the amplification branch by

$$\mathcal{C}_n^p := \mathcal{Q}_n[K_n^p].$$

Pairs with $\nu - \mu \not\vdash m_n - n$ use $\mathcal{E}_{\mu,\nu}(X) = \text{Tr}(X)I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu$ without changing the sampled label ν . Together with the common convention in Section 1.1, this defines the complete $U(d)$ -covariant sequence $(\mathcal{C}_n^p)_n$, whose output Young law on $\rho_{p,g}^{\otimes n}$ is $P_{m_n,p}$ for every n and g .

2.4 Unknown spectrum: randomized rounding

For $N \in \mathbb{N}$, let

$$\mathbb{L}_N = \left\{ x \in \mathbb{Z}^d : \sum_i x_i = N \right\}.$$

For every n such that $m_n > n$, set $\nu_n^{\text{fb}} = (m_n, 0, \dots, 0)$. Given $\mu \in \Upsilon_n$, draw independent variables $U_1, \dots, U_{d-1}$, each uniform on $[-1/2, 1/2)$, and set

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + U_i) + \frac{1}{2} \right\rfloor \quad (1 \leq i < d), \quad \tilde{\nu}_d = m_n - \sum_{i < d} \tilde{\nu}_i. \quad (14)$$

This is a spectrum-independent lattice version of the dilation $\mu \mapsto \gamma_n \mu$: it sends typical labels near np to labels near $m_n p$, with $\tilde{\nu} - \mu$ typically a partition. The uniform shifts randomize the rounding error, avoiding lattice bias in the Gaussian limit. Then $\tilde{\nu} \in \mathbb{L}_{m_n}$. Let $D_n(\tilde{\nu} \mid \mu)$ denote its distribution. Define $K_n^{\text{rnd}} : \mathbb{Y}_n \rightsquigarrow \mathbb{Y}_{m_n}$ by

$$K_n^{\text{rnd}}(\nu \mid \mu) = \begin{cases} D_n(\nu \mid \mu), & \nu \neq \nu_n^{\text{fb}} \text{ and } \nu - \mu \vdash m_n - n, \\ 1 - \sum_{\substack{\lambda \in \mathbb{Y}_{m_n}, \\ \lambda \neq \nu_n^{\text{fb}}, \\ \lambda - \mu \vdash m_n - n}} D_n(\lambda \mid \mu), & \nu = \nu_n^{\text{fb}}, \\ 0, & \text{otherwise.} \end{cases} \quad (15)$$

Under (15), every rounded label outside $\mathbb{Y}_{m_n}$, and every $\tilde{\nu} \in \mathbb{Y}_{m_n}$ for which $\tilde{\nu} - \mu \not\vdash m_n - n$, is replaced by ν_n^{fb} , which also retains any mass rounded onto it. Define the amplification branch by

$$\mathcal{C}_n^{\text{univ}} := \mathcal{Q}_n[K_n^{\text{rnd}}].$$

The rounding variables, (15), and the maps $\mathcal{E}_{\mu, \nu}$ are spectrum independent, while $\mathcal{Q}_n[K_n^{\text{rnd}}]$ is $U(d)$ -covariant. Together with the common convention in Section 1.1, this defines the complete spectrum-independent, $U(d)$ -covariant sequence $(\mathcal{C}_n^{\text{univ}})_n$.

2.5 Sector and label asymptotics

The following proposition is the only representation-theoretic asymptotic estimate used in the cloning protocols.

Proposition 2.3 (Uniform Cartan-sector fidelity). *Let $K \in \Delta_d^\circ$ and let $\delta_n \geq 0$ satisfy $\delta_n \rightarrow 0$. For all sufficiently large n , every $p \in K$, $\mu \in \mathbb{Y}_n$, and $\nu \in \mathbb{Y}_{m_n}$ satisfying*

$$\left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \quad \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n$$

also satisfies $\nu - \mu \vdash m_n - n$. Moreover,

$$F(\mathcal{C}_{\mu, \nu}(\rho_{p, \mu}), \rho_{p, \nu}) = F_{\text{orb}}(\gamma, p) + o(1)$$

uniformly over all such p, μ, ν .

The mechanism is already visible on states obtained by a fixed number of lowering operations. Writing $\omega = \nu - \mu$, the Cartan intertwiner satisfies

$$V_{\mu, \omega} F_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)}) V_{\mu, \omega},$$

where $F_\alpha^{(\lambda)}$ is the lowering operator for $\alpha = e_i - e_j$ on $\mathcal{H}_\lambda$. After normalization, each excitation splits binomially between the two factors, with fraction

$$s_{\alpha, n} = \frac{\mu_i - \mu_j}{\nu_i - \nu_j} = \frac{1}{\gamma} + o(1)$$

entering the μ -factor. Thus the conditional thermal parameter $q_{ij} = p_j/p_i$ is transformed into

$$1 - s_{\alpha, n} + s_{\alpha, n} q_{ij} \longrightarrow 1 - \frac{1 - q_{ij}}{\gamma} =: q_{ij}^{(\gamma)}.$$

The low-weight matrices of the sector output and ideal sector therefore approach those of $\tau_p^{(\gamma)}$ and τ_p , whose fidelity is $F_{\text{orb}}(\gamma, p)$. This is a statement about conditional states for typical label pairs, not convergence of the varying channels $\mathcal{C}_{\mu, \nu}$ to the Gaussian amplifier. The exact PBW calculation and uniform thermal-tail argument that complete the proof are given in Appendix B.

The second estimate concerns only K_n^{rnd} .

Proposition 2.4 (Randomized Young-label dilation). *There is a constant $c_{\text{rnd}} < \infty$, depending only on d and the fixed sequence (γ_n) , such that, for every compact $K \Subset \Delta_d^\circ$ and all sufficiently large n , every $p \in K$ and $\mu \in \mathbb{T}_n(p)$ satisfy*

$$\text{supp } K_n^{\text{rnd}}(\cdot \mid \mu) \subseteq \{\nu \in \mathbb{Y}_{m_n} : \nu - \mu \vdash m_n - n, \quad \|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}\}.$$

For $m_n > n$, let

$$Q_{m_n, p}(\nu) := \sum_{\mu \in \mathbb{Y}_n} P_{n, p}(\mu) K_n^{\text{rnd}}(\nu \mid \mu), \quad \nu \in \mathbb{Y}_{m_n},$$

be the induced output-label law. Then, for every compact $K \Subset \Delta_d^\circ$,

$$F(Q_{m_n, p}, P_{m_n, p}) = F_{\text{cl}}(\gamma, d) + o(1) \quad \text{uniformly for } p \in K. \quad (16)$$

The uniform dither in (14) makes the label kernel a discrete realization of continuous dilation. After centering at Np and scaling by $\sqrt{N}$, embed a label law as a piecewise-constant density on the fundamental cells of the label lattice. Before fallback replacement, the intersection-volume identity (127) identifies the embedded rounded law with the target-cell average of the $\sqrt{\gamma_n}$ -dilation of the embedded input law. Write

$$\Sigma_p := (\text{diag}(p) - pp^\top) \Big|_{\{x \in \mathbb{R}^d : \sum_i x_i = 0\}}.$$

The Young local limit is $\mathbf{N}_{0, \Sigma_p}$, so randomized rounding changes the limit to $\mathbf{N}_{0, \gamma \Sigma_p}$, whereas the ideal m_n -label law still converges to $\mathbf{N}_{0, \Sigma_p}$. Their Hellinger affinity is

$$F(\mathbf{N}_{0, \gamma \Sigma_p}, \mathbf{N}_{0, \Sigma_p}) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} = F_{\text{cl}}(\gamma, d).$$

The support check and fallback rule affect only vanishing probability on typical labels.

The proofs of Propositions 2.3 and 2.4 are given in Appendices B and C, respectively.

3 Gaussian and LAN ingredients on the positive support

We next isolate the Gaussian dilation and LAN tools needed by the fixed-rank converse. As in the preceding section, d here denotes a generic positive-support dimension. The Gaussian result treats the faithful thermal modes within that support, while the unified model in Section 4.3 combines them with the support-kernel vacuum modes and performs the ambient fixed-rank transfer. We first introduce the local coordinates, then solve the Gaussian problems in Section 3.2, and finally record the general fidelity transfer used with fixed-rank LAN in Section 3.3.

The LAN transport common to these converse arguments is summarized in Fig. 2. The upper path is the induced Gaussian-model competitor; the lower path is the ideal target against which its fidelity is measured.

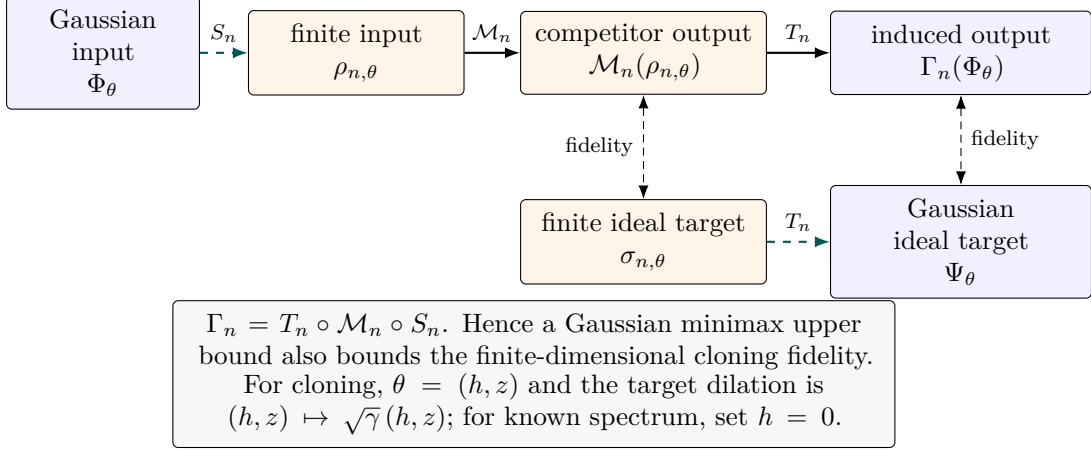


Figure 2: Transporting an arbitrary finite-dimensional cloner through two-way local asymptotic normality. A dashed arrow identifies its displayed endpoint up to uniform $o(1)$ trace-norm error on compact local parameter sets; the solid T_n arrow defines the induced output exactly. The LAN maps are not exact inverses, and the induced channels Γ_n need not be Gaussian or converge. The converse follows from the optimal Gaussian dilation and amplification bounds.

3.1 Local coordinates

Let

$$\mathbf{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad z \in \mathbb{C}^{r_d},$$

and, for $p \in \Delta_d^\circ$, define the source-normalized generator

$$X_p(z) = \sum_{j < k} \frac{z_{jk} E_{kj} - \bar{z}_{jk} E_{jk}}{\sqrt{p_j - p_k}}, \quad g_{n,z}^{(p)} = \exp\left(\frac{X_p(z)}{\sqrt{n}}\right),$$

where E_{jk} are the standard matrix units. In $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}$, the coordinates h and z describe local changes of eigenvalues and eigenbasis, respectively. Since $X_p(z)^* = -X_p(z)$, one has $g_{n,z}^{(p)} \in \mathbf{U}(d)$. For $L > 0$, set

$$\Omega_L = \{(h, z) \in \mathbf{H} \times \mathbb{C}^{r_d} : \|h\|_\infty \leq L, |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

For every fixed $p \in \Delta_d^\circ$ and $L > 0$, the vector $p + h/\sqrt{n}$ belongs to Δ_d° for every $(h, z) \in \Omega_L$ once n is sufficiently large. Under LAN, h becomes a $(d-1)$ -dimensional classical Gaussian shift, while z becomes a collection of displaced oscillator coordinates. The Gaussian theorem below optimizes the two corresponding fidelity losses jointly.

3.2 Gaussian amplification

Let $\mathcal{F}$ be the one-mode Fock space with number basis $\{|k\rangle : k \geq 0\}$, creation and annihilation operators a^*, a , and number operator $\hat{N} = a^*a$. We use the Weyl convention

$$D(z) = \exp(za^* - \bar{z}a). \quad (17)$$

For $0 < q < 1$, set

$$\tau_q = (1-q) \sum_{k \geq 0} q^k |k\rangle\langle k|.$$

Two thermal states commute, and hence

$$F(\tau_q, \tau_{q'}) = \frac{\sqrt{(1-q)(1-q')}}{1 - \sqrt{qq'}}. \quad (18)$$

For $G \geq 1$, put $q^{(G)} = 1 - (1-q)/G$. The quantum-limited phase-insensitive amplifier $\mathcal{A}_G$ satisfies

$$\mathcal{A}_G(D(z)\tau_q D(z)^*) = D(\sqrt{G}z)\tau_{q^{(G)}} D(\sqrt{G}z)^*. \quad (19)$$

Equivalently, $\mathcal{A}_G(X) = \text{Tr}_b[S_r(X \otimes |0\rangle\langle 0|_b)S_r^*]$, where S_r is a two-mode squeezing unitary with $\cosh^2 r = G$. Squeezing supplies the gain $\sqrt{G}$, while tracing out the vacuum idler produces the minimum added noise encoded by $q^{(G)}$.

For $p \in \Delta_d^\circ$, define on $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$

$$\tau_p = \bigotimes_{i < j} \tau_{q_{ij}}, \quad \tau_p^{(\gamma)} = \bigotimes_{i < j} \tau_{q_{ij}^{(\gamma)}}.$$

With $D(z) = \bigotimes_{i < j} D(z_{ij})$, the three relevant states are

$$\begin{aligned} \text{input: } \Phi_z^{(p)} &= D(z)\tau_p D(z)^*, \\ \text{ideal gain-}\sqrt{\gamma} \text{ target: } \Psi_z^{(p, \gamma)} &= D(\sqrt{\gamma}z)\tau_p D(\sqrt{\gamma}z)^*, \\ \text{amplifier output: } \mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}) &= D(\sqrt{\gamma}z)\tau_p^{(\gamma)} D(\sqrt{\gamma}z)^*. \end{aligned}$$

Thus the ideal target retains the thermal state τ_p ; only the actual amplifier output contains the additional noise $\tau_p^{(\gamma)}$. By (18),

$$F(\tau_{q^{(\gamma)}}, \tau_q) = f_\gamma(q). \quad (20)$$

Thus, for every $z \in \mathbb{C}^{r_d}$, the product-amplifier output and the corresponding ideal state satisfy

$$F\left(\mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}), \Psi_z^{(p, \gamma)}\right) = F_{\text{orb}}(\gamma, p).$$

Equip $\mathbf{H}$ with its induced Euclidean measure and recall

$$\Sigma_p = (\text{diag}(p) - pp^\top)|_{\mathbf{H}} > 0.$$

Write $\mathbf{N}_{h, \Sigma_p}$ for the Gaussian law on $\mathbf{H}$ with mean h and covariance Σ_p . A classical-quantum state has the form

$$R(x) = r(x)\rho_x, \quad r(x) \geq 0, \quad \int_{\mathbf{H}} r(x) dx = 1,$$

where x and r are classical, while ρ_x is a quantum state on $\mathcal{F}_{\text{orb}}$. A channel Λ is completely positive and preserves $\int \text{Tr } R(x) dx$. Here complete positivity means positivity of $\text{id}_{M_k} \otimes \Lambda$ for every $k \geq 1$, with the matrix factor acting on the quantum trace-class fibers; the classical algebra is abelian. If $S(x) = s(x)\sigma_x$, then

$$F(R, S) = \int_{\mathbf{H}} F(R(x), S(x)) dx = \int_{\mathbf{H}} \sqrt{r(x)s(x)} F(\rho_x, \sigma_x) dx.$$

With $\|R\|_1 := \int_{\mathbf{H}} \|R(x)\|_1 dx$, the usual monotonicity and joint-concavity properties remain valid, and the first equality above is the continuous direct-sum identity for fidelity.

Set

$$\Theta_{h, z}^{\text{in}} = \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)}, \quad \Theta_{h, z}^{\text{tar}} = \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p, \gamma)}.$$

Let

$$\mathbf{Z}_L = \{z \in \mathbb{C}^{r_d} : |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

Theorem 3.1 (Gaussian amplification optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p, \gamma)}) \, dz = F_{\text{orb}}(\gamma, p), \quad (21)$$

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}) \, dh \, dz = F_{\text{univ}}(\gamma, p). \quad (22)$$

The first supremum is over all quantum channels on $\mathcal{F}_{\text{orb}}$; the second is over all classical–quantum channels. No Gaussianity, covariance, or product-structure assumption is imposed on the competing channels in either supremum.

The achievability mechanism is explicit. The product amplifier $\mathcal{A}_\gamma^{\otimes r_d}$ attains $F_{\text{orb}}(\gamma, p)$ in the orbital model. In the hybrid classical–quantum model, tensor it with the classical pushforward $x \mapsto \sqrt{\gamma}x$. The resulting Gaussian affinity is $F_{\text{cl}}(\gamma, d)$, so multiplicativity gives $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) = F_{\text{univ}}(\gamma, p)$.

For the converse, average a near-optimal channel over translates of the expanding flat-prior boxes. The vanishing boundary-to-volume error yields a displacement-covariant limiting map. A weighted-fidelity witness then turns the orbital fidelity into a noise moment, which the least-noise inequality bounds by $F_{\text{orb}}(\gamma, p)$. In the hybrid model, covariance forces the classical trace kernel to have convolution form, and the corresponding Gaussian witness contributes at most $F_{\text{cl}}(\gamma, d)$. The two bounds therefore multiply to $F_{\text{univ}}(\gamma, p)$. The compactness, possible lost-mass, and witness arguments are given in [Proposition D.5](#) and [Appendix D](#).

3.3 LAN transfer

The two-way LAN statement used in the cloning converse is stated once, for every $1 \leq r \leq d$, in [Theorem 4.4](#). Its $r = d$ specialization is ordinary interior LAN. We use throughout the source-normalized convention

$$\exp(X_p(z)/\sqrt{n}) \longleftrightarrow D(z)\tau_p D(z)^*.$$

Other physical rotation conventions are related to this one by invertible, spectrum-dependent linear maps. Such a change must be applied to both the finite-dimensional family and the Gaussian displacement; we do not identify coordinates belonging to two different conventions. The fixed-rank theorem is the result of Lahiry–Nussbaum, whose $r = d$ case recovers interior LAN; Kahn–Guță provides its faithful-state antecedent [\[KG09, LN24\]](#).

We will use the following continuity estimate here and in the appendices. If A, C, A', C' are positive trace-class operators with trace at most one, then

$$|F(A, C) - F(A', C')| \leq \|A - A'\|_1^{1/2} + \|C - C'\|_1^{1/2}. \quad (23)$$

Indeed, the Powers–Størmer inequality bounds $\|\sqrt{A} - \sqrt{A'}\|_2$ by $\|A - A'\|_1^{1/2}$, and similarly for C, C' . Adding and subtracting $\sqrt{A'}\sqrt{C}$, the reverse triangle inequality and Schatten Hölder give [\(23\)](#), since $\|\sqrt{C}\|_2, \|\sqrt{A'}\|_2 \leq 1$.

The following lemma formalizes this LAN fidelity transfer.

Lemma 3.2 (LAN fidelity transfer). *Let $\{\rho_{n,\theta}\}_{\theta \in C}$ and $\{\sigma_{n,\theta}\}_{\theta \in C}$ be input and target models on a compact parameter set C . Let $\{\Phi_\theta\}$ and $\{\Psi_\theta\}$ be limiting models, and suppose channels S_n, T_n satisfy*

$$\sup_{\theta \in C} \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \rightarrow 0, \quad \sup_{\theta \in C} \|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \rightarrow 0.$$

For every channel $\mathcal{M}_n$, put $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$. Then

$$\inf_{\theta \in C} F(\Gamma_n(\Phi_\theta), \Psi_\theta) \geq \inf_{\theta \in C} F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) - o(1). \quad (24)$$

Proof. Let the two uniform approximation errors be δ_n and η_n . Since $T_n \circ \mathcal{M}_n$ is a channel, uniformly in θ , $\|\Gamma_n(\Phi_\theta) - T_n \mathcal{M}_n(\rho_{n,\theta})\|_1 \leq \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \leq \delta_n$. Also $\|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \leq \eta_n$. Together with (23), these bounds give

$$F(\Gamma_n(\Phi_\theta), \Psi_\theta) \geq F(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) - \sqrt{\delta_n} - \sqrt{\eta_n}.$$

Finally, monotonicity of fidelity under the channel T_n gives

$$F(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) \geq F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}).$$

Taking the infimum over θ proves (24). $\square$

Conditional sector limits versus LAN. Although the same thermal state τ_p appears in both arguments, it represents different limiting operations. The conditional-sector proof fixes the input and output Young labels and, on fixed-height cutoffs, identifies the ideal target with τ_p and the Cartan-sector output with $\tau_p^{(\gamma)}$; see (124) and (123). This comparison directly explains the achievable orbital factor $F_{\text{orb}}(\gamma, p)$. The unified fixed-rank LAN experiment instead retains the Young-label fluctuation as $\mathbf{N}_{0, \Sigma_p}$, while orbital LAN marginalizes this classical factor. Thus the common occurrence of τ_p reflects conditioning on Young labels in the sector calculation and marginalizing their fluctuation in orbital LAN. The fixed-rank proof below combines these ingredients with exact support compression and the vacuum-mode boundary argument.

4 Fixed-rank cloning in ambient dimension

Fix $1 \leq r \leq d$ and $p \in \Delta_r^\circ$. We now prove the main theorems directly on this rank- r stratum. Exact compression separates the positive-support calculation from the support-kernel contribution; rank-aware label processing gives achievability; and one fixed-rank LAN model, containing both thermal and vacuum modes, gives the converses. The case $r = d$ is obtained by the same proof with no support-kernel modes, rather than by a separate argument. If $d = r = 1$, the model is a singleton and every fidelity below is one; we therefore assume $d \geq 2$ in the nontrivial proof.

4.1 Exact rank compression

We decorate representation spaces and maps by the ambient dimension when needed. For a partition λ with at most r rows, let

$$J_\lambda : \mathcal{H}_\lambda^{(r)} \longrightarrow \mathcal{H}_\lambda^{(d)}$$

be the isometric $U(r)$ -intertwiner that maps the unit highest-weight vector to the unit highest-weight vector. Its range is the $U(r)$ -submodule generated by that vector, and we set $P_\lambda = J_\lambda J_\lambda^*$.

For a compatible pair of r -row partitions $\mu \vdash n$ and $\nu \vdash m$, write $\omega = \nu - \mu \vdash m - n$. In ambient dimension $s \in \{r, d\}$, the Cartan-sector channel is

$$\mathcal{C}_{\mu, \nu}^{(s)}(X) = \frac{\dim \mathcal{H}_\mu^{(s)}}{\dim \mathcal{H}_\nu^{(s)}} (V_{\mu, \omega}^{(s)})^* (X \otimes I_{\mathcal{H}_\omega^{(s)}}) V_{\mu, \omega}^{(s)}.$$

For a probability vector $x \in \mathbb{R}^s$, including one with zero entries, we use the conditional Schur state

$$\rho_{x, \lambda}^{(s)} = \frac{\pi_\lambda^{(s)}(\text{diag } x)}{s_\lambda^{(s)}(x)}$$

whenever the denominator is nonzero.

Lemma 4.1 (Exact rank-compression identity). *For every $p = (p_1, \dots, p_r)$ with $p_i > 0$ and $\sum_i p_i = 1$, and every r -row partition λ ,*

$$\rho_{\bar{p}, \lambda}^{(d)} = J_\lambda \rho_{p, \lambda}^{(r)} J_\lambda^*. \quad (25)$$

For every compatible r -row pair μ, ν and every trace-class operator X on $\mathcal{H}_\mu^{(r)}$,

$$P_\nu \mathcal{C}_{\mu, \nu}^{(d)}(J_\mu X J_\mu^*) P_\nu = a_{\mu, \nu}^{(r, d)} J_\nu \mathcal{C}_{\mu, \nu}^{(r)}(X) J_\nu^*, \quad (26)$$

where

$$a_{\mu, \nu}^{(r, d)} = \frac{\dim \mathcal{H}_\mu^{(d)} / \dim \mathcal{H}_\nu^{(d)}}{\dim \mathcal{H}_\mu^{(r)} / \dim \mathcal{H}_\nu^{(r)}} \quad (27)$$

$$= \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\mu_i + j - i}{\nu_i + j - i}. \quad (28)$$

Consequently, there is the exact finite-label fidelity factorization

$$\begin{aligned} F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{\bar{p}, \mu}^{(d)}, \rho_{\bar{p}, \nu}^{(d)})\right) \\ = \sqrt{a_{\mu, \nu}^{(r, d)}} F\left(\mathcal{C}_{\mu, \nu}^{(r)}(\rho_{p, \mu}^{(r)}, \rho_{p, \nu}^{(r)})\right). \end{aligned} \quad (29)$$

Proof. The singular specialization identity for Schur polynomials is

$$s_\lambda^{(d)}(p_1, \dots, p_r, 0, \dots, 0) = s_\lambda^{(r)}(p_1, \dots, p_r),$$

and it vanishes when λ has more than r rows. Polynomial functoriality for the standard inclusion $\iota : \mathbb{C}^r \hookrightarrow \mathbb{C}^d$ gives

$$\pi_\lambda^{(d)}(\iota \operatorname{diag}(p) \iota^*) = J_\lambda \pi_\lambda^{(r)}(\operatorname{diag} p) J_\lambda^*.$$

Thus the ambient operator is supported on the highest-generated $U(r)$ -submodule and restricts there to the r -dimensional polynomial representation. Normalizing by the Schur-polynomial identity proves (25).

The two maps

$$V_{\mu, \omega}^{(d)} J_\nu, \quad (J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}$$

are $U(r)$ -intertwiners from $\mathcal{H}_\nu^{(r)}$ to $\mathcal{H}_\mu^{(d)} \otimes \mathcal{H}_\omega^{(d)}$, and both send the unit highest-weight vector to the same vector $v_\mu \otimes v_\omega$. Generation from the highest-weight vector gives the exact restriction identity

$$V_{\mu, \omega}^{(d)} J_\nu = (J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}. \quad (30)$$

Inserting (30) into the two sector-channel formulas proves (26), with the scalar in (27). The Weyl dimension formula, and the fact that all rows after r vanish, give (28); the positive–positive Weyl factors cancel and the kernel–kernel factors are identically one.

If $A, B \geq 0$ and $PBP = B$, then

$$F(A, B) = \operatorname{Tr} \sqrt{\sqrt{B} A \sqrt{B}} = F(PAP, B).$$

Apply this with $P = P_\nu$, then use (25), (26), invariance under J_ν , and $F(aA, B) = \sqrt{a} F(A, B)$. This proves (29). Taking traces in (26) also gives

$$\operatorname{Tr} \left[P_\nu \mathcal{C}_{\mu, \nu}^{(d)}(J_\mu X J_\mu^*) P_\nu \right] = a_{\mu, \nu}^{(r, d)} \operatorname{Tr} X.$$

In particular, compatibility forces $a_{\mu, \nu}^{(r, d)} \leq 1$. □

When $r = d$, every J_λ is the identity and the product in (28) is empty, so $a_{\mu,\nu}^{(d,d)} = 1$. Thus exact compression is part of the common proof and becomes tautological when $r = d$.

Corollary 4.2 (Uniform ambient fixed-rank sector limit). *Let $K \in \Delta_r^\circ$, let $\delta_n \rightarrow 0$, and let $\mu \vdash n$, $\nu \vdash m_n$ be compatible r -row labels satisfying*

$$\left\| \frac{(\mu_1, \dots, \mu_r)}{n} - p \right\|_\infty \leq \delta_n, \quad \left\| \frac{(\nu_1, \dots, \nu_r)}{m_n} - p \right\|_\infty \leq \delta_n, \quad p \in K.$$

Then

$$\begin{aligned} F\left(\mathcal{C}_{\mu,\nu}^{(d)}(\rho_{\bar{p},\mu}^{(d)}, \rho_{\bar{p},\nu}^{(d)})\right) \\ = \prod_{(i,j) \in \mathcal{I}_{r,d}} f_\gamma(q_{ij}) + o(1) = F_{\text{orb}}^{(r,d)}(\gamma, p) + o(1), \end{aligned}$$

uniformly over the displayed family.

Proof. Every factor in (28) tends uniformly to $1/\gamma$, so

$$a_{\mu,\nu}^{(r,d)} = \gamma^{-r(d-r)} + o(1).$$

The remaining fidelity in (29) is the intrinsic positive-support sector calculation in dimension r . If $r \geq 2$, Proposition 2.3, applied with effective dimension r , gives the product over $1 \leq i < j \leq r$. If $r = 1$, all three $U(1)$ -modules are one-dimensional and the lower-dimensional sector fidelity is exactly one. In this case the result reduces to $\gamma^{-(d-1)/2}$, the proportional limit of Werner's pure-state cloning root fidelity. $\square$

This proof also makes the stabilizer bookkeeping transparent. The positive–positive roots are inherited from $U(r)$; the support–kernel roots occur only in the relative Weyl-dimension ratio; and kernel–kernel roots cancel exactly.

4.2 Rank-aware label processing and achievability

For every N , the ambient d -dimensional Young law of a rank- r state reduces exactly to the r -dimensional law:

$$P_{N,\bar{p}}^{(d)}(\lambda) = \begin{cases} P_{N,p}^{(r)}(\lambda), & \ell(\lambda) \leq r, \\ 0, & \ell(\lambda) > r. \end{cases} \quad (31)$$

Indeed, this is the same Schur-polynomial specialization used in Lemma 4.1; the Specht dimension is independent of the ambient unitary dimension.

Proposition 4.3 (Ambient lifted-kernel bound at fixed rank). *Fix $p \in \Delta_r^\circ$, set*

$$\mathbf{Y}_N^{(r)} = \{\lambda \vdash N : \ell(\lambda) \leq r\},$$

and let $K_n(\nu \mid \mu)$ be a Markov kernel from $\mathbf{Y}_n^{(r)}$ to $\mathbf{Y}_{m_n}^{(r)}$, with labels padded by zeros in ambient dimension d . Extend it arbitrarily to the remaining ambient input labels; for example, when $\ell(\mu) > r$, send the label deterministically to $(m_n, 0, \dots, 0)$. Set

$$\Pi_{n,p}^{(r)}(\mu, \nu) = P_{n,p}^{(r)}(\mu) K_n(\nu \mid \mu), \quad Q_{m_n,p}^{(r)}(\nu) = \sum_{\mu} \Pi_{n,p}^{(r)}(\mu, \nu).$$

Suppose $G \subset Y_n^{(r)} \times Y_{m_n}^{(r)}$, $b \in [0, 1]$, and every pair in G is compatible and has ambient sector fidelity at least b . On the other pairs, define the lift with any covariant channel into the prescribed ν -block, such as the maximally mixed replacement channel. Then the ambient Schur–Cartan lift $\mathcal{Q}_n^{(d)}[K_n]$ satisfies, for every $g \in U(d)$,

$$F\left(\mathcal{Q}_n^{(d)}[K_n](\rho_{\bar{p},g}^{\otimes n}, \rho_{\bar{p},g}^{\otimes m_n}\right) \geq b \left[F(Q_{m_n,p}^{(r)}, P_{m_n,p}^{(r)}) - \sqrt{\Pi_{n,p}^{(r)}(G^c)} \right].$$

Proof. The proof of [Proposition 2.2](#) uses only the direct-sum fidelity identity, covariance, joint concavity, and Cauchy–Schwarz. By [\(31\)](#), all blocks with more than r rows have zero input and target weight. Omitting those blocks therefore gives the displayed fixed-rank statement verbatim. $\square$

4.2.1 Known spectrum

Sample ν independently from the exact law $P_{m_n,p}^{(r)}$, pad both measured μ and sampled ν by zeros, and use the ambient d -dimensional Cartan-sector channel whenever $\nu - \mu$ is a partition. Otherwise, use the covariant replacement channel

$$X \mapsto \text{Tr}(X) \frac{I_{\mathcal{H}_\nu^{(d)}}}{\dim \mathcal{H}_\nu^{(d)}}$$

into the same sampled ν -block. Thus the fallback does not alter the sampled output label. On input labels with more than r rows, define the kernel by the deterministic extension in [Proposition 4.3](#); these branches have zero probability for the states under consideration. Because p_r and all adjacent positive gaps are bounded below on compact subsets of Δ_r° , two independent typical labels obey

$$\nu - \mu \vdash m_n - n$$

with probability $1 - o(1)$. The output-label law is exactly the target law, so its Hellinger factor is one. Combining Young concentration, [Corollary 4.2](#), and [Proposition 4.3](#) proves the lower bound in [Theorem 1.1](#).

4.2.2 Unknown spectrum of known rank

Define the randomized label dilation directly on the r active rows. For $i < r$, draw independent $U_i \sim \text{Unif}[-1/2, 1/2)$ and set

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + U_i) + \frac{1}{2} \right\rfloor, \quad \tilde{\nu}_r = m_n - \sum_{i < r} \tilde{\nu}_i.$$

Pad by $d - r$ zeros and use the compatibility check and fallback replacement of [Section 2.4](#). When $r = 1$, simply set $\tilde{\nu} = (m_n)$. On measured labels with more than r rows, choose the concrete fallback output $(m_n, 0, \dots, 0)$; these branches have zero probability on rank- r inputs. Call this padded kernel $K_n^{\text{rnd},(r)}$, and denote the resulting ambient lifted channel by

$$\mathcal{C}_n^{(r)} := \mathcal{Q}_n^{(d)}[K_n^{\text{rnd},(r)}].$$

It depends on r, d, n, m_n , but not on the unknown positive spectrum p .

The dimension- r Young local limit gives, uniformly for $p \in K \Subset \Delta_r^\circ$,

$$F(Q_{m_n,p}^{(r)}, P_{m_n,p}^{(r)}) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(r-1)/2} + o(1).$$

For $r = 1$, no local-limit theorem is needed: both laws are the point mass at the one-row label, so the fidelity is exactly one. Typical rounded pairs are compatible, and [Corollary 4.2](#) supplies the common sector factor. [Proposition 4.3](#) therefore gives (3) and proves the achievability assertion in [Theorem 1.2](#).

4.3 Fixed-rank LAN and the converses

The same local experiment covers every $1 \leq r \leq d$. For $r = d$ it reduces to ordinary interior LAN. For $r < d$, its support–kernel coordinates are genuine vacuum boundary modes supplied directly by fixed-rank LAN, rather than by a singular limit from faithful states [[LN24](#)].

Put

$$\mathbf{H}_r = \left\{ h \in \mathbb{R}^r : \sum_{i=1}^r h_i = 0 \right\}, \quad \Sigma_p^{(r)} = (\text{diag}(p) - pp^\top) \Big|_{\mathbf{H}_r}.$$

Recall the active root set $\mathcal{I}_{r,d}$ and the convention $p_j = 0$ for $j > r$ from (1). Use the anti-Hermitian local generator

$$X_p(z) = \sum_{(i,j) \in \mathcal{I}_{r,d}} \frac{z_{ij} E_{ji} - \bar{z}_{ij} E_{ij}}{\sqrt{p_i - p_j}}. \quad (32)$$

For fixed $(h, z) \in \mathbf{H}_r \times \mathbb{C}^{\mathcal{I}_{r,d}}$, define the single-copy local state

$$\rho_{n,h,z} = e^{X_p(z)/\sqrt{n}} \text{diag}\left(p + \frac{h}{\sqrt{n}}, 0_{d-r}\right) e^{-X_p(z)/\sqrt{n}}. \quad (33)$$

It is a rank- r state for every sufficiently large n , uniformly when (h, z) ranges over a fixed compact set.

Let

$$q_{ij} = \begin{cases} p_j/p_i, & j \leq r, \\ 0, & j > r, \end{cases} \quad \mathcal{F}_{r,d} = \bigotimes_{(i,j) \in \mathcal{I}_{r,d}} \mathcal{F}_{ij},$$

and put

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|, \quad \tau_p^{(r,d)} = \bigotimes_{(i,j) \in \mathcal{I}_{r,d}} \tau_{q_{ij}}, \quad D(z) = \bigotimes_{(i,j) \in \mathcal{I}_{r,d}} D(z_{ij}).$$

Thus $\tau_0 = |0\rangle\langle 0|$. The input and ideal target limit experiments are

$$\Phi_{h,z}^{(r,d)} = \mathbf{N}_{h, \Sigma_p^{(r)}} \otimes D(z) \tau_p^{(r,d)} D(z)^*, \quad (34)$$

$$\Psi_{h,z}^{(r,d,\gamma)} = \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p^{(r)}} \otimes D(\sqrt{\gamma}z) \tau_p^{(r,d)} D(\sqrt{\gamma}z)^*. \quad (35)$$

When $r < d$, the support–kernel coordinates are shifted pure Gaussian states, not thermal states with a small positive temperature. When $r = d$, that collection is empty.

Theorem 4.4 (Two-way LAN at fixed rank). *Fix d, r , and $p_1 > \dots > p_r > 0$. There are channels*

$$\begin{aligned} T_N^{(r)} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) &\longrightarrow L^1(\mathbf{H}_r; \mathcal{T}_1(\mathcal{F}_{r,d})), \\ S_N^{(r)} : L^1(\mathbf{H}_r; \mathcal{T}_1(\mathcal{F}_{r,d})) &\longrightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \end{aligned}$$

such that, for every compact $C \subset \mathbf{H}_r \times \mathbb{C}^{\mathcal{I}_{r,d}}$,

$$\sup_{(h,z) \in C} \left\| T_N^{(r)}(\rho_{n,h,z}^{\otimes n}) - \Phi_{h,z}^{(r,d)} \right\|_1 \longrightarrow 0, \quad (36)$$

$$\sup_{(h,z) \in C} \left\| S_n^{(r)}(\Phi_{h,z}^{(r,d)}) - \rho_{n,h,z}^{\otimes n} \right\|_1 \longrightarrow 0, \quad (37)$$

$$\sup_{(h,z) \in C} \left\| T_{m_n}^{(r)}(\rho_{n,h,z}^{\otimes m_n}) - \Psi_{h,z}^{(r,d,\gamma)} \right\|_1 \longrightarrow 0. \quad (38)$$

Proof. For every $1 \leq r \leq d$, the first two assertions are the two-way fixed-rank LAN theorem of Lahiry and Nussbaum [LN24, Theorem 3.3], after the coordinate identification in Remark 4.5. They explicitly note that $r = d$ recovers ordinary interior LAN. Their result is pointwise in the fixed base spectrum p , which is exactly what the converse below needs. In fact, they prove uniformity on expanding boxes: the spectral coordinate may grow as n^a , $a < 1/4$, and the rotation coordinate as n^β , $\beta < 1/9$ (their symbol for a is γ , renamed here to avoid conflict with the cloning ratio). We use only the weaker compact-set consequence. At $r = d$, this is ordinary interior LAN in the same source-normalized coordinate convention.

In both source normalizations, the spectral coordinate $u = (u_1, \dots, u_{r-1}) \in \mathbb{R}^{r-1}$ is identified with

$$h = \iota(u) := \left(u_1, \dots, u_{r-1}, -\sum_{i < r} u_i \right) \in \mathbf{H}_r.$$

If $V_p = (\delta_{ij}p_i - p_i p_j)_{i,j < r}$ is their multinomial covariance and A is the matrix of ι , then

$$AV_p A^\top = \text{diag}(p) - pp^\top.$$

Consequently, their classical normal experiment pushes forward to $\mathbf{N}_{h, \Sigma_p^{(r)}}$, as used above.

For the third assertion, observe the exact reparametrization

$$\rho_{n,h,z} = \rho_{m_n, \sqrt{\gamma_n}h, \sqrt{\gamma_n}z}, \quad \gamma_n = m_n/n.$$

Apply the forward LAN channel at sample size m_n . A fixed compact set contains all the scaled parameters for large n , and $\gamma_n \rightarrow \gamma$. Translation of the Gaussian density and Weyl displacement are uniformly trace-norm continuous on that compact set, which gives (38). $\square$

Remark 4.5 (Coordinate normalization). The normalization in (32) is chosen so that the finite-dimensional parameter z is also the oscillator displacement in (34). This is the source-normalized convention of the cited fixed-rank LAN theorem and the convention introduced in Section 3.1. Other physical rotation coordinates are related by invertible, spectrum-dependent linear maps, but the Gaussian displacement must then be reparametrized by the same map. We use $X_p(z) \leftrightarrow D(z)$ throughout, including when $r = d$.

4.3.1 Gaussian amplification with thermal and vacuum modes

For $q \in [0, 1)$, the quantum-limited amplifier obeys

$$\mathcal{A}_\gamma(D(z)\tau_q D(z)^*) = D(\sqrt{\gamma}z)\tau_{q^{(\gamma)}} D(\sqrt{\gamma}z)^*, \quad q^{(\gamma)} = 1 - \frac{1-q}{\gamma}.$$

Its fidelity against the ideal target is

$$F(\tau_{q^{(\gamma)}}, \tau_q) = f_\gamma(q).$$

At the boundary this reads

$$q = 0, \quad q^{(\gamma)} = \frac{\gamma-1}{\gamma}, \quad F(\tau_{(\gamma-1)/\gamma}, |0\rangle\langle 0|) = \gamma^{-1/2}.$$

The product amplifier therefore gives achievability uniformly across the thermal and vacuum modes. The converse requires one explicit boundary step because the vacuum target is not faithful. If $\hat{N}$ is the number operator, the positive-temperature one-mode witness is

$$W_q = \sqrt{\frac{1-q}{1-q^{(\gamma)}}} \left(\frac{q}{q^{(\gamma)}} \right)^{\hat{N}/2}, \quad q > 0.$$

Retain this injective witness for every thermal mode and use

$$W_0 = \sqrt{\gamma} |0\rangle\langle 0|$$

for every coherent mode. Moreover,

$$\|W_q - W_0\| = \sqrt{\gamma} \sqrt{\frac{\gamma q}{\gamma - 1 + q}} \rightarrow 0. \quad (39)$$

Let P_{vac} be the identity on the thermal modes tensored with the joint-vacuum projection on all coherent modes, and take the tensor product W of the displayed one-mode witnesses. Write

$$\tau = \tau_p^{(r,d)}, \quad \tau^{(\gamma)} = \bigotimes_{(i,j) \in \mathcal{I}_{r,d}} \tau_{q_{ij}}^{(\gamma)}, \quad B = F_{\text{orb}}^{(r,d)}(\gamma, p).$$

The operator W is compact: every positive-temperature factor is compact and every zero-temperature factor is rank one. The ideal zero-displacement target τ is supported on P_{vac} , so, for every positive output R ,

$$F(R, \tau) = F(P_{\text{vac}} R P_{\text{vac}}, \tau).$$

On that support W is injective. Modewise,

$$\text{Tr}(\tau_{(\gamma-1)/\gamma} W_0) = \text{Tr}(|0\rangle\langle 0| (W_0|_{\mathbb{C}|0})^{-1}) = \gamma^{-1/2}.$$

Together with the corresponding $q > 0$ identities, this gives

$$\text{Tr}(\tau^{(\gamma)} W) = \text{Tr}(\tau(W|_{P_{\text{vac}} \mathcal{F}_{r,d}})^{-1}) = B. \quad (40)$$

The weighted-fidelity lemma from Appendix D, applied after support compression, therefore yields

$$F(R, \tau)^2 \leq B \text{Tr}(RW). \quad (41)$$

The least-noise moment inequality also extends to the vacuum modes. Indeed, replace every zero temperature by $q > 0$, apply the positive-temperature inequality, and let $q \downarrow 0$. Here $\tau_q \rightarrow |0\rangle\langle 0|$ and $\tau_{q^{(\gamma)}} \rightarrow \tau_{(\gamma-1)/\gamma}$ in trace norm, while (39) gives operator-norm convergence of the witnesses. Hence every completely positive, trace-nonincreasing, gain- $\sqrt{\gamma}$ displacement-covariant map Γ satisfies

$$\text{Tr}[\Gamma(\tau)W] \leq \text{Tr}[\Gamma(\tau)] \text{Tr}(\tau^{(\gamma)} W) \leq B. \quad (42)$$

Combining (41) and (42) gives $F(\Gamma(\tau), \tau) \leq B$, including when a compact-limit map loses trace.

The corresponding fiberwise statement will be used for the hybrid model. If $R(y)$ is a positive trace-class-valued density, g is a probability density on $\mathbf{H}_r$, and $\chi > 0$, then compression followed by the hybrid weighted-fidelity lemma gives

$$F(R, g\tau)^2 \leq \left(\int_{\mathbf{H}_r} \chi(y) \text{Tr}[R(y)W] dy \right) \left(\int_{\mathbf{H}_r} \frac{g(y)}{\chi(y)} dy \right) B. \quad (43)$$

For $L > 0$, let

$$\begin{aligned} \mathbf{Z}_L^{(r,d)} &= \{z : |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for every } (i,j) \in \mathcal{I}_{r,d}\}, \\ \mathbf{H}_{r,L} &= \{h \in \mathbf{H}_r : \|h\|_\infty \leq L\}, \quad \Omega_L^{(r,d)} = \mathbf{H}_{r,L} \times \mathbf{Z}_L^{(r,d)}. \end{aligned}$$

Here $\mathbf{H}_r$ carries its induced Lebesgue measure, complex coordinates carry their standard real Lebesgue measure, and the zero-dimensional volume is understood to be one when $r = 1$.

Proposition 4.6 (Mixed thermal-coherent flat-prior optima). *With $\phi_z = D(z)\tau D(z)^*$ and $\psi_z = D(\sqrt{\gamma}z)\tau D(\sqrt{\gamma}z)^*$,*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\mathbf{Z}_L^{(r,d)}|} \int_{\mathbf{Z}_L^{(r,d)}} F(\Lambda(\phi_z), \psi_z) \, dz = B, \quad (44)$$

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L^{(r,d)}|} \int_{\Omega_L^{(r,d)}} F\left(\Lambda(\Phi_{h,z}^{(r,d)}), \Psi_{h,z}^{(r,d,\gamma)}\right) \, dh \, dz = c_\gamma^{(r-1)/2} B. \quad (45)$$

No Gaussianity, covariance, or product assumption is imposed on the competing channels. In (44), the supremum is over quantum channels on $\mathcal{T}_1(\mathcal{F}_{r,d})$. In (45), it is over classical-quantum channels on $L^1(\mathbf{H}_r; \mathcal{T}_1(\mathcal{F}_{r,d}))$.

Proof. For achievability, tensor the quantum-limited amplifiers over all active modes. On a coherent mode this is the usual amplifier with vacuum input; its added thermal noise gives $f_\gamma(0) = \gamma^{-1/2}$. For the hybrid model, tensor this channel with the classical pushforward $x \mapsto \sqrt{\gamma}x$. Multiplicativity gives the two displayed lower bounds.

For the converse, the Følner averaging and compact-map argument of Appendix D produces a gain- $\sqrt{\gamma}$ displacement-covariant, trace-nonincreasing limiting map. Equations (41)–(42) give the orbital bound. In the hybrid argument, compress each output fiber $R(y) \mapsto P_{\text{vac}} R(y) P_{\text{vac}}$ before applying the hybrid weighted-fidelity lemma. This does not change fidelity with the target, and it makes the restricted witness injective. The scalar convolution part of the proof is otherwise unchanged and contributes c_γ^{r-1} at the squared-fidelity level, hence $c_\gamma^{(r-1)/2}$ at root fidelity. This proves (45). $\square$

We now spell out the finite-dimensional transfer. Define

$$u_n^{(r)}(p) = \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right). \quad (46)$$

For a fixed L , restrict the orbit to the states $\rho_{n,0,z}$, $z \in \mathbf{Z}_L^{(r,d)}$. Discard the parameter-independent classical factor in $T_N^{(r)}$ and reinsert it in $S_N^{(r)}$. The general LAN fidelity-transfer lemma, Lemma 3.2, turns every nearly optimal finite cloner into a Gaussian channel Γ_n satisfying

$$u_n^{(r)}(p) \leq \inf_{z \in \mathbf{Z}_L^{(r,d)}} F(\Gamma_n(\phi_z), \psi_z) + o(1).$$

The infimum is no larger than the flat-prior average. First take $\limsup_n$, then the supremum over Gaussian-model channels, and finally $L \rightarrow \infty$. Equation (44) gives $\limsup_n u_n^{(r)}(p) \leq B$. Together with the finite-dimensional lower bound, this proves Theorem 1.1.

For all sufficiently large n , define the local fixed-rank minimax value

$$v_{n,L}^{(r)}(p) = \sup_{\mathcal{M}_n} \inf_{(h,z) \in \Omega_L^{(r,d)}} F\left(\mathcal{M}_n(\rho_{n,h,z}^{\otimes n}), \rho_{n,h,z}^{\otimes m_n}\right). \quad (47)$$

Proposition 4.7 (Pointwise fixed-rank local minimax limit). *For every $p \in \Delta_r^\circ$,*

$$\lim_{L \rightarrow \infty} \limsup_{n \rightarrow \infty} v_{n,L}^{(r)}(p) = \lim_{L \rightarrow \infty} \liminf_{n \rightarrow \infty} v_{n,L}^{(r)}(p) = F_{\text{univ}}^{(r,d)}(\gamma, p).$$

Proof. For fixed L , all spectra $p + h/\sqrt{n}$ lie in one compact subset of Δ_r° for large n . The fixed-rank channel constructed above and continuity of (3) therefore give the lower bound uniformly on the local family.

For the converse, choose a channel within $1/n$ of $v_{n,L}^{(r)}(p)$ and sandwich it between $S_n^{(r)}$ and $T_{m_n}^{(r)}$. The three approximations in Theorem 4.4 and fidelity transfer give

$$v_{n,L}^{(r)}(p) \leq \inf_{(h,z) \in \Omega_L^{(r,d)}} F\left(\Gamma_n(\Phi_{h,z}^{(r,d)}), \Psi_{h,z}^{(r,d,\gamma)}\right) + o(1).$$

Bound the infimum by the average and use (45), first letting $n \rightarrow \infty$ and then $L \rightarrow \infty$. This yields the upper bound. The outer limits exist because both the limsup and liminf are nonincreasing in L . $\square$

Finally, for a compact set K as in Theorem 1.2, write

$$w_n^{(r)}(K) = \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \text{U}(d)}} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right).$$

If p_0 lies in the interior of K relative to the affine hyperplane $\sum_i p_i = 1$, every fixed local spectral box around p_0 lies in K for all sufficiently large n . Hence $w_n^{(r)}(K) \leq v_{n,L}^{(r)}(p_0)$. Let $n \rightarrow \infty$, then $L \rightarrow \infty$, and finally take the infimum over such p_0 . The assumption that K is the closure of this relative interior and the continuity of $F_{\text{univ}}^{(r,d)}$ give

$$\limsup_n w_n^{(r)}(K) \leq \inf_{p \in K} F_{\text{univ}}^{(r,d)}(\gamma, p).$$

The uniform rank-aware construction above supplies the reverse inequality and completes Theorem 1.2.

5 The Grassmannian projector orbit

The flat rank- r state is $\rho_P = P/r$, with $P \in \text{Gr}(r, d)$. For $r > 1$, its positive eigenvalue has multiplicity r , so it is not covered by the simple-positive-spectrum theorem. For $r = 1$, it is the pure-state orbit already covered by Theorem 1.1; the direct calculation below provides an independent derivation. We prove Theorem 1.3 without invoking degenerate LAN.

5.1 Exact sector calculation

Let

$$D_s^\lambda = \dim \mathcal{H}_\lambda^{(s)}, \quad R_\lambda = \frac{D_d^\lambda}{D_r^\lambda} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\lambda_i + j - i}{j - i}. \quad (48)$$

For the standard projector P_0 onto the first r coordinates, the conditional state in the λ -sector is

$$\rho_{P_0, \lambda} = J_\lambda \frac{I_{\mathcal{H}_\lambda^{(r)}}}{D_r^\lambda} J_\lambda^*.$$

The r -dimensional Cartan channel is $U(r)$ -covariant. It therefore maps the maximally mixed state on the irreducible input module exactly to the maximally mixed state on the irreducible output module. The compression identity (29) consequently becomes, for every compatible pair,

$$F\left(\mathcal{C}_{\mu,\nu}^{(d)}(\rho_{P_0,\mu}), \rho_{P_0,\nu}\right) = \sqrt{\frac{R_\mu}{R_\nu}}. \quad (49)$$

For $\mu_i/n \rightarrow 1/r$ and $\nu_i/m_n \rightarrow 1/r$, this tends to the right-hand side of (8).

5.2 A compatible exact-marginal coupling of flat Schur laws

Let

$$\mathbf{P}_N^{(r)}(\lambda) = \frac{\dim \mathcal{S}_\lambda D_r^\lambda}{r^N}, \quad \lambda \vdash N, \quad \ell(\lambda) \leq r, \quad (50)$$

be the Schur–Weyl law of the maximally mixed r -level state. The complete Schur decomposition is

$$(P/r)^{\otimes N} = \bigoplus_{\ell(\lambda) \leq r} \mathbf{P}_N^{(r)}(\lambda) \frac{\Pi_{\lambda,P}}{D_r^\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}. \quad (51)$$

Here $\Pi_{\lambda,P}$ is the rank- D_r^λ support projection defined intrinsically by the subspace $P \subset \mathbb{C}^d$. The Schur–Weyl/random-word central limit theorem [Kup02, Joh01] states that

$$X_N := \frac{\lambda - (N/r)\mathbf{1}_r}{\sqrt{N}} \Rightarrow X, \quad (52)$$

where X is a fixed scaling of the ordered eigenvalue vector of a traceless $r \times r$ GUE matrix. In particular,

$$\Pr(X_1 > \dots > X_r) = 1.$$

Both X_n and X_{m_n} converge to the same law. Their Prokhorov distance therefore tends to zero, so Strassen’s coupling theorem supplies a coupling $\Pi_n(\mu, \nu)$ of the *exact* marginals $\mathbf{P}_n^{(r)}$ and $\mathbf{P}_{m_n}^{(r)}$ such that

$$\left\| \frac{\mu - (n/r)\mathbf{1}_r}{\sqrt{n}} - \frac{\nu - (m_n/r)\mathbf{1}_r}{\sqrt{m_n}} \right\|_\infty \rightarrow 0 \quad \text{in probability.} \quad (53)$$

Disintegrate this finite coupling to obtain a label kernel $K_n(\nu \mid \mu)$.

Put $\omega = \nu - \mu$. Tightness in (53) gives $\omega_i > 0$ for every i with probability $1 - o(1)$, because its deterministic leading term is $(m_n - n)/r$. For adjacent rows,

$$\begin{aligned} \frac{\omega_i - \omega_{i+1}}{\sqrt{n}} &= \sqrt{\gamma_n} \left[\frac{\nu_i - \nu_{i+1}}{\sqrt{m_n}} \right] - \left[\frac{\mu_i - \mu_{i+1}}{\sqrt{n}} \right] \\ &\Rightarrow (\sqrt{\gamma} - 1)(X_i - X_{i+1}) > 0. \end{aligned}$$

Thus ω is a partition with probability $1 - o(1)$. Use the ambient Cartan-sector channel on this good event and, otherwise, the covariant replacement channel into the same sampled ν -block. The output-label marginal is exactly the target law, and (49) holds uniformly on the typical part. If G_n is the intersection of compatibility and the two typical-label events, the direct-sum fidelity formula and joint concavity give explicitly

$$F\left(\mathcal{Q}_n^{(d)}[K_n]((P/r)^{\otimes n}), (P/r)^{\otimes m_n}\right) \geq \sum_{(\mu,\nu) \in G_n} \Pi_n(\mu, \nu) \sqrt{\frac{R_\mu}{R_\nu}} \rightarrow \gamma^{-r(d-r)/2}.$$

Indeed, $\Pi_n(G_n^c) \rightarrow 0$, while the summand converges uniformly on the typical part. On every compatible pair, $\nu_i \geq \mu_i$ and hence

$$0 \leq \sqrt{\frac{R_\mu}{R_\nu}} \leq 1.$$

Thus convergence in probability under the coupling also implies convergence of the displayed expectations (equivalently, the summands are uniformly integrable). This proves achievability in [Theorem 1.3](#).

5.3 Direct finite-dimensional converse

We give the argument because it avoids any appeal to degenerate-spectrum LAN. Twirling a cloner over $U(d)$, and over input and output permutations, cannot decrease its worst-case Grassmannian fidelity. After Schur pinching, let

$$\Phi_{\nu\mu} : \mathcal{T}_1(\mathcal{H}_\mu^{(d)}) \longrightarrow \mathcal{T}_1(\mathcal{H}_\nu^{(d)})$$

be its $U(d)$ -equivariant completely positive subchannels. Schur's lemma gives

$$\Phi_{\nu\mu}^*(I_{\mathcal{H}_\nu^{(d)}}) = K(\nu \mid \mu) I_{\mathcal{H}_\mu^{(d)}}, \quad \sum_\nu K(\nu \mid \mu) = 1.$$

Thus K is a state-independent label transition. For every branch with $K(\nu \mid \mu) > 0$, normalize it to the covariant channel

$$\mathcal{E}_{\mu,\nu} = \frac{\Phi_{\nu\mu}}{K(\nu \mid \mu)}.$$

Zero-probability branches may be defined arbitrarily.

Let $\Pi_{\lambda,P}$ be the rank- D_r^λ support projection of the conditional λ -sector state. Covariance and irreducibility give

$$\mathcal{E}_{\mu,\nu}(I_{\mathcal{H}_\mu^{(d)}}) = \frac{D_d^\mu}{D_\nu^\nu} I_{\mathcal{H}_\nu^{(d)}}.$$

The unnormalized ν -block of the actual output is therefore

$$Y_{\nu,P} = \sum_\mu \mathbf{P}_n^{(r)}(\mu) K(\nu \mid \mu) \mathcal{E}_{\mu,\nu} \left(\frac{\Pi_{\mu,P}}{D_r^\mu} \right),$$

whereas the target block is $\mathbf{P}_{m_n}^{(r)}(\nu) \Pi_{\nu,P} / D_r^\nu$. Since $\Pi_{\mu,P} / D_r^\mu \leq I / D_r^\mu$, positivity gives the uniform support-overlap bound

$$\mathrm{Tr} \left[\Pi_{\nu,P} \mathcal{E}_{\mu,\nu} \left(\frac{\Pi_{\mu,P}}{D_r^\mu} \right) \right] \leq \frac{R_\mu}{R_\nu}. \quad (54)$$

For any positive, possibly subnormalized A ,

$$F \left(A, \frac{\Pi_{\nu,P}}{D_r^\nu} \right) \leq \sqrt{\mathrm{Tr}(\Pi_{\nu,P} A)}, \quad (55)$$

because $\mathrm{Tr} \sqrt{B} \leq \sqrt{\mathrm{rank}(B) \mathrm{Tr} B}$. Using the direct-sum fidelity formula, then (54)–(55), and finally Cauchy–Schwarz over ν , gives the finite- n upper bound

$$F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n})$$

$$\begin{aligned}
&\leq \sum_{\nu} \sqrt{\frac{P_{m_n}^{(r)}(\nu)}{R_{\nu}}} \sqrt{\sum_{\mu} P_n^{(r)}(\mu) K(\nu | \mu) R_{\mu}} \\
&\leq \left[\left(\sum_{\mu} P_n^{(r)}(\mu) R_{\mu} \right) \left(\sum_{\nu} \frac{P_{m_n}^{(r)}(\nu)}{R_{\nu}} \right) \right]^{1/2}.
\end{aligned} \tag{56}$$

Here and in this display, the ν -sums may be restricted to $\ell(\nu) \leq r$, since the target law vanishes otherwise. Any output mass in longer diagrams only strengthens the final inequality.

Let $k = r(d - r)$ and

$$C_{r,d} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{1}{r(j-i)}.$$

Young concentration at the uniform r -spectrum and (48) give

$$\frac{1}{n^k} \sum_{\mu} P_n^{(r)}(\mu) R_{\mu} \longrightarrow C_{r,d}, \tag{57}$$

$$m_n^k \sum_{\nu} \frac{P_{m_n}^{(r)}(\nu)}{R_{\nu}} \longrightarrow C_{r,d}^{-1}. \tag{58}$$

For the first limit, $n^{-k} R_{\mu}$ is uniformly bounded. For the second, restrict first to $\nu_r \geq m_n/(2r)$, where the rescaled inverse is uniformly bounded; the complement has exponentially small probability, and globally $m_n^k/R_{\nu} \leq m_n^k$ because $R_{\nu} \geq 1$. Thus the exponential Schur large-deviation bound gives uniform integrability. Substitution into (56) yields

$$\limsup_n F \leq \left(\frac{n}{m_n} \right)^{k/2} \longrightarrow \gamma^{-r(d-r)/2},$$

which completes the converse.

A Strict comparison with purify–clone–trace

This appendix upgrades the comparison from PCT’s published data-processing guarantee to the actual mixed-system output after the purifying registers are discarded. For every fixed $\gamma > 1$, without assuming a near-unit cloning fidelity, we obtain an explicit upper bound on that output and prove that it is strictly below the fidelity achieved by our cloner when $r > 1$. For a rank- r state with simple positive spectrum, the exact support factorization of [Lemma A.5](#) separates the ambient support-embedding cost from a PCT channel intrinsic to the positive support. We analyze that intrinsic channel first and then apply the factorization to every ambient dimension. Flat projector states are treated separately by a commuting-statistics argument. Thus the comparison is directly between the fixed-gain asymptotic fidelities of the two cloners, not merely between their guarantees near fidelity one.

A.1 Simple spectra on the positive support

In this subsection d denotes the intrinsic positive-support dimension, not a larger ambient dimension. Fix $d \geq 2$ and

$$p_1 > \cdots > p_d > 0, \quad \sum_{i=1}^d p_i = 1, \quad \rho_p = \text{diag}(p_1, \dots, p_d),$$

and let $m_n/n \rightarrow \gamma > 1$. We use root fidelity $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$. Put

$$q_{ij} = \frac{p_j}{p_i}, \quad h_q(x) = \frac{\sqrt{(1-q)(1-x)}}{1 - \sqrt{qx}},$$

and define

$$q_{ij}^{\text{PCT}} = \frac{\gamma - 1 + \gamma q_{ij}}{\gamma + (\gamma - 1)q_{ij}}. \quad (59)$$

The resulting upper bound on the actual post-trace purify–clone–trace output is

$$B_{\text{PCT}}(\gamma, p) = \left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(d-1)/2} \prod_{i < j} h_{q_{ij}}(q_{ij}^{\text{PCT}}). \quad (60)$$

Let $\sigma_{n, m_n}^{\text{PCT}}(p)$ be the mixed-system output obtained by random purification in $\mathbb{C}^d \otimes \mathbb{C}^d$, Werner’s pure-state $n \rightarrow m_n$ cloner in dimension d^2 , and partial trace over every purifying register.

Theorem A.1 (Strict comparison on a simple positive support). *For fixed d , p , and $\gamma > 1$ as above,*

$$\limsup_{n \rightarrow \infty} F(\sigma_{n, m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) \leq B_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p) < F_{\text{orb}}(\gamma, p). \quad (61)$$

Here

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_\gamma(q_{ij}), \quad F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} F_{\text{orb}}(\gamma, p),$$

where

$$f_\gamma(q) = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q}.$$

We first identify the limiting channel obtained by composing partial trace with interior LAN, and then feed into that channel the Fock-space limit of Werner’s cloner.

A.1.1 Partial trace followed by interior LAN

Let $H = E = \mathbb{C}^d$, set

$$S = \text{diag}(\sqrt{p_1}, \dots, \sqrt{p_d}), \quad |\psi_p\rangle = \sum_{i=1}^d \sqrt{p_i} |ii\rangle,$$

and identify a vector in $H \otimes E$ with its coefficient matrix:

$$|Z\rangle = \sum_{i,j} Z_{ij} |ij\rangle.$$

The one-particle purification-tangent space is

$$K_p := \psi_p^\perp = K_p^{\text{diag}} \oplus \bigoplus_{i < j} \text{span}\{|ij\rangle, |ji\rangle\},$$

where

$$K_p^{\text{diag}} = \left\{ \sum_i z_i |ii\rangle : \sum_i \sqrt{p_i} z_i = 0 \right\}.$$

Let $\Gamma_s(K_p)$ be the symmetric Fock space over K_p , and let $b(u)$ denote the annihilation operator in the one-particle direction $u \in K_p$. We abbreviate $b_{ij} = b(|ij\rangle)$ for $i \neq j$.

For a matrix tangent $Z \in K_p$, the differential of partial trace at the canonical purification is

$$D_p(Z) := \left. \frac{d}{dt} \right|_{t=0} \text{Tr}_E |\psi_p + tZ\rangle \langle \psi_p + tZ| = ZS + SZ^*. \quad (62)$$

Its diagonal and positive-CCR orbital coordinates are

$$h_i(Z) = 2\sqrt{p_i} \Re Z_{ii}, \quad (63)$$

$$\beta_{ij}(Z) = \frac{D_p(Z)_{ji}}{\sqrt{p_i - p_j}} = \frac{\sqrt{p_i} Z_{ji} + \sqrt{p_j} \overline{Z_{ij}}}{\sqrt{p_i - p_j}}, \quad i < j. \quad (64)$$

Notice that $\sum_i h_i(Z) = 0$. Set

$$\mathbf{H}_0 = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad \bar{\Sigma}_p = \text{diag}(p) - pp^\top, \quad \Sigma_p = \bar{\Sigma}_p|_{\mathbf{H}_0}.$$

The explicit limiting channel. We now construct a classical-quantum Gaussian channel

$$\Lambda_p : \mathcal{T}_1(\Gamma_s(K_p)) \longrightarrow L^1\left(\mathbf{H}_0; \mathcal{T}_1(\mathcal{F}^{\otimes \binom{d}{2}})\right).$$

For the diagonal block, define

$$v_i = \sqrt{p_i} |ii\rangle - p_i |\psi_p\rangle \in K_p^{\text{diag}}, \quad X_i = b(v_i) + b(v_i)^*. \quad (65)$$

These fields commute. Indeed,

$$\langle v_i, v_j \rangle = p_i \delta_{ij} - p_i p_j = (\bar{\Sigma}_p)_{ij} \quad (66)$$

is real, so $[X_i, X_j] = 0$. Moreover $\sum_i X_i = 0$. The diagonal part of Λ_p is the joint spectral measurement of $(X_1, \dots, X_d)$, regarded as a classical variable in $\mathbf{H}_0$.

For each $i < j$, put $\Delta_{ij} = p_i - p_j$ and define

$$a_{ij} = \frac{\sqrt{p_i} b_{ji} + \sqrt{p_j} b_{ij}^*}{\sqrt{\Delta_{ij}}}, \quad (67)$$

$$c_{ij} = \frac{\sqrt{p_i} b_{ij} + \sqrt{p_j} b_{ji}^*}{\sqrt{\Delta_{ij}}}. \quad (68)$$

A direct CCR calculation gives

$$[a_{ij}, a_{ij}^*] = [c_{ij}, c_{ij}^*] = 1, \quad [a_{ij}, c_{ij}] = [a_{ij}, c_{ij}^*] = 0.$$

Thus $(b_{ij}, b_{ji}) \mapsto (a_{ij}, c_{ij})$ is a two-mode squeezing transformation. The (i, j) -part of Λ_p applies this transformation and discards the complementary mode c_{ij} , retaining a_{ij} . Different unordered pairs use disjoint input modes and commute with the diagonal fields, so these blocks form a product channel.

Let $|Z\rangle$ now denote the Fock coherent vector of amplitude $Z \in K_p$. The preceding construction gives

$$\Lambda_p(|Z\rangle\langle Z|) = \mathbf{N}_{h(Z), \Sigma_p} \otimes \bigotimes_{i < j} D(\beta_{ij}(Z)) \tau_{q_{ij}} D(\beta_{ij}(Z))^*, \quad (69)$$

where

$$\tau_q = (1 - q) \sum_{r \geq 0} q^r |r\rangle\langle r|.$$

To verify the diagonal part, use $\langle X_i \rangle_Z = 2\Re\langle v_i, Z \rangle = h_i(Z)$ and (66). For the off-diagonal part, $\langle a_{ij} \rangle_Z = \beta_{ij}(Z)$, while on the input vacuum

$$\langle a_{ij}^* a_{ij} \rangle = \frac{p_j}{p_i - p_j}.$$

The corresponding geometric parameter is therefore $p_j/p_i = q_{ij}$.

Strong convergence of the finite channels. Let R_M project onto Fock number vectors of total occupation at most M , and let J_M map such a number vector to the normalized symmetric occupation vector with all remaining particles in ψ_p . Define the trace-preserving occupation embedding

$$\mathcal{J}_M(X) = J_M R_M X R_M J_M^* + \text{Tr}[(I - R_M)X] |\psi_p\rangle\langle\psi_p|^{\otimes M}. \quad (70)$$

Let T_M^p be an interior classical-quantum LAN extraction channel centered at ρ_p , with its orbital normalization and phase chosen so that its output annihilator is the LAN limit of the positive-CCR fluctuation

$$\frac{1}{\sqrt{M(p_i - p_j)}} \sum_{\ell=1}^M E_{ij}^{(\ell)}, \quad i < j.$$

Define

$$\Lambda_{p,M} = T_M^p \circ \text{Tr}_{E^M} \circ \mathcal{J}_M. \quad (71)$$

Lemma A.2 (Interior partial-trace/LAN limit). *For every trace-class operator X on $\Gamma_s(K_p)$,*

$$\|\Lambda_{p,M}(X) - \Lambda_p(X)\|_1 \longrightarrow 0. \quad (72)$$

Proof. First take a coherent amplitude $Z \in K_p$. Put

$$|\psi_{Z,M}\rangle = \frac{|\psi_p\rangle + M^{-1/2}|Z\rangle}{\sqrt{1 + \|Z\|^2/M}}.$$

The elementary multinomial-to-Poisson limit gives, uniformly for Z in bounded subsets of K_p ,

$$\left\| \mathcal{J}_M(|Z\rangle\langle Z|) - |\psi_{Z,M}\rangle\langle\psi_{Z,M}|^{\otimes M} \right\|_1 \rightarrow 0. \quad (73)$$

After the purifying systems are traced out, the second state in (73) becomes the product state $\rho_{Z,M}^{\otimes M}$, where

$$\rho_{Z,M} = \rho_p + M^{-1/2}(ZS + SZ^*) + O_Z(M^{-1}). \quad (74)$$

To make the local chart explicit, define the anti-Hermitian matrix

$$G(\beta) = \sum_{i < j} \frac{\beta_{ij}E_{ji} - \overline{\beta_{ij}}E_{ij}}{\sqrt{p_i - p_j}}. \quad (75)$$

It satisfies $[G(\beta), \rho_p]_{ji} = \sqrt{p_i - p_j} \beta_{ij}$, so (64) is precisely the off-diagonal coordinate of $D_p(Z)$. Because p has a simple spectrum, smooth eigenvalue/eigenvector perturbation theory, with the eigenvector phases fixed continuously at the standard basis, gives

$$\rho_{Z,M} = g_M \text{diag}(p + M^{-1/2}h_M)g_M^*, \quad h_M \rightarrow h(Z), \quad \sqrt{M} \log g_M \rightarrow G(\beta(Z)).$$

Thus $\rho_{Z,M}$ lies in the usual interior-LAN local chart with bounded coordinates converging to $(h(Z), \beta(Z))$. The finite-dimensional LAN theorem [KG09], uniformly on bounded local parameter sets, therefore gives

$$\Lambda_{p,M}(|Z\rangle\langle Z|) \rightarrow \mathbf{N}_{h(Z), \Sigma_p} \otimes \bigotimes_{i < j} D(\beta_{ij}(Z))\tau_{q_{ij}}D(\beta_{ij}(Z))^*$$

in trace norm. This is exactly (69).

It remains to extend the convergence to arbitrary trace-class inputs. The complex linear span of coherent projectors is trace-norm dense in the trace class. Indeed, a bounded operator annihilating that span has identically zero Husimi symbol; analyticity then makes every number-basis matrix element zero, and Hahn–Banach gives density. Both $\Lambda_{p,M}$ and Λ_p are trace-norm contractions. If Y is a finite linear combination of coherent projectors, then

$$\limsup_M \|\Lambda_{p,M}(X) - \Lambda_p(X)\|_1 \leq 2\|X - Y\|_1.$$

Approximating X by such Y proves (72). $\square$

Remark A.3 (Coordinate convention). The proof uses the positive-CCR displacement β_{ij} in (64). The generator $G(\beta)$ in (75) is exactly the source-normalized generator $X_p(\beta)$ of Section 3.1. Thus the same parameter β appears in the finite-dimensional local family and in the Gaussian displacement $D(\beta)$.

Action on the thermal tangent state. Let τ_ϑ be a one-mode thermal state with mean photon number

$$N = \frac{\vartheta}{1 - \vartheta}.$$

Because the product thermal state is invariant under passive changes of one-particle basis, it factorizes according to the diagonal/off-diagonal decomposition of K_p . The joint fields in (65) have covariance $(2N + 1)\Sigma_p$. For each $i < j$, the retained mode has mean photon number

$$N_{ij}^{\text{out}} = \frac{p_i N + p_j (N + 1)}{p_i - p_j}. \quad (76)$$

The input is centered Gaussian and gauge invariant, and $\langle a_{ij}^2 \rangle = 0$, so the retained one-mode Gaussian state is thermal, rather than merely having the displayed mean photon number. Consequently,

$$\frac{N_{ij}^{\text{out}}}{N_{ij}^{\text{out}} + 1} = \frac{N + (N + 1)q_{ij}}{N + 1 + Nq_{ij}}. \quad (77)$$

Taking

$$\vartheta = \frac{\gamma - 1}{\gamma}, \quad N = \gamma - 1,$$

turns (77) into (59). We have proved

$$\Lambda_p(\tau_\vartheta^{\otimes(d^2-1)}) = \mathbf{N}_{0, (2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}^{\text{PCT}}}. \quad (78)$$

A.1.2 The Fock limit of Werner cloning

We record explicitly the arbitrary-dimensional occupation calculation used below. Let $D \geq 2$, fix a unit vector $\psi \in \mathbb{C}^D$, and choose an orthonormal basis $e_1, \dots, e_{D-1}$ of $\psi^\perp$. For $\mathbf{k} = (k_1, \dots, k_{D-1})$, let $|\mathbf{k}\rangle_m$ be the normalized symmetric occupation vector with k_j particles in e_j and the remaining $m - |\mathbf{k}|$ particles in ψ .

Lemma A.4 (Werner occupation law). *Let $\mathcal{W}_{n,m}^{(D)}$ be Werner's universal pure-state cloner in dimension D . Its output on $|\psi\rangle\langle\psi|^{\otimes n}$ is diagonal in the occupation basis, with eigenvalue*

$$\pi_{n,m}^{(D)}(\mathbf{k}) = \frac{\binom{m-|\mathbf{k}|}{n}}{\binom{m+D-1}{n+D-1}}, \quad |\mathbf{k}| \leq m - n. \quad (79)$$

If $m_n/n \rightarrow \gamma > 1$, then under the occupation-Fock embedding

$$\mathcal{W}_{n,m_n}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n}) \longrightarrow \tau_{(\gamma-1)/\gamma}^{\otimes(D-1)} \quad (80)$$

in trace norm.

Proof. Werner's channel [Wer98] is

$$\mathcal{W}_{n,m}^{(D)}(X) = \frac{D[n]}{D[m]} P_m(X \otimes I^{\otimes(m-n)}) P_m, \quad D[r] = \binom{r+D-1}{D-1},$$

where P_m projects onto the symmetric subspace. On an occupation vector with total transverse occupation $t = |\mathbf{k}|$, the expectation of $P_m(|\psi\rangle\langle\psi|^{\otimes n} \otimes I) P_m$ is $\binom{m-t}{n} / \binom{m}{n}$. Multiplying by $D[n]/D[m]$

gives (79). Irreducibility of $\text{Sym}^t(\mathbb{C}^{D-1})$ under the stabilizer $U(D-1)$ makes the operator scalar on each fixed- t subspace, while the stabilizer's central phase eliminates cross- t blocks. Thus these are all of its eigenvalues. Their normalization also follows from Vandermonde's identity.

For fixed $\mathbf{k}$,

$$\pi_{n,m_n}^{(D)}(\mathbf{k}) \longrightarrow \gamma^{-(D-1)} \left(\frac{\gamma-1}{\gamma} \right)^{|\mathbf{k}|}.$$

The right side is the product of $D-1$ normalized geometric laws. Since both sides are probability masses on the countable occupation lattice, the discrete Scheffé lemma upgrades pointwise convergence to ℓ^1 convergence, which is exactly the trace-norm convergence (80). $\square$

A.1.3 Proof of the strict comparison

Proof of Theorem A.1. Use the canonical purification ψ_p . Every other minimal purification has the form $(I \otimes U)\psi_p$, with $U \in U(d)$. Covariance of Werner's cloner and invariance of partial trace give, for every U ,

$$\begin{aligned} & \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(d^2)}(((I \otimes U)|\psi_p\rangle\langle\psi_p|(I \otimes U^*))^{\otimes n}) \\ &= \text{Tr}_{E^m} \left[(I \otimes U)^{\otimes m} \mathcal{W}_{n,m}^{(d^2)}(|\psi_p\rangle\langle\psi_p|^{\otimes n})(I \otimes U^*)^{\otimes m} \right] \\ &= \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(d^2)}(|\psi_p\rangle\langle\psi_p|^{\otimes n}). \end{aligned}$$

Thus the Haar average in random purification disappears exactly after the purifying registers are discarded:

$$\sigma_{n,m}^{\text{PCT}}(p) = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(d^2)}(|\psi_p\rangle\langle\psi_p|^{\otimes n}). \quad (81)$$

Pull the Werner output in (81) back to $\Gamma_s(K_p)$ using the occupation embedding, and call the resulting Fock state $\omega_{n,m}$. Lemma A.4, with $D = d^2$, gives

$$\left\| \omega_{n,m_n} - \tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)} \right\|_1 \longrightarrow 0. \quad (82)$$

The definition of $\Lambda_{p,m}$ and (81) give the exact identity

$$T_{m_n}^p(\sigma_{n,m_n}^{\text{PCT}}(p)) = \Lambda_{p,m_n}(\omega_{n,m_n}).$$

By contractivity, Lemma A.2, and (82),

$$\begin{aligned} & \left\| \Lambda_{p,m_n}(\omega_{n,m_n}) - \Lambda_p\left(\tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)}\right) \right\|_1 \\ & \leq \left\| \omega_{n,m_n} - \tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)} \right\|_1 + \left\| \Lambda_{p,m_n}\left(\tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)}\right) - \Lambda_p\left(\tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)}\right) \right\|_1 \longrightarrow 0. \end{aligned}$$

Using (78), the LAN limit of the actual PCT output is therefore

$$\Theta_p^{\text{PCT}} = \mathbf{N}_{0,(2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}}^{\text{PCT}}. \quad (83)$$

Ordinary interior LAN applied to the ideal target gives

$$T_{m_n}^p(\rho_p^{\otimes m_n}) \longrightarrow \Theta_p := \mathbf{N}_{0,\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}} \quad (84)$$

in trace norm.

Fidelity is nondecreasing under $T_{m_n}^p$ and continuous under trace-norm convergence. Hence

$$\limsup_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) \leq F(\Theta_p^{\text{PCT}}, \Theta_p). \quad (85)$$

The fidelity on the right factorizes. In $d - 1$ real dimensions,

$$\begin{aligned} F(\mathbf{N}_{0,(2\gamma-1)\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) &= \left(\frac{2\sqrt{2\gamma-1}}{1+(2\gamma-1)} \right)^{(d-1)/2} \\ &= \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(d-1)/2}. \end{aligned} \quad (86)$$

For each thermal pair,

$$F(\tau_{q_{ij}}^{\text{PCT}}, \tau_{q_{ij}}) = h_{q_{ij}}(q_{ij}^{\text{PCT}}).$$

Equations (85) and (86) prove the first inequality in (61).

It remains to prove strictness. The optimal one-mode amplifier has thermal parameter

$$q_{ij}^{\text{opt}} = \frac{\gamma - 1 + q_{ij}}{\gamma}, \quad f_\gamma(q_{ij}) = h_{q_{ij}}(q_{ij}^{\text{opt}}).$$

Moreover,

$$q_{ij}^{\text{opt}} - q_{ij} = \frac{(\gamma - 1)(1 - q_{ij})}{\gamma} > 0.$$

Direct calculation gives

$$q_{ij}^{\text{PCT}} - q_{ij}^{\text{opt}} = \frac{q_{ij}(\gamma - 1)(1 - q_{ij})}{\gamma[\gamma + (\gamma - 1)q_{ij}]} > 0. \quad (87)$$

For fixed $q \in (0, 1)$,

$$\frac{d}{dx} \log h_q(x) = \frac{\sqrt{q} - \sqrt{x}}{2\sqrt{x}(1-x)(1-\sqrt{qx})} < 0, \quad x > q.$$

Thus every quantum factor in B_{PCT} is strictly smaller than the corresponding $f_\gamma(q_{ij})$. The classical factors obey

$$\frac{\sqrt{2\gamma-1}}{\gamma} < \frac{2\sqrt{\gamma}}{1+\gamma}, \quad \gamma > 1, \quad (88)$$

because, after squaring and clearing positive denominators, the difference is proportional to $(\gamma - 1)^2(2\gamma + 1) > 0$. Multiplying the strict modewise inequalities proves $B_{\text{PCT}} < F_{\text{univ}}$. Finally $2\sqrt{\gamma}/(1 + \gamma) < 1$, so $F_{\text{univ}} < F_{\text{orb}}$. $\square$

A.2 Exact support factorization

For every fixed-rank stratum with simple positive spectrum, the ambient PCT fidelity factors into the support-embedding contribution and the intrinsic positive-support fidelity analyzed above. For $s \geq 1$, write

$$\mathbf{d}_s(N) = \dim \text{Sym}^N(\mathbb{C}^s) = \binom{N + s - 1}{s - 1}.$$

Let $\sigma_{n,m}^{\text{PCT},(d,r)}(\bar{p})$ be the mixed-system PCT output for the rank- r state $\rho_{\bar{p}}$ on $\mathbb{C}^d$, using a minimal purifying register $\mathbb{C}^r$. Let $\sigma_{n,m}^{\text{PCT},(r,r)}(p)$ denote the corresponding output when the positive support itself is regarded as the system.

Lemma A.5 (Exact PCT support factorization). *Fix an isometry $V : \mathbb{C}^r \rightarrow \mathbb{C}^d$, put $P = VV^*$, and identify $\rho_{\bar{p}} = V\rho_pV^*$. Then*

$$\begin{aligned} F\left(\sigma_{n,m}^{\text{PCT},(d,r)}(\bar{p}), \rho_{\bar{p}}^{\otimes m}\right) \\ = \sqrt{s_{n,m}^{(r,d)}} F\left(\sigma_{n,m}^{\text{PCT},(r,r)}(p), \rho_p^{\otimes m}\right), \end{aligned} \quad (89)$$

where

$$s_{n,m}^{(r,d)} = \frac{\mathbf{d}_{rd}(n)/\mathbf{d}_{rd}(m)}{\mathbf{d}_{r^2}(n)/\mathbf{d}_{r^2}(m)} = \prod_{j=r^2}^{rd-1} \frac{n+j}{m+j}. \quad (90)$$

Proof. Choose the canonical minimal purification $|\psi_{\bar{p}}\rangle = (V \otimes I_r) \sum_i \sqrt{p_i} |ii\rangle$, and put $Q = P \otimes I_r$. The Haar average in random purification disappears after the purifying registers are traced out, by covariance of Werner's cloner. The projection Q has rank r^2 , contains $\psi_{\bar{p}}$, and commutes tensorwise with the symmetric projection. Compressing Werner's channel to $Q^{\otimes m}$ therefore gives

$$Q^{\otimes m} \mathcal{W}_{n,m}^{(rd)}(|\psi_{\bar{p}}\rangle\langle\psi_{\bar{p}}|^{\otimes n}) Q^{\otimes m} = s_{n,m}^{(r,d)} (V \otimes I_r)^{\otimes m} \mathcal{W}_{n,m}^{(r^2)}(|\psi_p\rangle\langle\psi_p|^{\otimes n}) (V^* \otimes I_r)^{\otimes m}.$$

After tracing the purifying systems, the target is supported on $P^{\otimes m}$. Fidelity is unchanged by compressing its first argument to that support, and $F(aA, B) = \sqrt{a}F(A, B)$, proving the claim. $\square$

For a rank- r state with simple positive spectrum, define

$$q_{ij}^{\text{PCT}} = \frac{\gamma - 1 + \gamma q_{ij}}{\gamma + (\gamma - 1)q_{ij}}, \quad (91)$$

$$B_{\text{PCT}}^{(r,d)}(\gamma, p) = \gamma^{-r(d-r)/2} \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(r-1)/2} \prod_{1 \leq i < j \leq r} h_{q_{ij}}(q_{ij}^{\text{PCT}}). \quad (92)$$

Corollary A.6 (Strict PCT comparison on every simple rank stratum). *For $2 \leq r \leq d$, fixed $p \in \Delta_r^\circ$, and fixed $\gamma > 1$,*

$$\limsup_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT},(d,r)}(\bar{p}), \rho_{\bar{p}}^{\otimes m_n}\right) \leq B_{\text{PCT}}^{(r,d)}(\gamma, p) < F_{\text{univ}}^{(r,d)}(\gamma, p) < F_{\text{orb}}^{(r,d)}(\gamma, p).$$

For $r = 1$, PCT is Werner's optimal pure-state cloner and equality holds with $F_{\text{orb}}^{(1,d)} = F_{\text{univ}}^{(1,d)} = \gamma^{-(d-1)/2}$.

Proof. The factor in (90) tends to $\gamma^{-r(d-r)}$. Apply Theorem A.1 in the positive support, with its dimension denoted there by $d = r$, and combine it with Lemma A.5. Multiplying its two strict inequalities by $\gamma^{-r(d-r)/2}$ gives the result. $\square$

A.3 Projector states

The purify-clone-trace (PCT) cloner combines Werner's pure-state cloner [Wer98] with random purification [TWZ25], then discards the purifying registers [LTHC26, FGSS26, JSO26]. On the projector orbit its actual mixed-system fidelity, not only its data-processing guarantee, can be compared directly with Theorem 1.3.

For $s \geq 1$, write

$$\mathbf{d}_s(N) = \dim \text{Sym}^N(\mathbb{C}^s) = \binom{N+s-1}{s-1},$$

and let $\Pi_{\text{sym},N}^{(s)}$ be the projection onto this symmetric subspace. For $m \geq n$ and an input X supported on $\text{Sym}^n(\mathbb{C}^s)$, Werner's pure-state n -to- m cloner in dimension s is

$$\mathcal{W}_{n,m}^{(s)}(X) = \frac{\mathbf{d}_s(n)}{\mathbf{d}_s(m)} \Pi_{\text{sym},m}^{(s)}(X \otimes I_s^{\otimes(m-n)}) \Pi_{\text{sym},m}^{(s)}. \quad (93)$$

Fix $P \in \text{Gr}(r, d)$, choose an isometry $V_P : \mathbb{C}^r \rightarrow \mathbb{C}^d$ with $V_P V_P^* = P$, and put

$$|\Phi_P\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r V_P |i\rangle \otimes |i\rangle, \quad |\Phi_r\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r |ii\rangle.$$

Let $\sigma_{n,m}^{\text{PCT}}(P)$ denote the system output of PCT on $(P/r)^{\otimes n}$, using a purifying register $\mathbb{C}^r$.

Proposition A.7 (PCT is strictly suboptimal on the projector orbit). *Define*

$$s_{n,m}^{(r,d)} = \frac{\mathbf{d}_{rd}(n)/\mathbf{d}_{rd}(m)}{\mathbf{d}_{r^2}(n)/\mathbf{d}_{r^2}(m)} = \prod_{j=r^2}^{rd-1} \frac{n+j}{m+j}. \quad (94)$$

There is an exact finite-sample factorization

$$F(\sigma_{n,m}^{\text{PCT}}(P), (P/r)^{\otimes m}) = \sqrt{s_{n,m}^{(r,d)}} F(\zeta_{n,m}^{(r)}, (I_r/r)^{\otimes m}), \quad (95)$$

where

$$\zeta_{n,m}^{(r)} = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(r^2)}(|\Phi_r\rangle\langle\Phi_r|^{\otimes n}). \quad (96)$$

If $m_n/n \rightarrow \gamma > 1$, then

$$\limsup_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n}) \leq \gamma^{-r(d-r)/2} \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(r-1)/2}. \quad (97)$$

Consequently, for every $r > 1$ and fixed $\gamma > 1$, PCT is strictly below the optimal Grassmannian value (8). For $r = 1$, the internal factor in (95) is one and PCT reduces to Werner's optimal pure-state cloner.

Proof. Every purification in the random-purification output has the form $(I \otimes U)\Phi_P$, with $U \in \text{U}(r)$. Unitary covariance of $\mathcal{W}_{n,m}^{(rd)}$ and invariance of the partial trace give

$$\begin{aligned} & \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(rd)} \left(((I \otimes U)|\Phi_P\rangle\langle\Phi_P|(I \otimes U^*))^{\otimes n} \right) \\ &= \text{Tr}_{E^m} \left[(I \otimes U)^{\otimes m} \mathcal{W}_{n,m}^{(rd)} (|\Phi_P\rangle\langle\Phi_P|^{\otimes n}) (I \otimes U^*)^{\otimes m} \right] \\ &= \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(rd)} (|\Phi_P\rangle\langle\Phi_P|^{\otimes n}). \end{aligned}$$

Thus the Haar average disappears exactly after the purifying registers are discarded.

Let $Q = P \otimes I_r$ on $\mathbb{C}^d \otimes \mathbb{C}^r$. Its rank is r^2 , its range contains Φ_P , and $Q^{\otimes m}$ commutes with the symmetric projection. Compressing (93) therefore gives

$$\begin{aligned} & Q^{\otimes m} \mathcal{W}_{n,m}^{(rd)} (|\Phi_P\rangle\langle\Phi_P|^{\otimes n}) Q^{\otimes m} \\ &= s_{n,m}^{(r,d)} (V_P \otimes I_r)^{\otimes m} \mathcal{W}_{n,m}^{(r^2)} (|\Phi_r\rangle\langle\Phi_r|^{\otimes n}) (V_P^* \otimes I_r)^{\otimes m}. \end{aligned} \quad (98)$$

Taking the partial trace over E^m yields

$$P^{\otimes m} \sigma_{n,m}^{\text{PCT}}(P) P^{\otimes m} = s_{n,m}^{(r,d)} V_P^{\otimes m} \zeta_{n,m}^{(r)} (V_P^*)^{\otimes m}. \quad (99)$$

The target is supported on $P^{\otimes m}$. Hence root fidelity is unchanged by compressing the first argument to that support, and its homogeneity proves (95).

It remains to bound the internal fidelity. Assume $r > 1$, put

$$u_r = \frac{1}{r} \mathbf{1}_r, \quad \Sigma_r = (\text{diag}(u_r) - u_r u_r^\top) \Big|_{\mathbf{H}_r},$$

and measure every system output in the computational basis. If $C_m = (C_{m,1}, \dots, C_{m,r})$ is the resulting count vector, set

$$Z_m = \frac{C_m - m u_r}{\sqrt{m}} \in \mathbf{H}_r.$$

Under the target $(I_r/r)^{\otimes m}$, the multinomial central limit theorem gives

$$Z_m \implies \mathbf{N}_{0, \Sigma_r}. \quad (100)$$

We next show that under $\zeta_{n,m_n}^{(r)}$,

$$Z_{m_n} \implies \mathbf{N}_{0, (2\gamma-1)\Sigma_r}. \quad (101)$$

For this purpose put $R = r^2$ and use the symmetric occupation basis about the condensate Φ_r . Under the corresponding embedding into the $(R-1)$ -mode Fock space, the purified Werner output is diagonal with mass

$$\pi_{n,m}(\mathbf{k}) = \frac{\binom{m-|\mathbf{k}|}{n}}{\binom{m+R-1}{n+R-1}}, \quad \mathbf{k} \in \mathbb{N}^{R-1}, \quad |\mathbf{k}| \leq m-n. \quad (102)$$

For every fixed $\mathbf{k}$,

$$\pi_{n,m_n}(\mathbf{k}) \longrightarrow \gamma^{-(R-1)} \left(\frac{\gamma-1}{\gamma} \right)^{|\mathbf{k}|}.$$

Extend π_{n,m_n} by zero to all of $\mathbb{N}^{R-1}$. Both sides are normalized probability masses (normalization follows also from Vandermonde's identity), so the discrete Scheffé lemma gives trace-norm convergence to

$$\tau_\vartheta^{\otimes(R-1)}, \quad \tau_\vartheta = (1-\vartheta) \sum_{j \geq 0} \vartheta^j |j\rangle\langle j|, \quad \vartheta = \frac{\gamma-1}{\gamma}, \quad (103)$$

whose mean occupation in every mode is $N = \gamma - 1$.

To read off the count fluctuations, fix $h \in \mathbf{H}_r$ and set

$$H_h = \text{diag}(h) \otimes I_r, \quad v_h = H_h \Phi_r \in \Phi_r^\perp.$$

Then

$$\langle v_g, v_h \rangle = \frac{1}{r} \sum_{i=1}^r g_i h_i = g^\top \Sigma_r h. \quad (104)$$

Let J_m be the condensate occupation embedding, $\mathcal{N}$ the Fock number operator, $b(v)$ the annihilation operator in direction v , $d\Gamma(A)$ the second quantization of A , and $P_\perp = I - |\Phi_r\rangle\langle\Phi_r|$. The standard condensate fluctuation identity gives, on each finite-particle sector,

$$J_m^* \left[\frac{1}{\sqrt{m}} \sum_{\ell=1}^m H_h^{(\ell)} \right] J_m$$

$$= b^*(v_h) \sqrt{1 - \frac{\mathcal{N}}{m}} + \sqrt{1 - \frac{\mathcal{N}}{m}} b(v_h) + \frac{1}{\sqrt{m}} d\Gamma(P_\perp H_h P_\perp) \longrightarrow b^*(v_h) + b(v_h). \quad (105)$$

The fields on the right commute for $h \in \mathbf{H}_r$, because the inner products in (104) are real. The pulled-back operators may be extended to the entire Fock space by replacing $\sqrt{1 - \mathcal{N}/m}$ with $\sqrt{(1 - \mathcal{N}/m)_+}$ and compressing the last term to $\{\mathcal{N} \leq m\}$. The finite-particle vectors form a common analytic core, so (105) implies strong-resolvent convergence and hence strong convergence of the bounded fluctuation exponentials. Together with the trace-norm state convergence in (103), this gives convergence of the joint characteristic functions. In the gauge-invariant thermal state of mean N ,

$$\mathrm{Tr} \left[\tau_\vartheta^{\otimes(R-1)} e^{it(b^*(v_h) + b(v_h))} \right] = \exp \left(-\frac{t^2}{2} (2N + 1) \|v_h\|^2 \right).$$

Since measuring the system counts after tracing out E^m is the same as measuring H_h before that trace, the Cramér–Wold theorem, $2N + 1 = 2\gamma - 1$, and (104) prove (101).

Weak convergence suffices after an arbitrarily small common classical smoothing. For $t > 0$, add to Z_m an independent $\mathbf{N}_{0,t\Sigma_r}$ random vector on $\mathbf{H}_r$. This is a classical channel, so fidelity monotonicity bounds the internal quantum fidelity by the root fidelity of the two smoothed count laws. Convolution with the fixed nondegenerate Gaussian has densities that converge pointwise under weak convergence, and hence in L^1 by Scheffé’s lemma. The limiting laws have covariances $(1+t)\Sigma_r$ and $(2\gamma - 1 + t)\Sigma_r$. The L^1 convergence of both densities implies convergence of their root fidelity, and the limiting Gaussian root fidelity is

$$\left(\frac{\sqrt{(1+t)(2\gamma - 1 + t)}}{\gamma + t} \right)^{(r-1)/2}.$$

It follows, after $t \downarrow 0$, that

$$\limsup_{n \rightarrow \infty} F \left(\zeta_{n,m_n}^{(r)}, (I_r/r)^{\otimes m_n} \right) \leq \left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(r-1)/2}. \quad (106)$$

Finally, (94) gives

$$s_{n,m_n}^{(r,d)} \longrightarrow \gamma^{-r(d-r)}.$$

Combining this with (95) and (106) proves (97). Its right-hand side is strictly smaller than $\gamma^{-r(d-r)/2}$ when $r > 1$, because $2\gamma - 1 < \gamma^2$. The $r = 1$ assertion follows directly from (96). $\square$

The preceding proposition is a direct comparison at every fixed $\gamma > 1$, and it remains informative when the limiting fidelity is bounded away from one. By contrast, the two recent sample-complexity analyses use the PCT data-processing guarantee for their fidelity-threshold objective. In their squared-fidelity convention $F_{\mathrm{sq}} := F^2$,

$$F_{\mathrm{sq}} \left(\sigma_{n,m}^{\mathrm{PCT}}(P), (P/r)^{\otimes m} \right) \geq \frac{d_{rd}(n)}{d_{rd}(m)} \longrightarrow \gamma^{-(rd-1)} \quad [\text{FGSS26, JSO26}].$$

Our Grassmannian theorem instead gives the exact squared-fidelity value $\gamma^{-r(d-r)}$. At the limiting squared-fidelity level, this is larger than the published PCT guarantee by the factor γ^{r^2-1} for $r > 1$; that comparison of guarantees alone would not order the actual protocols. The actual strict ordering is supplied by (97).

Remark A.8 (The near-unit-gain boundary). The strict comparison itself holds for every fixed $\gamma > 1$ and requires no high-fidelity assumption. As $\gamma \downarrow 1$, however, the particular strictness certificate above becomes second order. Indeed, writing $\gamma = 1 + \delta$, the additional factor detected by the count statistic satisfies

$$\left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(r-1)/2} = 1 - \frac{r-1}{4}\delta^2 + O(\delta^3).$$

It therefore does not by itself settle the leading sample-complexity comparison in a joint near-unit-gain limit. The proof uses only the $r - 1$ commuting diagonal count directions; a sharper analysis of all internal directions would require a degenerate central-limit or LAN statement and is not claimed here.

B Exact-weight PBW calculation of the sector fidelity

This appendix proves [Proposition 2.3](#) by comparing the target sector $\rho_{p,\nu}$ with the Cartan-sector output $\mathcal{C}_{\mu,\nu}(\rho_{p,\mu})$ on finitely many weight spaces near their highest weights, then controlling the remaining target mass. Throughout this appendix, d is a generic intrinsic support dimension; the ambient rank- r construction applies the result with $d = r$ and then uses [Lemma 4.1](#). [Lemma B.1](#) shows that normalized lowering-operator vectors become asymptotically orthonormal on these spaces, while [Lemma B.2](#) gives the uniform limit $p^\lambda/s_\lambda(p) \rightarrow \prod_{i < j} (1 - p_j/p_i)$. The Cartan inclusion $V_{\mu,\nu-\mu}$ then splits each lowering operation between $\mathcal{H}_\mu$ and $\mathcal{H}_{\nu-\mu}$ with asymptotically binomial weights, producing the thermal limits τ_p and $\tau_p^{(\gamma)}$ and hence the fidelity $F_{\text{orb}}(\gamma, p)$.

Because the calculations involve only weights of height bounded in terms of L , $O_L(\cdot)$ may depend on L, d, c, C in $cN \leq \Delta_\alpha^\lambda \leq CN$ for $\alpha \in \mathbb{R}_+$, but is uniform in N and all dominant λ satisfying these inequalities.

B.1 PBW vectors on complete weight spaces

Identify the positive roots of $U(d)$ with

$$\mathbb{R}_+ = \{e_i - e_j : 1 \leq i < j \leq d\}.$$

For $\alpha = e_i - e_j$, set

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij}, \quad H_\alpha = [E_\alpha, F_\alpha] = E_{ii} - E_{jj}, \quad \Delta_\alpha^\lambda = \lambda_i - \lambda_j.$$

Fix a total order on $\mathbb{R}_+$. For $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbb{R}_+}$, write

$$F^\mathbf{r} = \prod_{\alpha \in \mathbb{R}_+}^< F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_\alpha r_\alpha.$$

For a nonnegative simple-root combination η , let $\text{ht}(\eta)$ be its simple-root height and put

$$\mathcal{P}(\eta) = \left\{ \mathbf{r} : \sum_{\alpha \in \mathbb{R}_+} r_\alpha \alpha = \eta \right\}.$$

The set $\mathcal{P}(\eta)$ is finite, and the PBW theorem implies that $\{F^\mathbf{r} v_\lambda : \mathbf{r} \in \mathcal{P}(\eta)\}$ spans $\mathcal{H}_\lambda[\lambda - \eta]$ [[FH91](#)]. Define

$$\mathcal{K}_{\lambda,L} = \bigoplus_{\text{ht}(\eta) \leq L} \mathcal{H}_\lambda[\lambda - \eta]. \quad (107)$$

Let $P_{\lambda,L}$ denote the orthogonal projection onto $\mathcal{K}_{\lambda,L}$.

Lemma B.1 (PBW Gram matrices at fixed height). *Let $\lambda = \lambda^{(N)}$ range over dominant weights satisfying*

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in \mathbf{R}_+)$$

for fixed $0 < c < C < \infty$. Define

$$\tilde{e}_\mathbf{r}^\lambda = \prod_{\alpha \in \mathbf{R}_+}^< \frac{F_\alpha^{r_\alpha}}{\sqrt{r_\alpha!} (\Delta_\alpha^\lambda)^{r_\alpha/2}} v_\lambda. \quad (108)$$

For every fixed L , uniformly over $\text{ht}(\eta) \leq L$ and $\mathbf{r}, \mathbf{s} \in \mathbf{P}(\eta)$,

$$\langle \tilde{e}_\mathbf{r}^\lambda, \tilde{e}_\mathbf{s}^\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} + O_L(N^{-1/2}). \quad (109)$$

Consequently, for all sufficiently large N , the complete Gram matrix $G_{\lambda, \eta}$ on every such exact-weight block is invertible. If

$$(e_\mathbf{r}^\lambda)_{\mathbf{r} \in \mathbf{P}(\eta)} = (\tilde{e}_\mathbf{r}^\lambda)_{\mathbf{r} \in \mathbf{P}(\eta)} G_{\lambda, \eta}^{-1/2}, \quad (110)$$

then these vectors form an orthonormal basis of $\mathcal{H}_\lambda[\lambda - \eta]$, and

$$e_\mathbf{r}^\lambda = \tilde{e}_\mathbf{r}^\lambda + O_L(N^{-1/2}) \quad (111)$$

uniformly for $\text{ht}(\eta) \leq L$ and $\mathbf{r} \in \mathbf{P}(\eta)$.

Proof. For $\mathbf{r} \in \mathbf{P}(\eta)$, every positive root has positive simple-root height, so $|\mathbf{r}| \leq \text{ht}(\eta)$. Put

$$\hat{F}_\alpha = \frac{F_\alpha}{\sqrt{\Delta_\alpha^\lambda}}, \quad \hat{E}_\alpha = \frac{E_\alpha}{\sqrt{\Delta_\alpha^\lambda}}.$$

Each unnormalized Gram entry $\langle v_\lambda, (\hat{F}^\mathbf{r})^* \hat{F}^\mathbf{s} v_\lambda \rangle$ is therefore a word of length at most $2L$. Commuting root operators cannot increase the word length, and each root has height at most $d - 1$. Hence every intermediate vector lies in

$$\mathcal{K}_{\lambda, R_L}, \quad R_L := 2L(d - 1).$$

For $0 \leq R \leq R_L$, the restrictions

$$\hat{E}_\alpha : \mathcal{K}_{\lambda, R} \rightarrow \mathcal{K}_{\lambda, R}, \quad \hat{F}_\alpha : \mathcal{K}_{\lambda, R} \rightarrow \mathcal{K}_{\lambda, R+d-1}$$

satisfy $\|\hat{E}_\alpha\| + \|\hat{F}_\alpha\| = O_L(1)$. To see this, decompose $\mathcal{H}_\lambda$ orthogonally into irreducible $\mathfrak{sl}_2(\alpha)$ -summands. If a summand meets the weight space of $\lambda - \eta$, where $\text{ht}(\eta) \leq R$, let t be its level there and M its highest H_α -eigenvalue. The top of this string, of weight $\lambda - \eta + t\alpha$, is again a weight of $\mathcal{H}_\lambda$; hence $\eta - t\alpha$ is a nonnegative simple-root combination and $t \text{ht}(\alpha) \leq \text{ht}(\eta) \leq R$. Moreover,

$$M = \Delta_\alpha^\lambda - \eta(H_\alpha) + 2t = \Delta_\alpha^\lambda + O_L(1).$$

At level t , the F_α - and E_α -coefficients are $\sqrt{(t+1)(M-t)}$ and $\sqrt{t(M-t+1)}$. After division by $\sqrt{\Delta_\alpha^\lambda}$, both are $O_L(1)$, proving the norm bound. All operator norms below are taken after restriction to $\mathcal{K}_{\lambda, R_L}$, with the codomain norm inherited from $\mathcal{H}_\lambda$.

The matrix-unit commutator formula now gives

$$[\hat{E}_\alpha, \hat{F}_\beta] = \delta_{\alpha\beta} I + O_L(N^{-1/2}). \quad (112)$$

For $\alpha = \beta$, the left side is $H_\alpha/\Delta_\alpha^\lambda = I + O_L(N^{-1})$ on the cutoff. For $\alpha \neq \beta$, the unnormalized commutator is zero or, up to sign, a root operator E_γ or F_γ . In the nonzero case, the preceding bound gives $\|[\widehat{E}_\alpha, \widehat{F}_\beta]\| = O_L(\sqrt{\Delta_\gamma^\lambda}/\sqrt{\Delta_\alpha^\lambda \Delta_\beta^\lambda}) = O_L(\sqrt{N}/N) = O_L(N^{-1/2})$.

Commute every $\widehat{E}_\alpha$ to the right in $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$. The scalar commutators in (112) contribute only when $\mathbf{r} = \mathbf{s}$, in which case they give $\prod_\alpha r_\alpha!$. Every other term vanishes or contains an $O_L(N^{-1/2})$ remainder. Only $O_L(1)$ terms occur, and all remaining operator factors have norm $O_L(1)$; consequently,

$$\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} \prod_\alpha r_\alpha! + O_L(N^{-1/2}).$$

Dividing by the factorials in (108) proves (109).

The number and sizes of the exact-weight blocks are bounded in terms of L , so the entrywise estimate gives $G_{\lambda, \eta} = I + O_L(N^{-1/2})$ in operator norm. Thus $G_{\lambda, \eta}$ is invertible for large N , and the PBW spanning vectors form a basis. Finally, functional calculus gives $G_{\lambda, \eta}^{-1/2} = I + O_L(N^{-1/2})$; hence (110) is an orthonormal basis and satisfies (111). $\square$

B.2 Character normalization and fixed-height matrices

For $a = (a_1, \dots, a_d)$, write $p^a = \prod_i p_i^{a_i}$.

Lemma B.2 (Uniform character normalization). *Let $K \in \Delta_d^\circ$, let $\delta_N \rightarrow 0$, and suppose $p \in K$, $\lambda \vdash N$, and $\|\lambda/N - p\|_\infty \leq \delta_N$. Then, uniformly over this family,*

$$\frac{p^\lambda}{s_\lambda(p)} = \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right) + o(1). \quad (113)$$

Proof. Write the numerator in the Weyl character formula as

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma, \quad A_\sigma = \prod_i p_i^{\lambda_{\sigma(i)} + d - \sigma(i)}.$$

For $\sigma \neq \text{id}$,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} = \sum_i \frac{\lambda_i}{N} \log p_i - \sum_i \frac{\lambda_{\sigma(i)}}{N} \log p_i + O_K(N^{-1}).$$

For $x \in \mathbb{R}^d$, put

$$D_\sigma(x, p) = \sum_i x_i \log p_i - \sum_i x_{\sigma(i)} \log p_i.$$

Strict rearrangement gives $D_\sigma(p, p) > 0$. Since K is compact, the permutations are finite, and the logarithms are uniformly bounded on K , the assumptions $\|\lambda/N - p\|_\infty \leq \delta_N$ and $\delta_N \rightarrow 0$ give $c_K > 0$ such that, for every $\sigma \neq \text{id}$ and all sufficiently large N ,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} \geq c_K, \quad A_\sigma \leq e^{-c_K N} A_{\text{id}}.$$

Consequently,

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma = A_{\text{id}}(1 + O_K(e^{-c_K N})), \quad A_{\text{id}} = p^\lambda \prod_i p_i^{d-i}.$$

Since

$$\det(p_i^{d-j})_{i,j=1}^d = \prod_i p_i^{d-i} \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right),$$

the Weyl character formula gives, uniformly,

$$s_\lambda(p) = p^\lambda \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right)^{-1} (1 + O_K(e^{-c_K N})).$$

This proves (113). $\square$

For the remainder of the appendix, index the number basis of $\mathcal{F}_{\text{orb}}$ by $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbf{R}_+}$, and write it as $\{|\mathbf{r}\rangle\}$. Define the finite-rank projection

$$P_L^{\mathcal{F}} = \sum_{\text{ht}(\sum_{\alpha} r_{\alpha} \alpha) \leq L} |\mathbf{r}\rangle\langle \mathbf{r}|. \quad (114)$$

For every fixed L and all sufficiently large N , each highest weight λ satisfying the gap bounds in Lemma B.1 has the orthonormal basis $e_{\mathbf{r}}^{\lambda}$ of $\mathcal{K}_{\lambda,L}$ constructed there. Define

$$J_{\lambda,L} : \mathcal{K}_{\lambda,L} \longrightarrow P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}, \quad J_{\lambda,L} e_{\mathbf{r}}^{\lambda} = |\mathbf{r}\rangle. \quad (115)$$

By Lemma B.1, the two spaces have bases indexed by the same $\mathbf{r}$, so $J_{\lambda,L}$ is unitary onto $P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}$. It is defined only on $\mathcal{K}_{\lambda,L}$, and it depends on λ and L .

Proof of Proposition 2.3. Fix K and δ_n as in the proposition, write $m = m_n$, and put $\omega = \nu - \mu$. Since $m/n = \gamma_n \rightarrow \gamma > 1$, we have $m/n - 1 \geq (\gamma - 1)/2$ for all sufficiently large n . Uniformly over the allowed parameters,

$$\begin{aligned} \left\| \frac{\omega}{m-n} - p \right\|_{\infty} &= \left\| \frac{m/n}{m/n-1} \left(\frac{\nu}{m} - p \right) - \frac{1}{m/n-1} \left(\frac{\mu}{n} - p \right) \right\|_{\infty} \\ &\leq \frac{m/n+1}{m/n-1} \delta_n = o(1). \end{aligned}$$

Every coordinate and every adjacent gap is uniformly positive on K . Hence $\omega \vdash m - n$ for all sufficiently large n . For $\alpha = e_i - e_j$, set

$$\begin{aligned} s_{\alpha,n} &= \frac{\Delta_{\alpha}^{\mu}}{\Delta_{\alpha}^{\nu}} = \frac{n}{m} \frac{\mu_i/n - \mu_j/n}{\nu_i/m - \nu_j/m} = \frac{1}{\gamma} + o(1), \\ 1 - s_{\alpha,n} &= \frac{\Delta_{\alpha}^{\omega}}{\Delta_{\alpha}^{\nu}}. \end{aligned} \quad (116)$$

Consequently, there are constants $0 < c_K < C_K < \infty$, depending only on K and γ , such that, for all large n ,

$$c_K n \leq \Delta_{\alpha}^{\mu}, \Delta_{\alpha}^{\nu}, \Delta_{\alpha}^{\omega} \leq C_K n \quad (\alpha \in \mathbf{R}_+). \quad (117)$$

Let $V = V_{\mu,\omega} : \mathcal{H}_{\nu} \rightarrow \mathcal{H}_{\mu} \otimes \mathcal{H}_{\omega}$, and write $F_{\alpha}^{(\lambda)} = d\pi_{\lambda}(F_{\alpha})$ for the action of the root operator on $\mathcal{H}_{\lambda}$. Intertwining and the highest-weight normalization give

$$VF_{\alpha}^{(\nu)} = (F_{\alpha}^{(\mu)} \otimes I + I \otimes F_{\alpha}^{(\omega)})V.$$

Expanding each $(F_{\alpha}^{(\mu)} \otimes I + I \otimes F_{\alpha}^{(\omega)})^{r_{\alpha}}$ binomially gives the exact identity

$$V\tilde{e}_{\mathbf{r}}^{\nu} = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \tilde{e}_{\mathbf{a}}^{\mu} \otimes \tilde{e}_{\mathbf{r}-\mathbf{a}}^{\omega}, \quad (118)$$

where

$$\beta_{\mathbf{r},\mathbf{a}}^{(n)} = \prod_{\alpha \in \mathbf{R}_+} \sqrt{\binom{r_\alpha}{a_\alpha} s_{\alpha,n}^{a_\alpha} (1 - s_{\alpha,n})^{r_\alpha - a_\alpha}}. \quad (119)$$

If $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$, then (118) has $O_L(1)$ terms, all of height at most L , with $0 \leq \beta_{\mathbf{r},\mathbf{a}}^{(n)} \leq 1$. Since $m/n \rightarrow \gamma > 1$, the sizes $n, m, m - n$ are uniformly comparable, so (117) and Lemma B.1 give $e_{\mathbf{b}}^\lambda = \tilde{e}_{\mathbf{b}}^\lambda + O_L(n^{-1/2})$ for $\lambda \in \{\mu, \nu, \omega\}$ and all relevant $\mathbf{b}$. Substituting these relations into (118) and using $\|Vx\| = \|x\|$ gives

$$V e_{\mathbf{r}}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} e_{\mathbf{a}}^\mu \otimes e_{\mathbf{r}-\mathbf{a}}^\omega + O_L(n^{-1/2}) \quad (120)$$

uniformly in norm for $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$.

Put

$$\sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \quad c_\lambda(p) = \frac{p^\lambda}{s_\lambda(p)}, \quad q_\alpha = \frac{p_j}{p_i} \quad (\alpha = e_i - e_j).$$

For $\lambda \in \{\mu, \nu\}$, the weight of $e_{\mathbf{b}}^\lambda$ is $\lambda - \sum_\alpha b_\alpha \alpha$, and hence

$$\rho_{p,\lambda} e_{\mathbf{b}}^\lambda = c_\lambda(p) \prod_\alpha q_\alpha^{b_\alpha} e_{\mathbf{b}}^\lambda.$$

The sector channel gives the local formula

$$\sigma_{\mu,\nu} = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V^*(\rho_{p,\mu} \otimes I_{\mathcal{H}_\omega}) V.$$

All asymptotic estimates below are uniform over the allowed family. The character estimate, Lemma B.2, gives

$$c_\lambda(p) = \prod_\alpha (1 - q_\alpha) + o(1) \quad (\lambda = \mu, \nu),$$

while the Weyl dimension formula, (117), and (116) give

$$\frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} = \prod_{\alpha=e_i-e_j} \frac{\Delta_\alpha^\mu + j - i}{\Delta_\alpha^\nu + j - i} = \prod_{\alpha \in \mathbf{R}_+} (s_{\alpha,n} + O(n^{-1})) = \gamma^{-r_d} + o(1).$$

Thus, substituting (120) and the weight action above into this formula, and using $\|\rho_{p,\mu}\| \leq 1$, gives, for fixed L ,

$$\begin{aligned} \langle e_{\mathbf{r}}^\nu, \sigma_{\mu,\nu} e_{\mathbf{s}}^\nu \rangle &= \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} \sum_{\substack{\mathbf{a} \leq \mathbf{r} \\ \mathbf{b} \leq \mathbf{s}}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \beta_{\mathbf{s},\mathbf{b}}^{(n)} \langle e_{\mathbf{a}}^\mu, \rho_{p,\mu} e_{\mathbf{b}}^\mu \rangle \langle e_{\mathbf{r}-\mathbf{a}}^\omega, e_{\mathbf{s}-\mathbf{b}}^\omega \rangle + O_L(n^{-1/2}) \\ &= \delta_{\mathbf{r},\mathbf{s}} \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} c_\mu(p) \prod_{\alpha \in \mathbf{R}_+} (1 - s_{\alpha,n} + s_{\alpha,n} q_\alpha)^{r_\alpha} + O_L(n^{-1/2}), \end{aligned} \quad (121)$$

$$\langle e_{\mathbf{r}}^\nu, \rho_{p,\nu} e_{\mathbf{s}}^\nu \rangle = \delta_{\mathbf{r},\mathbf{s}} c_\nu(p) \prod_{\alpha \in \mathbf{R}_+} q_\alpha^{r_\alpha}. \quad (122)$$

Recall the following notation from Section 3.2:

$$q_\alpha^{(\gamma)} = 1 - \frac{1 - q_\alpha}{\gamma}, \quad \tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle \langle k|, \quad \tau_p = \bigotimes_{\alpha \in \mathbf{R}_+} \tau_{q_\alpha}, \quad \tau_p^{(\gamma)} = \bigotimes_{\alpha \in \mathbf{R}_+} \tau_{q_\alpha^{(\gamma)}}.$$

Since $s_{\alpha,n} = \gamma^{-1} + o(1)$, (121) and (122) now give

$$\begin{aligned}\langle e_{\mathbf{r}}^\nu, \sigma_{\mu,\nu} e_{\mathbf{s}}^\nu \rangle &= \delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} (1 - q_{\alpha}^{(\gamma)}) (q_{\alpha}^{(\gamma)})^{r_{\alpha}} + o(1), \\ \langle e_{\mathbf{r}}^\nu, \rho_{p,\nu} e_{\mathbf{s}}^\nu \rangle &= \delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} (1 - q_{\alpha}) q_{\alpha}^{r_{\alpha}} + o(1).\end{aligned}$$

These are precisely the number-basis matrix elements of $\tau_p^{(\gamma)}$ and τ_p . For fixed L , the cutoff space has fixed finite dimension, so the uniform entrywise limits imply convergence in trace norm:

$$\left\| J_{\nu,L} P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0, \quad (123)$$

$$\left\| J_{\nu,L} P_{\nu,L} \rho_{p,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0. \quad (124)$$

Both $\sigma_{\mu,\nu}$ and $\rho_{p,\nu}$ commute with the diagonal torus. The second assertion is immediate; the first follows from torus invariance of $\rho_{p,\mu}$ and covariance of $\mathcal{C}_{\mu,\nu}$. Consequently, $P_{\nu,L}$ reduces both states, and the direct-sum identity gives

$$\begin{aligned}F(\sigma_{\mu,\nu}, \rho_{p,\nu}) &= F(P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) \\ &\quad + F((I - P_{\nu,L}) \sigma_{\mu,\nu} (I - P_{\nu,L}), (I - P_{\nu,L}) \rho_{p,\nu} (I - P_{\nu,L})).\end{aligned}$$

Schatten Hölder gives $F(A, C) \leq \sqrt{\text{Tr } A \text{ Tr } C}$, so the second term is at most

$$\sqrt{\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}]}, \quad (125)$$

because $\text{Tr}[(I - P_{\nu,L}) \sigma_{\mu,\nu}] \leq 1$.

Taking traces in (124) gives, for fixed L ,

$$\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}] = \text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p] + o(1)$$

uniformly as $n \rightarrow \infty$. Since K is compact in Δ_d° , there is $b_K < 1$ such that $q_{\alpha} \leq b_K$ for every $p \in K$ and every positive root α . The occupation numbers under τ_p are independent geometric variables with parameters at most b_K . If their total occupation is s , then their total root height is at most $(d-1)s$. Moreover, the probability of any fixed occupation vector of total occupation s is at most b_K^s , and there are $\binom{s+r_d-1}{r_d-1}$ such vectors. Therefore

$$\sup_{p \in K} \text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p] \leq \sum_{s > L/(d-1)} \binom{s+r_d-1}{r_d-1} b_K^s \longrightarrow 0 \quad (L \rightarrow \infty). \quad (126)$$

The same direct-sum decomposition for $F(\tau_p^{(\gamma)}, \tau_p)$ bounds its complementary part by the square root of the same target thermal tail. Consequently,

$$\begin{aligned}& \left| F(\sigma_{\mu,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \\ & \leq \left| F(P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) - F(P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}}, P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}}) \right| \\ & \quad + \sqrt{\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}]} + \sqrt{\text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p]}.\end{aligned}$$

By (23) and (123)–(124), for every fixed L ,

$$\begin{aligned}& F(P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) \\ & \longrightarrow F(P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}}, P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}})\end{aligned}$$

uniformly. First let $n \rightarrow \infty$ at fixed L , and then let $L \rightarrow \infty$ using (125)–(126). This proves

$$\left| F(\sigma_{\mu, \nu}, \rho_{p, \nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \longrightarrow 0$$

uniformly. Finally,

$$F(\tau_p^{(\gamma)}, \tau_p) = \prod_{i < j} F(\tau_{q_{ij}}^{(\gamma)}, \tau_{q_{ij}}) = F_{\text{orb}}(\gamma, p).$$

Here multiplicativity of fidelity follows directly from multiplicativity of the trace norm under tensor products, and the final equality uses (18) and (20). $\square$

C Young local limit and randomized rounding

This appendix proves Proposition 2.4 by expressing the randomized rounding law $D_n(\tilde{\nu} \mid \mu)$ as the normalized intersection volume of the dilated source cell around $\gamma_n \mu$ with the target cell labeled by $\tilde{\nu}$. Here d is again the active label dimension; in the fixed-rank construction it is instantiated as r , independently of the ambient Hilbert-space dimension. Lemma C.1 shows that the centered and rescaled target label law $P_{N,p}$ converges uniformly to $\mathbf{N}_{0, \Sigma_p}$, while Lemma C.2 shows that sending $P_{n,p}$ through D_n changes this limit to $\mathbf{N}_{0, \gamma \Sigma_p}$ and yields the label affinity $F_{\text{cl}}(\gamma, d)$. The final proof checks that typical rounded labels satisfy $\nu - \mu \vdash m_n - n$ and that replacement by ν_n^{fb} moves only vanishing mass.

Define the half-open fundamental cell

$$\mathbf{C} = \left\{ \sum_{i=1}^{d-1} t_i (e_i - e_d) : -\frac{1}{2} \leq t_i < \frac{1}{2} \right\} \subset \mathbf{H}.$$

Volumes are taken with the induced $(d-1)$ -dimensional Lebesgue measure. The rounding rule (14) is equivalent, outside a null set of cell boundaries, to choosing the unique $\tilde{\nu} \in \mathbf{L}_{m_n}$ for which

$$\gamma_n(\mu + U) \in \tilde{\nu} + \mathbf{C}, \quad U = \sum_{i=1}^{d-1} U_i (e_i - e_d).$$

For $i < d$, this is precisely the nearest-integer rule because $\lfloor x + \frac{1}{2} \rfloor$ is the unique integer in $(x - \frac{1}{2}, x + \frac{1}{2}]$; the d -th coordinate is then automatic because both sides sum to m_n . Since U is uniform on $\mathbf{C}$, the change of variables $x = \gamma_n(\mu + u)$ gives $dx = \gamma_n^{d-1} du$ and sends the event domain to $\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})$. Consequently,

$$D_n(\tilde{\nu} \mid \mu) = \frac{|\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})|}{\gamma_n^{d-1} |\mathbf{C}|}. \quad (127)$$

All label laws below are regarded as functions on $\mathbf{L}_N$: if a law R is originally defined on $\mathbf{Y}_N$, set $R(\lambda) = 0$ for every $\lambda \in \mathbf{L}_N \setminus \mathbf{Y}_N$.

For $\lambda \in \mathbf{L}_N$, define the recentered and rescaled lattice cell

$$\mathbf{C}_{N,p}(\lambda) = \frac{\lambda - Np + \mathbf{C}}{\sqrt{N}}.$$

For any $R : \mathbf{L}_N \rightarrow \mathbb{R}$ satisfying $\sum_{\lambda \in \mathbf{L}_N} |R(\lambda)| < \infty$, define

$$(\mathcal{E}_{N,p} R)(x) = \frac{N^{(d-1)/2}}{|\mathbf{C}|} R(\lambda) \quad (x \in \mathbf{C}_{N,p}(\lambda)).$$

The cells $C_{N,p}(\lambda)$, $\lambda \in \mathbb{L}_N$, tile $\mathbb{H}$, and their boundaries have zero induced Lebesgue measure. The map $\mathcal{E}_{N,p}$ preserves ℓ^1 -distance, $\|\mathcal{E}_{N,p}R - \mathcal{E}_{N,p}S\|_1 = \|R - S\|_{\ell^1}$, and, for nonnegative R, S , Hellinger affinity, $\int_{\mathbb{H}} \sqrt{(\mathcal{E}_{N,p}R)(\mathcal{E}_{N,p}S)} dx = \sum_{\lambda \in \mathbb{L}_N} \sqrt{R(\lambda)S(\lambda)}$. Indeed, each $C_{N,p}(\lambda)$ has volume $|C|N^{-(d-1)/2}$, which cancels the prefactor in both cellwise integrals.

Lemma C.1 (Uniform Young local limit). *For every compact $K \Subset \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{N,p}P_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (128)$$

Proof. Let $M_{N,p}$ be the multinomial law on nonnegative integer compositions of N , extended by zero to $\mathbb{L}_N$. We first show

$$\sup_{p \in K} \|P_{N,p} - M_{N,p}\|_{\ell^1} \longrightarrow 0. \quad (129)$$

Set $W_{N,p} = \{\lambda \in \mathbb{L}_N : \|\lambda - Np\|_\infty \leq N^{2/3}\}$. For all large N , $W_{N,p} \cap \text{supp } M_{N,p} \subset Y_N$, since $\lambda_i - \lambda_{i+1} \geq N(p_i - p_{i+1}) - 2N^{2/3} > 0$ for $i < d$, uniformly over $p \in K$.

For $\lambda \in Y_N$, the Weyl dimension and character formulas give

$$P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \dim \mathcal{S}_\lambda = N! \frac{\prod_{i < j} (\lambda_i - \lambda_j + j - i)}{\prod_i (\lambda_i + d - i)!},$$

$$s_\lambda(p) = \frac{\det[p_i^{\lambda_j + d - j}]_{i,j}}{\prod_{i < j} (p_i - p_j)}.$$

Uniformly on $W_{N,p} \cap \text{supp } M_{N,p}$,

$$s_\lambda(p) = \frac{\prod_i p_i^{\lambda_i + d - i}}{\prod_{i < j} (p_i - p_j)} (1 + O_K(e^{-c_K N}))$$

for some $c_K > 0$. Indeed, both $(\lambda_i + d - i)/N$ and $\log p_i$ are uniformly strictly decreasing there, so strict rearrangement makes every nonidentity determinant term exponentially smaller than the identity term.

Comparison with $M_{N,p}(\lambda) = N! \prod_i p_i^{\lambda_i} / \prod_i \lambda_i!$ gives

$$\frac{P_{N,p}(\lambda)}{M_{N,p}(\lambda)} = (1 + O_K(e^{-c_K N})) \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)} \prod_i \frac{\lambda_i! (Np_i)^{d-i}}{(\lambda_i + d - i)!}.$$

Writing the last product as $\prod_i \prod_{a=1}^{d-i} \frac{Np_i}{\lambda_i + a}$, each factor $\frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)}$, $i < j$, and $\frac{Np_i}{\lambda_i + a}$, $1 \leq a \leq d - i$, equals $1 + O_K(N^{-1/3})$. Therefore

$$\|P_{N,p} - M_{N,p}\|_{\ell^1} \leq \sup_{\lambda \in W_{N,p} \cap \text{supp } M_{N,p}} \left| \frac{P_{N,p}(\lambda)}{M_{N,p}(\lambda)} - 1 \right| + P_{N,p}(W_{N,p}^c) + M_{N,p}(W_{N,p}^c) \longrightarrow 0$$

uniformly for $p \in K$, since the last two terms vanish by Lemma 2.1 and Hoeffding's inequality.

By (129) and the ℓ^1 -isometry of $\mathcal{E}_{N,p}$, it remains to show

$$\sup_{p \in K} \|\mathcal{E}_{N,p}M_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (130)$$

Fix $\beta_0 \in (1/2, 2/3)$ and set

$$V_{N,p} = \{\lambda \in \mathbb{L}_N : \|\lambda - Np\|_\infty \leq N^{\beta_0}\}, \quad x_\lambda = \frac{\lambda - Np}{\sqrt{N}}.$$

Since $\inf_{p \in K} p_d > 0$, $V_{N,p} \subset \text{supp } M_{N,p}$ for all sufficiently large N , uniformly in $p \in K$. Uniformly for $p \in K$ and $\lambda \in V_{N,p}$,

$$\frac{N^{(d-1)/2}}{|\mathbb{C}|} M_{N,p}(\lambda) = \varphi_{0,\Sigma_p}(x_\lambda)(1 + o(1)). \quad (131)$$

To see this, uniform Stirling and third-order Taylor expansions give

$$\log M_{N,p}(\lambda) = -\frac{d-1}{2} \log(2\pi N) - \frac{1}{2} \sum_i \log p_i - \frac{1}{2} \sum_i \frac{x_{\lambda,i}^2}{p_i} + O_K(N^{3\beta_0-2} + N^{\beta_0-1} + N^{-1}).$$

This is the Gaussian exponent since $\langle x, \Sigma_p^{-1} x \rangle = \sum_i x_i^2/p_i$. The normalizing constants agree since $|\mathbb{C}|^2 = \det[\langle e_i - e_d, e_j - e_d \rangle]_{i,j < d} = d$, while $\ker \Sigma_p = \text{span}\{(1, \dots, 1)\}$ and $\text{cof}_{ii}(\Sigma_p) = \prod_j p_j$ give $\det_{\mathbb{H}} \Sigma_p = d \prod_j p_j$. The remainder is $o(1)$ since $1/2 < \beta_0 < 2/3$.

Set

$$U_{N,p} = \bigcup_{\lambda \in V_{N,p}} \mathbb{C}_{N,p}(\lambda).$$

The multinomial mass outside $V_{N,p}$ and the Gaussian mass outside $U_{N,p}$ are exponentially small: for some $c_K, C_K > 0$,

$$\sup_{p \in K} \left[M_{N,p}(V_{N,p}^c) + \int_{U_{N,p}^c} \varphi_{0,\Sigma_p}(x) dx \right] \leq C_K e^{-c_K N^{2\beta_0-1}}. \quad (132)$$

Indeed, if $\mathbb{C}_{N,p}(\lambda)$ contains x , then $\|x_\lambda\|_\infty \leq \|x\|_\infty + (d-1)/(2\sqrt{N})$. Thus

$$\{x : \|x\|_\infty \leq N^{\beta_0-1/2} - (d-1)/(2\sqrt{N})\} \subset U_{N,p}.$$

On the set at left $\|x_\lambda\|_\infty \leq N^{\beta_0-1/2}$, so $\lambda \in V_{N,p}$. Hoeffding controls $M_{N,p}(V_{N,p}^c)$; the complementary inclusion and the uniform eigenvalue bounds for Σ_p control the Gaussian tail on $U_{N,p}^c$.

The Gaussian density has vanishing relative oscillation on $\mathbb{C}_{N,p}(\lambda)$, $\lambda \in V_{N,p}$. Indeed, uniformly for $p \in K$, $\lambda \in V_{N,p}$, and $x, y, z \in \mathbb{C}_{N,p}(\lambda)$, $\|x - y\| = O(N^{-1/2})$ and $\|z\| = O(N^{\beta_0-1/2})$. Since $\nabla \log \varphi_{0,\Sigma_p}(z) = -\Sigma_p^{-1} z$,

$$\begin{aligned} \left| \log \frac{\varphi_{0,\Sigma_p}(x)}{\varphi_{0,\Sigma_p}(y)} \right| &\leq \sup_{z \in \mathbb{C}_{N,p}(\lambda)} \|\nabla \log \varphi_{0,\Sigma_p}(z)\| \|x - y\| \\ &\leq O_K(N^{\beta_0-1/2}) O(N^{-1/2}) = O_K(N^{\beta_0-1}), \\ \left| \frac{\varphi_{0,\Sigma_p}(x)}{\varphi_{0,\Sigma_p}(y)} - 1 \right| &= O_K(N^{\beta_0-1}) = o(1). \end{aligned} \quad (133)$$

The second estimate follows by exponentiation.

The embedded multinomial density is consequently uniformly accurate on $U_{N,p}$. Indeed, $0 \in \mathbb{C}$, so the point $x_\lambda = (\lambda - Np)/\sqrt{N}$ lies in $\mathbb{C}_{N,p}(\lambda)$. Uniformly for $p \in K$, $\lambda \in V_{N,p}$, and x in the same cell,

$$\begin{aligned} \frac{N^{(d-1)/2} M_{N,p}(\lambda)}{|\mathbb{C}| \varphi_{0,\Sigma_p}(x)} &= \frac{N^{(d-1)/2} M_{N,p}(\lambda)}{|\mathbb{C}| \varphi_{0,\Sigma_p}(x_\lambda)} \frac{\varphi_{0,\Sigma_p}(x_\lambda)}{\varphi_{0,\Sigma_p}(x)} \\ &= (1 + o(1))(1 + o(1)) = 1 + o(1). \end{aligned} \quad (134)$$

The two factors are $1 + o(1)$ by the multinomial local estimate (131) and the Gaussian cell-oscillation bound (133), respectively.

For each $p \in K$,

$$\begin{aligned} \|\mathcal{E}_{N,p}M_{N,p} - \varphi_{0,\Sigma_p}\|_1 &= \int_{U_{N,p}} |(\mathcal{E}_{N,p}M_{N,p})(x) - \varphi_{0,\Sigma_p}(x)| \, dx \\ &\quad + \int_{U_{N,p}^c} |(\mathcal{E}_{N,p}M_{N,p})(x) - \varphi_{0,\Sigma_p}(x)| \, dx \\ &\leq M_{N,p}(V_{N,p}^c) + \int_{U_{N,p}^c} \varphi_{0,\Sigma_p}(x) \, dx \\ &\quad + \sup_{\substack{\lambda \in V_{N,p} \\ x \in \mathbb{C}_{N,p}(\lambda)}} \left| \frac{N^{(d-1)/2} M_{N,p}(\lambda)}{|\mathbb{C}| \varphi_{0,\Sigma_p}(x)} - 1 \right|, \end{aligned}$$

Here the inequality uses $\int_{U_{N,p}} \varphi_{0,\Sigma_p}(x) \, dx \leq 1$ and the cell normalization $\int_{U_{N,p}^c} (\mathcal{E}_{N,p}M_{N,p})(x) \, dx = M_{N,p}(V_{N,p}^c)$. Estimates (132) and (134) make all three terms on the right vanish uniformly, proving (130).

Finally, the ℓ^1 -isometry of $\mathcal{E}_{N,p}$ gives

$$\begin{aligned} \sup_{p \in K} \|\mathcal{E}_{N,p}P_{N,p} - \varphi_{0,\Sigma_p}\|_1 &\leq \sup_{p \in K} \|P_{N,p} - M_{N,p}\|_{\ell^1} \\ &\quad + \sup_{p \in K} \|\mathcal{E}_{N,p}M_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0 \end{aligned}$$

by (129) and (130). □

Lemma C.2 (Continuous limit of randomized dilation). *Let $\tilde{Q}_{m_n,p} = P_{n,p}D_n$ denote the $\tilde{\nu}$ -marginal of the joint law obtained by composing the kernel D_n with $P_{n,p}$, before the replacement in (15). For every compact $K \Subset \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma\Sigma_p}\|_1 \longrightarrow 0, \quad (135)$$

$$\sup_{p \in K} |F(\tilde{Q}_{m_n,p}, P_{m_n,p}) - F_{\text{cl}}(\gamma, d)| \longrightarrow 0. \quad (136)$$

Proof. For $a > 0$ and $f \in L^1(\mathbb{H})$, set

$$(\text{dil}_a f)(y) = a^{-(d-1)/2} f(y/\sqrt{a}),$$

and define the target-cell averaging operator explicitly by

$$(\text{Av}_{m_n,p} f)(x) := \frac{m_n^{(d-1)/2}}{|\mathbb{C}|} \int_{\mathbb{C}_{m_n,p}(\nu)} f(y) \, dy, \quad x \in \mathbb{C}_{m_n,p}(\nu).$$

Since $m_n = \gamma_n n$, the cell definitions and (127) give

$$\begin{aligned} \gamma_n^{-1/2} \mathbb{C}_{m_n,p}(\nu) &= \frac{\gamma_n^{-1}(\nu + \mathbb{C}) - np}{\sqrt{n}}, \\ \left| \gamma_n^{-1/2} \mathbb{C}_{m_n,p}(\nu) \cap \mathbb{C}_{n,p}(\mu) \right| &= \frac{|\mathbb{C}|}{n^{(d-1)/2}} D_n(\nu \mid \mu). \end{aligned}$$

Thus, for $x \in \mathcal{C}_{m_n,p}(\nu)$, changing variables by $z = y/\sqrt{\gamma_n}$ and summing over the source cells gives

$$\begin{aligned} & [\text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p}P_{n,p})](x) \\ &= \frac{m_n^{(d-1)/2}}{|\mathcal{C}|} \sum_{\mu \in \mathcal{L}_n} \frac{n^{(d-1)/2}}{|\mathcal{C}|} P_{n,p}(\mu) \left| \gamma_n^{-1/2} \mathcal{C}_{m_n,p}(\nu) \cap \mathcal{C}_{n,p}(\mu) \right| \\ &= \frac{m_n^{(d-1)/2}}{|\mathcal{C}|} \sum_{\mu \in \mathcal{L}_n} P_{n,p}(\mu) D_n(\nu \mid \mu) = (\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p})(x). \end{aligned} \tag{137}$$

For $f, g \in L^1(\mathbf{H})$,

$$\| \text{Av}_{m_n,p} f - \text{Av}_{m_n,p} g \|_1 \leq \| f - g \|_1, \quad \| \text{dil}_{\gamma_n} f - \text{dil}_{\gamma_n} g \|_1 = \| f - g \|_1.$$

By [Lemma C.1](#), $\| \mathcal{E}_{n,p}P_{n,p} - \varphi_{0,\Sigma_p} \|_1 = o(1)$ uniformly on K . Since $\text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} = \varphi_{0,\gamma_n \Sigma_p}$, dilation isometry gives $\| \text{dil}_{\gamma_n}(\mathcal{E}_{n,p}P_{n,p}) - \varphi_{0,\gamma_n \Sigma_p} \|_1 = o(1)$. Moreover, $\| \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p} \|_1 = o(1)$ uniformly because $\gamma_n \rightarrow \gamma$. Cell averaging satisfies

$$\| \text{Av}_{m_n,p} f - f \|_1 \leq 2^{d-1} \sup_{\substack{h \in \mathbf{H} \\ \|h\| \leq \text{diam}(\mathcal{C})/\sqrt{m_n}}} \| f(\cdot + h) - f \|_1.$$

Since $\gamma_n \rightarrow \gamma$ and K is compact, $\sup_{p \in K, n \geq n_0} \|\nabla \varphi_{0,\gamma_n \Sigma_p}\|_1 < \infty$ for some n_0 . Together with $\|f(\cdot + h) - f\|_1 \leq \|h\| \|\nabla f\|_1$, this gives $\| \text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma_n \Sigma_p} \|_1 = o(1)$. Using the contraction of $\text{Av}_{m_n,p}$ and (137),

$$\begin{aligned} \| \mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma \Sigma_p} \|_1 &= \| \text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p}P_{n,p}) - \varphi_{0,\gamma \Sigma_p} \|_1 \\ &\leq \| \text{Av}_{m_n,p} [\text{dil}_{\gamma_n}(\mathcal{E}_{n,p}P_{n,p}) - \varphi_{0,\gamma_n \Sigma_p}] \|_1 \\ &\quad + \| \text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma_n \Sigma_p} \|_1 + \| \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p} \|_1 \\ &\leq \| \text{dil}_{\gamma_n}(\mathcal{E}_{n,p}P_{n,p}) - \varphi_{0,\gamma_n \Sigma_p} \|_1 \\ &\quad + \| \text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma_n \Sigma_p} \|_1 + \| \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p} \|_1 = o(1). \end{aligned}$$

The equality is (137). This proves (135).

Applying [Lemma C.1](#) at sample size m_n gives

$$\mathcal{E}_{m_n,p} P_{m_n,p} \longrightarrow \varphi_{0,\Sigma_p}$$

uniformly in L^1 . The embedding preserves Hellinger affinity, and (23) also holds for nonnegative L^1 densities. Therefore

$$F(\tilde{Q}_{m_n,p}, P_{m_n,p}) \longrightarrow F(\mathbf{N}_{0,\gamma \Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = F_{\text{cl}}(\gamma, d)$$

uniformly on K . □

Proof of Proposition 2.4. For $i < d$, the nearest-integer rule gives

$$|\tilde{\nu}_i - \gamma_n \mu_i| \leq \frac{\gamma_n + 1}{2}.$$

Since the coordinate sums of $\tilde{\nu}$ and $\gamma_n \mu$ are both m_n ,

$$|\tilde{\nu}_d - \gamma_n \mu_d| \leq \frac{(d-1)(\gamma_n + 1)}{2}.$$

Thus one may take

$$c_{\text{rnd}} = \frac{d-1}{2} \left(1 + \sup_n \gamma_n \right) < \infty.$$

Compactness of K gives $c_0 > 0$ such that, for all large n , every $p \in K$ and $\mu \in \mathbb{T}_n(p)$ satisfy

$$\mu_d \geq c_0 n, \quad \mu_i - \mu_{i+1} \geq c_0 n \quad (i < d).$$

For every $\tilde{\nu} \in \text{supp } D_n(\cdot \mid \mu)$,

$$(\tilde{\nu} - \mu)_d \geq (\gamma_n - 1)\mu_d - c_{\text{rnd}},$$

and

$$(\tilde{\nu}_i - \mu_i) - (\tilde{\nu}_{i+1} - \mu_{i+1}) \geq (\gamma_n - 1)(\mu_i - \mu_{i+1}) - 2c_{\text{rnd}}.$$

These quantities are positive for all large n , uniformly on K . Hence $\tilde{\nu} - \mu$ is a partition; since μ is a partition, so is $\tilde{\nu}$. Moreover, $\tilde{\nu}_d > \mu_d > 0$, so $\tilde{\nu} \neq \nu_n^{\text{fb}}$. Hence, for $\mu \in \mathbb{T}_n(p)$, the law $D_n(\cdot \mid \mu)$ is supported on $\mathbb{Y}_{m_n}$, assigns no mass to ν_n^{fb} , and undergoes no replacement in (15). Consequently,

$$K_n^{\text{rnd}}(\nu \mid \mu) = D_n(\nu \mid \mu), \quad \nu \in \mathbb{Y}_{m_n},$$

and every ν in this support satisfies $\nu - \mu \vdash m_n - n$ and $\|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}$.

Let $\varepsilon_{n,p}$ be the probability, under $\mu \sim P_{n,p}$ and $\tilde{\nu} \sim D_n(\cdot \mid \mu)$, that $\tilde{\nu}$ is replaced in (15). The preceding argument gives

$$\varepsilon_{n,p} \leq P_{n,p}(\mathbb{T}_n(p)^c),$$

which tends uniformly to zero by Lemma 2.1.

By (136), the unreplaced output law has affinity $F_{\text{cl}}(\gamma, d) + o(1)$ with $P_{m_n,p}$, uniformly on K . Moving the rejected mass to ν_n^{fb} changes the law by at most $2\varepsilon_{n,p}$ in ℓ^1 . Applying (23) to classical distributions therefore gives (16). $\square$

D Unified Gaussian flat-prior converse

This appendix proves the faithful thermal building block used in the two converse inequalities of Theorem 3.1 by averaging any candidate channel Λ over larger parameter boxes and testing the unavoidable noise in its output. The fixed-rank theorem uses this argument for all positive-temperature modes and the explicit vacuum-support compression of Section 4.3 when some $q_{ij} = 0$. Lemma D.1 turns fidelity into such a test expectation. Lemma D.2 shows that any bounded sequence of nearly displacement-covariant maps has a subsequence converging to an exactly covariant map with the same trace bound. Proposition D.3 shows that every covariant map with the required gain is at least as noisy as the amplifier, as measured by the number test, even allowing for lost trace. Lemma D.4 governs the classical-distribution part of a classical–quantum channel: translation covariance forces it to scale x to $\sqrt{\gamma}x$, add independent noise, and possibly lose mass. Proposition D.5 combines these bounds, giving $F_{\text{orb}}(\gamma, p)$ for $\Phi_z^{(p)} \mapsto \Psi_z^{(p,\gamma)}$ and $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p)$ for $\Theta_{h,z}^{\text{in}} \mapsto \Theta_{h,z}^{\text{tar}}$.

The classical parameter space has dimension $k = 0$ for the orbital problem and $k = d - 1$ for the hybrid classical–quantum problem. All trace-class-valued integrals below are Bochner integrals in trace norm.

D.1 Weighted fidelity and compact limits

Lemma D.1 (Weighted root-fidelity bound). *Let R, T be positive trace-class operators, and let W be bounded, positive, and injective. Define*

$$\mathrm{Tr}(TW^{-1}) := \lim_{\varepsilon \downarrow 0} \mathrm{Tr}(T(W + \varepsilon I)^{-1}) \in [0, \infty].$$

If this quantity is finite, then

$$F(R, T)^2 \leq \mathrm{Tr}(RW) \mathrm{Tr}(TW^{-1}). \quad (138)$$

Let $\mathbf{E}$ be a finite-dimensional Euclidean space, let $R \in L^1(\mathbf{E}; \mathcal{T}_1(\mathcal{F}_{\mathrm{orb}}))$ be positive, let g be a probability density on $\mathbf{E}$, and let $\chi > 0$ be measurable. For positive R, S in this L^1 -space, set $F(R, S) = \int_{\mathbf{E}} F(R(y), S(y)) \, dy$, extending the hybrid-state convention in [Section 3](#). The notation gT means the density $y \mapsto g(y)T$. Whenever the right-hand side below is finite,

$$F(R, gT)^2 \leq \left(\int_{\mathbf{E}} \chi(y) \mathrm{Tr}[R(y)W] \, dy \right) \left(\int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right) \mathrm{Tr}(TW^{-1}). \quad (139)$$

Proof. For $W_\varepsilon = W + \varepsilon I$, the Schatten Hölder inequality gives

$$F(R, T) = \|(R^{1/2}W_\varepsilon^{1/2})(W_\varepsilon^{-1/2}T^{1/2})\|_1 \leq \sqrt{\mathrm{Tr}(RW_\varepsilon)} \sqrt{\mathrm{Tr}(TW_\varepsilon^{-1})}.$$

Squaring and letting $\varepsilon \downarrow 0$, using $\mathrm{Tr}(RW_\varepsilon) = \mathrm{Tr}(RW) + \varepsilon \mathrm{Tr} R$ and monotone convergence for the second trace, proves [\(138\)](#). For the hybrid state,

$$F(R, gT) = \int_{\mathbf{E}} \sqrt{g(y)} F(R(y), T) \, dy.$$

Apply the first inequality pointwise and then Cauchy–Schwarz to the factors $\sqrt{\chi(y) \mathrm{Tr}[R(y)W]}$ and $\sqrt{g(y)/\chi(y)}$. This gives [\(139\)](#). $\square$

Lemma D.2 (Compact-observable limits of completely positive maps). *Let $\mathcal{H}_0, \mathcal{H}_1$ be separable Hilbert spaces and let $\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ be completely positive maps with $\sup_j \|\Gamma_j\|_{1 \rightarrow 1} < \infty$. There are a subsequence and a completely positive map $\Gamma : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ such that*

$$\mathrm{Tr}[\Gamma_j(X)K] \longrightarrow \mathrm{Tr}[\Gamma(X)K] \quad (140)$$

for every $X \in \mathcal{T}_1(\mathcal{H}_0)$ and every compact $K \in \mathcal{K}(\mathcal{H}_1)$. If $\mathrm{Tr} \Gamma_j(X) \leq c \mathrm{Tr} X$ for every $X \geq 0$, then $\mathrm{Tr} \Gamma(X) \leq c \mathrm{Tr} X$.

Suppose in addition that $U_g \in \mathrm{U}(\mathcal{H}_0)$ and $V_g \in \mathrm{U}(\mathcal{H}_1)$ are families of unitaries indexed by a set G , and that, for every fixed $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$,

$$\|\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*\|_1 \longrightarrow 0.$$

Then

$$\Gamma(U_g X U_g^*) = V_g \Gamma(X) V_g^*$$

for every $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$.

Proof. Let $M = \sup_j \|\Gamma_j\|_{1 \rightarrow 1}$, and let $\mathcal{D} \subset \mathcal{K}(\mathcal{H}_1)$ be a countable norm-dense $\mathbb{Q}(i)$ -linear $*$ -algebra of finite-rank operators. The predual $\mathcal{T}_1(\mathcal{H}_0)$ is separable, so a diagonal weak- $*$ compactness argument gives a subsequence for which $\Gamma_j^*(K)$ converges weak- $*$ in $\mathcal{B}(\mathcal{H}_0)$ for every $K \in \mathcal{D}$. The bound $\|\Gamma_j^*(K)\| \leq M\|K\|$ extends both the limit and the convergence to every compact K , producing a bounded complex-linear map $\Gamma^* : \mathcal{K}(\mathcal{H}_1) \rightarrow \mathcal{B}(\mathcal{H}_0)$.

To check complete positivity, fix r and $A \geq 0$ in $M_r(\mathcal{K}(\mathcal{H}_1))$. Since $A^{1/2}$ is compact, choose $B_\ell \in M_r(\mathcal{D})$ with $B_\ell \rightarrow A^{1/2}$. Since $\Gamma_j^{*(r)}(B_\ell^* B_\ell) \geq 0$, first letting $j \rightarrow \infty$ and then $\ell \rightarrow \infty$ gives $\Gamma^{*(r)}(A) \geq 0$.

Since $\mathcal{K}(\mathcal{H}_1)^* = \mathcal{T}_1(\mathcal{H}_1)$, define $\Gamma(X)$ by

$$\mathrm{Tr}[\Gamma(X)K] = \mathrm{Tr}[X\Gamma^*(K)] \quad (K \in \mathcal{K}(\mathcal{H}_1)).$$

Then Γ is completely positive and (140) holds. If $P_N \uparrow I$ are finite-rank projections and $X \geq 0$, then

$$\mathrm{Tr} \Gamma(X) = \sup_N \lim_j \mathrm{Tr}[\Gamma_j(X)P_N] \leq c \mathrm{Tr} X.$$

Finally, for fixed g, X and compact K , the operator $V_g^* K V_g$ is compact, and hence

$$\begin{aligned} & \mathrm{Tr}[(\Gamma(U_g X U_g^*) - V_g \Gamma(X) V_g^*) K] \\ &= \lim_j \mathrm{Tr}[(\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*) K] = 0. \end{aligned}$$

Compact operators separate trace-class operators, so the limiting map is covariant. $\square$

D.2 The least-noise oscillator moment

Fix $\gamma > 1$. For $s \geq 1$ and $q_1, \dots, q_s \in (0, 1)$, put

$$q_\ell^{(\gamma)} = 1 - \frac{1 - q_\ell}{\gamma}, \quad \tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell^{(\gamma)}}.$$

The proof below uses the following two external inputs, isolated here to make the dependence on [GM06] explicit.

Fact D.I (Covariant-amplifier representation). *Every displacement-covariant one-mode channel of amplitude gain $\sqrt{\gamma}$ admits an idler realization*

$$a_{\mathrm{out}} = \sqrt{\gamma} a_{\mathrm{in}} + \sqrt{\gamma - 1} b^*.$$

The idler state need not be Gaussian; if the channel is also phase covariant, it may be chosen number diagonal [GM06, Sec. II.C and Appendix; Sec. III for general gain].

Fact D.II (Vacuum-idler stochastic order). *For a thermal input τ_q and a number-diagonal idler, let p_k be the output number distribution of the amplifier in Fact D.I. Then*

$$\sum_{k=0}^m (1 - q^{(\gamma)}) (q^{(\gamma)})^k \geq \sum_{k=0}^m p_k \quad (m \geq 0).$$

Thus the vacuum-idler output $\tau_{q^{(\gamma)}}$ is stochastically smallest among these outputs [GM06, Lem. 2.3 and Sec. III].

Proposition D.3 (Least-noise moment). *Let Γ be a completely positive trace-nonincreasing map on the s -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

If

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^s w_{\ell}(\hat{N}_{\ell}),$$

where every w_{ℓ} is bounded, nonnegative, and nonincreasing on $\mathbb{Z}_{\geq 0}$, then

$$\mathrm{Tr}[\Gamma(\tau)W(\hat{\mathbf{N}})] \leq \mathrm{Tr}[\Gamma(\tau)] \mathrm{Tr}[\tau^{(\gamma)}W(\hat{\mathbf{N}})]. \quad (141)$$

Proof. Averaging Γ over independent phase rotations preserves its trace bound and displacement covariance, and it does not change either side of (141). We may therefore assume that Γ is phase covariant in every mode.

Consider one mode. Write $\mathrm{Tr} \Gamma(X) = \mathrm{Tr}(AX)$ with $0 \leq A \leq I$. Displacement covariance implies $D(z)^*AD(z) = A$; hence irreducibility of the Weyl representation gives $A = cI$, where $c = \mathrm{Tr} \Gamma(\tau_q)$. If $c > 0$, then $\Lambda = c^{-1}\Gamma$ is a phase-covariant amplifier of amplitude gain $\sqrt{\gamma}$. By Fact D.I, it has a number-diagonal idler realization. Fact D.II therefore gives

$$\sum_{n=0}^m (1 - q^{(\gamma)})(q^{(\gamma)})^n \geq \sum_{n=0}^m \langle n | \Lambda(\tau_q) | n \rangle \quad (m \geq 0). \quad (142)$$

The distribution on the left is the vacuum-idler output, namely that of $\tau_{q^{(\gamma)}}$, and both output states are number diagonal. For a bounded nonincreasing w , summation by parts writes its expectation as a nonnegative linear combination of these cumulative probabilities. Both distributions in (142) have total mass one, so the common limiting constant cancels. Hence

$$\mathrm{Tr}[\Lambda(\tau_q)w(\hat{N})] \leq \mathrm{Tr}[\tau_{q^{(\gamma)}}w(\hat{N})].$$

Multiplication by c proves the one-mode case; the case $c = 0$ is immediate.

For the multimode case, the claim is immediate if some w_{ℓ} vanishes identically; otherwise, divide each w_{ℓ} by $w_{\ell}(0)$ and assume $0 \leq w_{\ell} \leq 1$. Put

$$W_{\geq \ell} = I_1 \otimes \cdots \otimes I_{\ell-1} \otimes \bigotimes_{j=\ell}^s w_j(\hat{N}_j), \quad W_{\geq s+1} = I,$$

and, for an operator X on mode ℓ , define

$$\Gamma_{\ell}(X) = \mathrm{Tr}_{\neq \ell} \left[W_{\geq \ell+1}^{1/2} \Gamma(\tau_{q_1} \otimes \cdots \otimes \tau_{q_{\ell-1}} \otimes X \otimes \tau_{q_{\ell+1}} \otimes \cdots \otimes \tau_{q_s}) W_{\geq \ell+1}^{1/2} \right].$$

Because $0 \leq W_{\geq \ell+1} \leq I$ and this operator is the identity on mode ℓ , each Γ_{ℓ} is completely positive, trace nonincreasing, and displacement covariant in that mode. The phase-covariant reduction following (141) also makes Γ_{ℓ} phase covariant because the fixed thermal inputs and $W_{\geq \ell+1}$ are phase invariant. Moreover,

$$\mathrm{Tr}[\Gamma_{\ell}(\tau_{q_{\ell}})w_{\ell}(\hat{N}_{\ell})] = \mathrm{Tr}[\Gamma(\tau)W_{\geq \ell}], \quad \mathrm{Tr} \Gamma_{\ell}(\tau_{q_{\ell}}) = \mathrm{Tr}[\Gamma(\tau)W_{\geq \ell+1}].$$

Applying the one-mode inequality to Γ_{ℓ} and iterating over $\ell = 1, \dots, s$ gives (141). $\square$

Specialize to $s = r_d$, $q_\ell = q_{ij}$, $\tau = \tau_p$, and $\tau^{(\gamma)} = \tau_p^{(\gamma)}$. For $\mathbf{k} = (k_{ij})_{i < j} \in \mathbb{Z}_{\geq 0}^{r_d}$, let $t_{\mathbf{k}}$ and $t_{\mathbf{k}}^{(\gamma)}$ be the corresponding number-basis eigenvalues, and define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (143)$$

Equivalently,

$$W_\gamma = \prod_{i < j} w_{ij}(\hat{N}_{ij}), \quad w_{ij}(n) = \sqrt{\frac{1 - q_{ij}}{1 - q_{ij}^{(\gamma)}}} \left(\frac{q_{ij}}{q_{ij}^{(\gamma)}} \right)^{n/2}.$$

Here $\hat{N}_{ij}$ is the number operator of the mode indexed by $i < j$, and $|\mathbf{k}| := \sum_{i < j} k_{ij}$ is the total occupation number. Since $q_{ij} < q_{ij}^{(\gamma)}$, each $w_{ij}(n)$ is positive, bounded, and strictly decreasing in n , and the eigenvalues of W_γ tend to zero as $|\mathbf{k}| \rightarrow \infty$. Thus W_γ is bounded, injective, and compact. Its definition gives

$$\text{Tr}(\tau_p^{(\gamma)} W_\gamma) = \text{Tr}(\tau_p W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = F_{\text{orb}}(\gamma, p), \quad (144)$$

where the inverse is understood as in [Lemma D.1](#).

D.3 Classical trace kernels

Lemma D.4 (Translation-covariant scalar kernels). *Let $\mathbf{E}$ be a finite-dimensional Euclidean space and let $\mathcal{R} : L^1(\mathbf{E}) \rightarrow C_0(\mathbf{E})^*$ be a positive contraction. For a finite Radon measure λ , define $T_a \lambda(A) = \lambda(A - a)$ for Borel A . Suppose*

$$\mathcal{R}(u(\cdot - h)) = T_{\sqrt{\gamma}h} \mathcal{R}(u) \quad (u \in L^1(\mathbf{E}), h \in \mathbf{E}) \quad (145)$$

as an equality of finite Radon measures. Then there is a finite positive measure μ , with $\mu(\mathbf{E}) \leq 1$, such that

$$\mathcal{R}(u)(A) = \int_{\mathbf{E}} u(x) \mu(A - \sqrt{\gamma}x) dx \quad (146)$$

for every $u \in L^1(\mathbf{E})$ and every Borel set A .

Proof. If $\mathbf{E} = \{0\}$, take $\mu = \mathcal{R}(1)$. Otherwise choose a nonnegative approximate identity $u_j \in C_c(\mathbf{E})$ with $\int u_j = 1$ and $\text{supp } u_j \subset \{x : \|x\| \leq j^{-1}\}$. The positive measures $\mathcal{R}(u_j)$ lie in the unit ball of $C_0(\mathbf{E})^*$, so weak-* compactness and separability yield a positive subsequential weak-* limit μ with $\mu(\mathbf{E}) \leq 1$.

For $u \in L^1(\mathbf{E})$, the identity $u * u_j = \int_{\mathbf{E}} u(x) u_j(\cdot - x) dx$ holds as a Bochner integral in $L^1(\mathbf{E})$, so bounded linearity and covariance give

$$\mathcal{R}(u * u_j) = \int_{\mathbf{E}} u(x) T_{\sqrt{\gamma}x} \mathcal{R}(u_j) dx.$$

Pair with $\phi \in C_0(\mathbf{E})$. The approximate-identity property and contractivity give $\mathcal{R}(u * u_j) \rightarrow \mathcal{R}(u)$ in total variation. On the right, weak-* convergence applies pointwise because $y \mapsto \phi(y + \sqrt{\gamma}x)$ lies in $C_0(\mathbf{E})$, and the integrand is bounded by $|u(x)| \|\phi\|_\infty$. Dominated convergence gives

$$\langle \phi, \mathcal{R}(u) \rangle = \int_{\mathbf{E}} u(x) \langle \phi, T_{\sqrt{\gamma}x} \mu \rangle dx.$$

By definition of T_a , the right-hand side pairs ϕ with the finite Radon measure in (146), whose total variation is at most $\|u\|_1 \mu(\mathbf{E})$. Riesz uniqueness proves the claim. $\square$

D.4 One flat-prior converse for both Gaussian models

Proposition D.5 (Flat-prior converse inequality). *Let $k \in \{0, d-1\}$. For $k = 0$, let $\mathbf{E} = \{0\}$, omit the classical factor, and interpret h -integration as evaluation at 0. For $k = d-1$, let $\mathbf{E} = \mathbf{H}$ with its induced Euclidean measure. Let*

$$\mathbf{E}_L = \{h \in \mathbf{E} : \|h\|_\infty \leq L\}, \quad \Xi_L = \mathbf{E}_L \times \mathbf{Z}_L,$$

with $|\mathbf{E}_L| = 1$ when $k = 0$. Define

$$R_{0,z}^{(0)} = \Phi_z^{(p)}, \quad S_{0,z}^{(0)} = \Psi_z^{(p,\gamma)},$$

$$R_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{in}}, \quad S_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{tar}}, \quad c_k = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{k/2}.$$

For $k = 0$, the supremum below is over quantum channels on $\mathcal{F}_{\text{orb}}$; for $k = d-1$, it is over classical-quantum channels. Then

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_{h,z}^{(k)}), S_{h,z}^{(k)}) \, dh \, dz \leq c_k F_{\text{orb}}(\gamma, p). \quad (147)$$

Proof. For $k = 0$, set $f_1 = f_\gamma = \chi = 1$ on the one-point space. For $k = d-1$, let f_1 and f_γ be the centered Gaussian densities with covariances Σ_p and $\gamma\Sigma_p$, and set

$$\chi(y) = \sqrt{\frac{f_1(y)}{f_\gamma(y)}} = \gamma^{k/4} \exp\left[-\frac{\gamma-1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle\right].$$

This function is bounded and strictly positive; for $k = d-1$, it also belongs to $C_0(\mathbf{E})$. In both cases,

$$\int_{\mathbf{E}} \frac{f_1(y)}{\chi(y)} \, dy = \int_{\mathbf{E}} \sqrt{f_1(y)f_\gamma(y)} \, dy = c_k. \quad (148)$$

For $\xi = (h, z)$, define

$$(\mathcal{U}_\xi^{\text{in}} R)(x) = D(z)R(x-h)D(z)^*, \quad (\mathcal{U}_\xi^{\text{out}} R)(y) = D(\sqrt{\gamma}z)R(y-\sqrt{\gamma}h)D(\sqrt{\gamma}z)^*.$$

Then $R_\xi^{(k)} = \mathcal{U}_\xi^{\text{in}}(f_1\tau_p)$ and $S_\xi^{(k)} = \mathcal{U}_\xi^{\text{out}}(f_1\tau_p)$. For a channel Λ , let

$$\bar{\Lambda}_L = \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\eta}^{\text{out}} \circ \Lambda \circ \mathcal{U}_\eta^{\text{in}} \, d\eta.$$

Invariance of F under the isometries $\mathcal{U}_\xi^{\text{out}}$ and joint concavity, applied to the Bochner average, give

$$F_L(\Lambda) = \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_\xi^{(k)}), S_\xi^{(k)}) \, d\xi \leq F(\bar{\Lambda}_L(f_1\tau_p), f_1\tau_p). \quad (149)$$

Let $F_* = \limsup_{L \rightarrow \infty} \sup_{\Lambda} F_L(\Lambda)$. Choose $L_j \rightarrow \infty$ with $\sup_{\Lambda} F_{L_j}(\Lambda) \rightarrow F_*$, and channels $\tilde{\Lambda}_j$ satisfying $F_{L_j}(\tilde{\Lambda}_j) \geq \sup_{\Lambda} F_{L_j}(\Lambda) - j^{-1}$. Set

$$\Lambda_j = \overline{\tilde{\Lambda}_{jL_j}}, \quad R_j = \Lambda_j(f_1\tau_p).$$

Then

$$F_{L_j}(\tilde{\Lambda}_j) \rightarrow F_*, \quad F_{L_j}(\tilde{\Lambda}_j) \leq F(R_j, f_1\tau_p). \quad (150)$$

For $\eta \in \mathbb{E} \times \mathbb{C}^{r_d}$, write

$$A_{j,\eta} = \mathcal{U}_{-\eta}^{\text{out}} \circ \tilde{\Lambda}_j \circ \mathcal{U}_\eta^{\text{in}}, \quad \Lambda_j = \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,\eta} \, d\eta.$$

The group law and the change of variables $u = \eta + \xi$ give

$$\begin{aligned} \Lambda_j(\mathcal{U}_\xi^{\text{in}} R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j} + \xi} A_{j,u}(R) \, du, \\ \mathcal{U}_\xi^{\text{out}} \Lambda_j(R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,u}(R) \, du. \end{aligned}$$

The two integrals cancel on the intersection of the two averaging sets, and $\|A_{j,u}\|_{1 \rightarrow 1} \leq 1$. Therefore, for fixed ξ ,

$$\|\Lambda_j(\mathcal{U}_\xi^{\text{in}} R) - \mathcal{U}_\xi^{\text{out}} \Lambda_j(R)\|_1 \leq \frac{|(\Xi_{L_j} + \xi) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|} \|R\|_1 \longrightarrow 0. \quad (151)$$

Indeed, Ξ_1 is a bounded polytope and $\Xi_L = L\Xi_1$, so the ratio is $O_\xi(L^{-1})$.

Define

$$\Gamma_j^\chi(X) = \int_{\mathbb{E}} \chi(y) [\Lambda_j(f_1 X)](y) \, dy.$$

These maps are completely positive and satisfy $\|\Gamma_j^\chi\|_{1 \rightarrow 1} \leq \|\chi\|_\infty$. Taking $\xi = (0, z)$ in (151), multiplying by χ , and integrating gives

$$\begin{aligned} &\|\Gamma_j^\chi(D(z)X D(z)^*) - D(\sqrt{\gamma}z) \Gamma_j^\chi(X) D(\sqrt{\gamma}z)^*\|_1 \\ &\leq \|\chi\|_\infty \frac{|(\Xi_{L_j} + (0, z)) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|} \|X\|_1 \longrightarrow 0. \end{aligned}$$

Hence Lemma D.2 gives, after passing to a subsequence, a completely positive displacement-covariant limit Γ^χ such that

$$\text{Tr}[\Gamma_j^\chi(X)K] \longrightarrow \text{Tr}[\Gamma^\chi(X)K] \quad (152)$$

for trace-class X and compact K . With

$$M_j = \int_{\mathbb{E}} \chi(y) \text{Tr}[R_j(y)W_\gamma] \, dy = \text{Tr}[\Gamma_j^\chi(\tau_p)W_\gamma],$$

compactness of W_γ gives

$$M_j \longrightarrow \text{Tr}[\Gamma^\chi(\tau_p)W_\gamma]. \quad (153)$$

Moreover, Lemma D.1, (148), and (144) give

$$F(R_j, f_1 \tau_p)^2 \leq M_j c_k F_{\text{orb}}(\gamma, p). \quad (154)$$

If $k = 0$, then $\Gamma_j^\chi = \Lambda_j$ are channels. The trace-bound clause of Lemma D.2 with $c = 1$ makes the limit trace nonincreasing. By Proposition D.3,

$$\begin{aligned} \lim_j M_j &= \text{Tr}[\Gamma^\chi(\tau_p)W_\gamma] \\ &\leq \text{Tr} \Gamma^\chi(\tau_p) \text{Tr}(\tau_p^{(\gamma)} W_\gamma) \leq F_{\text{orb}}(\gamma, p). \end{aligned}$$

Combining (150) with (154) yields $F_* \leq F_{\text{orb}}(\gamma, p)$.

Assume henceforth that $k = d - 1$. For $u \in L^1(\mathbf{E})$, define

$$\mathcal{R}_j(u)(A) = \int_A \text{Tr}([\Lambda_j(u\tau_p)](y)) \, dy.$$

The maps $\mathcal{R}_j : L^1(\mathbf{E}) \rightarrow C_0(\mathbf{E})^*$ are positive contractions. Using countable dense subsets of $L^1(\mathbf{E})$ and $C_0(\mathbf{E})$, pass to a further subsequence converging weak-* on every u to a positive contraction $\mathcal{R}$. Taking $z = 0$ in (151), pairing with a function in $C_0(\mathbf{E})$, and passing to the limit gives

$$\mathcal{R}(u(\cdot - h)) = T_{\sqrt{\gamma}h} \mathcal{R}(u).$$

Let μ be the measure from Lemma D.4; then $\mu(\mathbf{E}) \leq 1$.

Finite-rank projections, (152), the weak-* convergence of $\mathcal{R}_j(f_1)$, and $\chi \in C_0(\mathbf{E})$ give

$$\begin{aligned} \text{Tr } \Gamma^\chi(\tau_p) &\leq \liminf_j \text{Tr } \Gamma_j^\chi(\tau_p) \\ &= \lim_j \int_{\mathbf{E}} \chi(y) \mathcal{R}_j(f_1)(dy) = \int_{\mathbf{E}} \chi(y) \mathcal{R}(f_1)(dy). \end{aligned}$$

By (146), the last integral equals

$$\int_{\mathbf{E}} \mu(dr) \int_{\mathbf{E}} f_1(x) \chi(r + \sqrt{\gamma}x) \, dx.$$

Completing the square gives

$$\int_{\mathbf{E}} f_1(x) \chi(r + \sqrt{\gamma}x) \, dx = c_k \exp\left[-\frac{\gamma - 1}{2\gamma(\gamma + 1)} \langle r, \Sigma_p^{-1}r \rangle\right].$$

Since $\mu(\mathbf{E}) \leq 1$, it follows that $\text{Tr } \Gamma^\chi(\tau_p) \leq c_k$.

Finally write $\text{Tr } \Gamma^\chi(X) = \text{Tr}(AX)$. Covariance implies that A commutes with every Weyl displacement, so $A = cI$, with

$$c = \text{Tr } \Gamma^\chi(\tau_p) \leq c_k \leq 1.$$

Thus Γ^χ is trace nonincreasing, and Proposition D.3 gives

$$\begin{aligned} \lim_j M_j &= \text{Tr}[\Gamma^\chi(\tau_p)W_\gamma] \\ &\leq \text{Tr } \Gamma^\chi(\tau_p) \text{Tr}(\tau_p^{(\gamma)}W_\gamma) \leq c_k F_{\text{orb}}(\gamma, p). \end{aligned}$$

Combining this with (150) and (154) yields

$$F_* \leq c_k F_{\text{orb}}(\gamma, p),$$

which proves the hybrid classical–quantum case. $\square$

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